

class05

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2026-04-13

TITLE HERE

Subtitle

Subsubtitle

for text in bold

text not in bold

text in italics

indent text

Q1: Make a ggplot plot for text indented and bolded

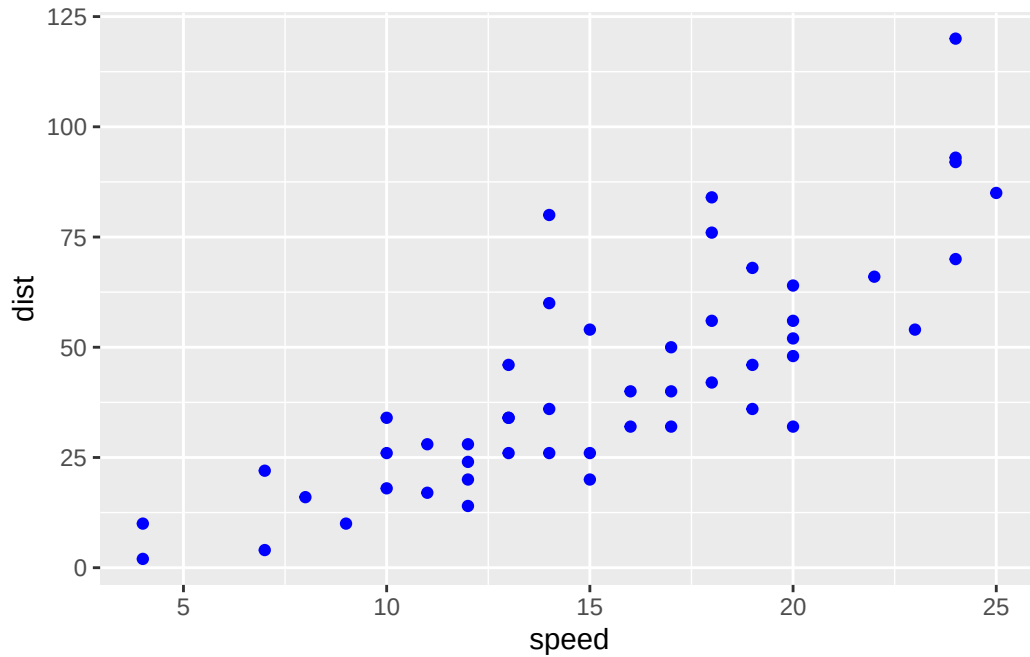
Answer Q1: meow

```
x <- 1:5  
x + 1
```

```
[1] 2 3 4 5 6
```

Let's make a graph

```
library(ggplot2)  
  
ggplot(cars) +  
  aes(x=speed, y=dist) + geom_point(col= "BLUE")
```



A bad code would look like this "four" + 2", this would not run because you cant ass non-numeric values with a binary operator

fixing bad code:

```
4+2
```

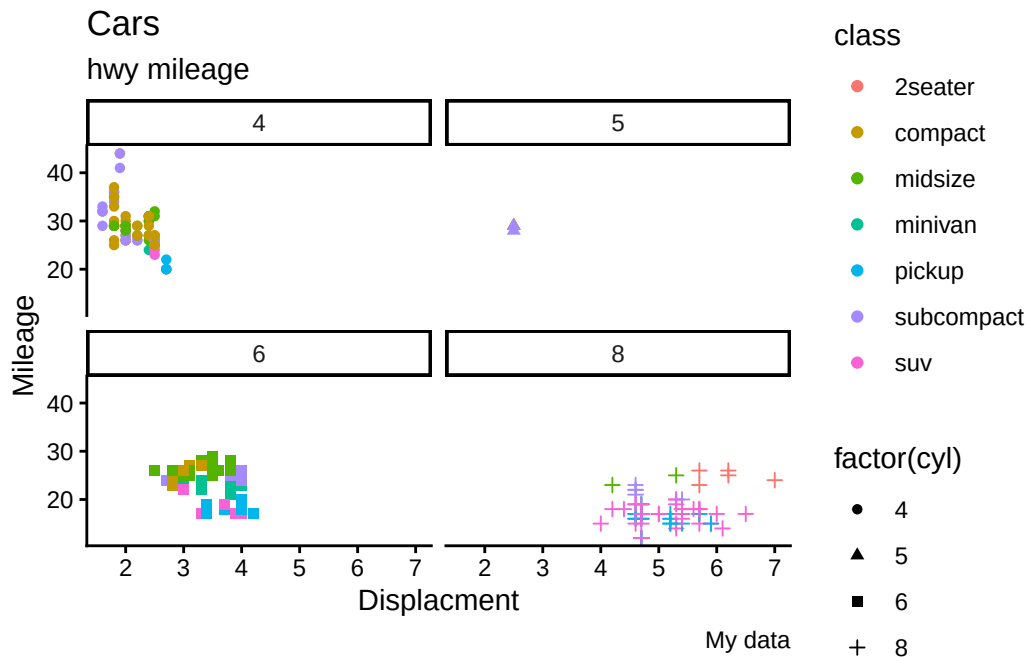
```
[1] 6
```

ggplot 2 is already installed on my laptop from previous classes

plots:

```
library(ggplot2)

ggplot(data=mpg)+
  aes(x=displ, y=hwy, color=class, shape=factor(cyl)) +
  geom_point() +
  theme_classic() +
  labs(title="Cars", subtitle= "hwy mileage", caption= "My data", x= "Displacment", y= "Miles")
  facet_wrap(~cyl)
```

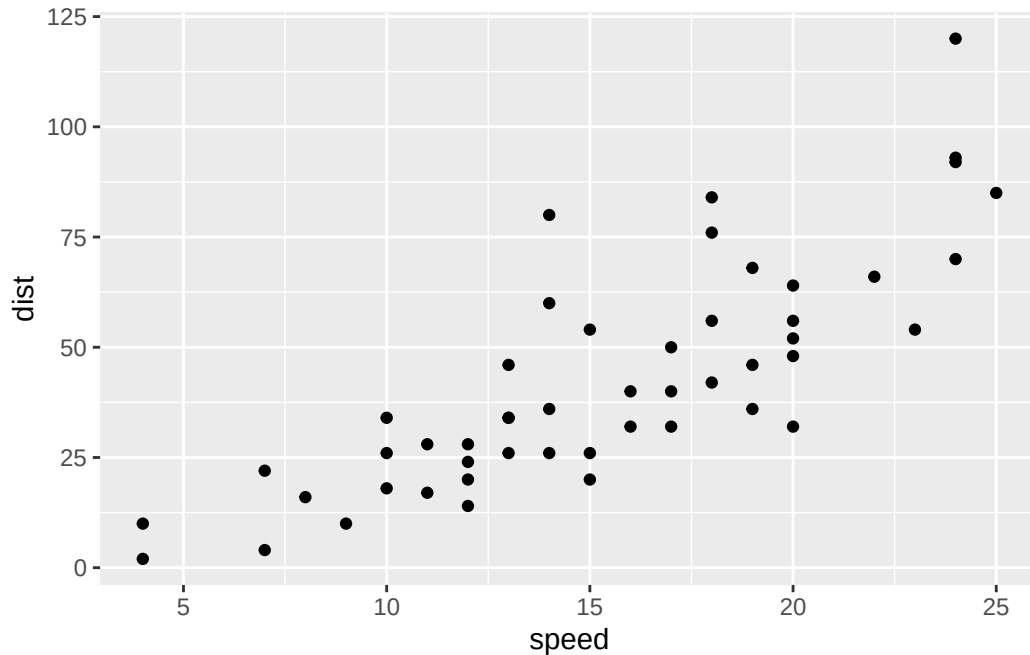


Section 6:

Specific aesthetic mappings with aes()

making the initial plot:

```
library(ggplot2)
ggplot(cars) +
  aes(x=speed, y=dist) +
  geom_point()
```



Question 1: Which geometric layer should be used to create scatter plots in ggplot2

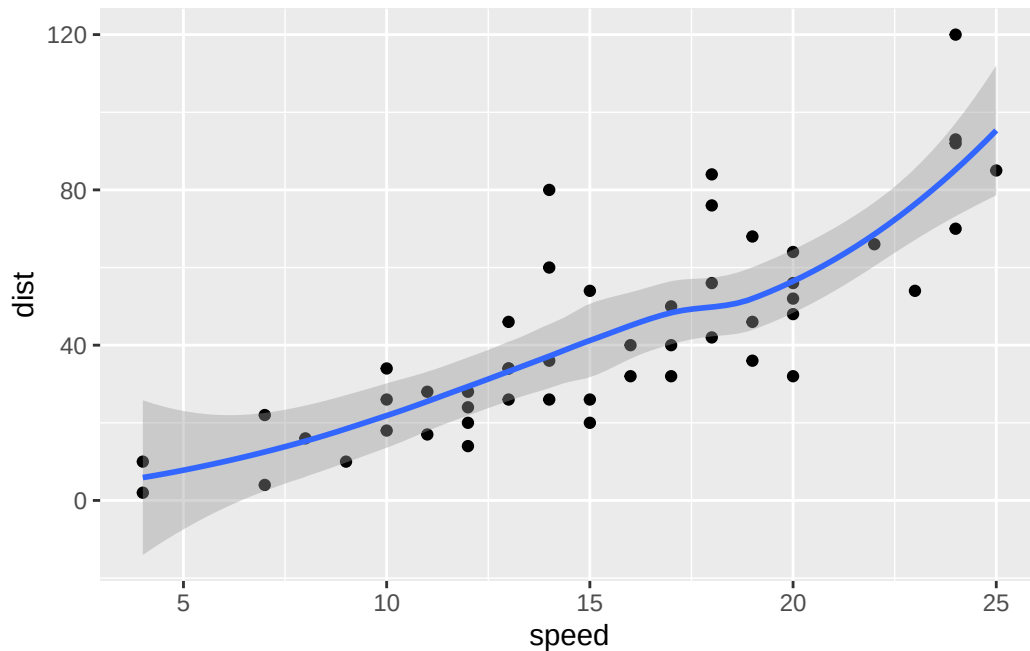
Answer 1: `geom_point` should be used to make scatter plots in ggplot2

Q2: In your own RStudio, can you add a trend line layer to help show the relationship between the plot variables with the `geom_smooth()` function?

A2: To do this, I added the code `geom_smooth()` at the end of the initial plot code

```
ggplot(cars) +  
  aes(x=speed, y=dist) +  
  geom_point() +  
  geom_smooth()
```

``geom_smooth()`` using `method = 'loess'` and `formula = 'y ~ x'`

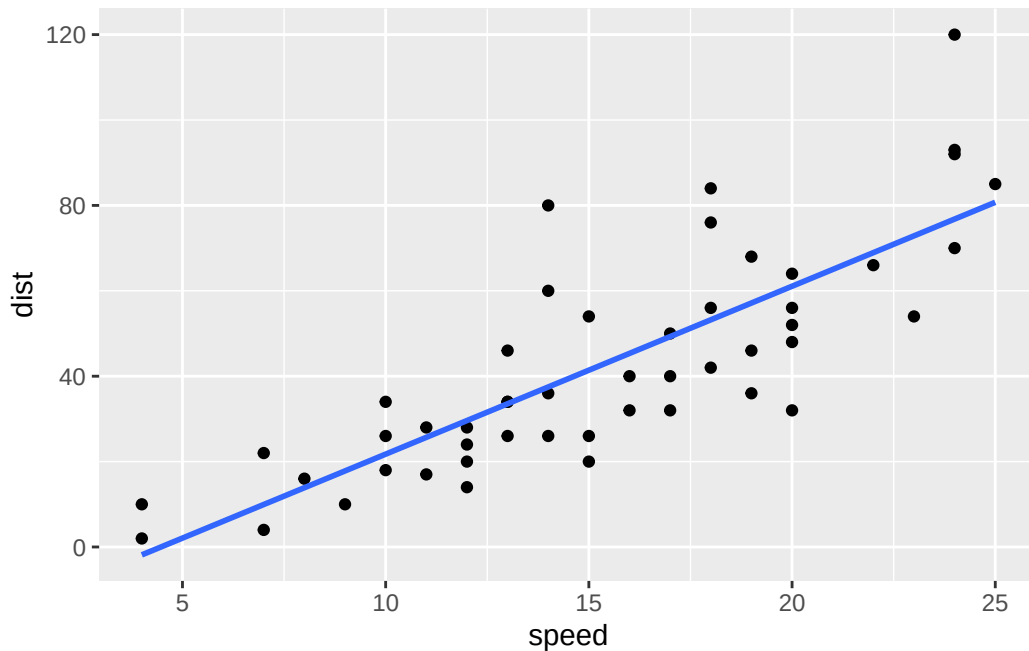


Q3: Argue with `geom_smooth()` to add a straight line from a linear model without the shaded standard error region?

A3: To do this, I added told `geom_smooth()` to use linear model (`lm`) as its method, and secondly, I set standard error to `FALSE` to indicate that I don't want it to be on the plot

```
ggplot(cars) +  
  aes(x=speed, y=dist) +  
  geom_point() +  
  geom_smooth(method="lm", se=FALSE)
```

``geom_smooth()`` using formula = 'y ~ x'



Q4: Can you finish this plot by adding various label annotations with the `labs()` function and changing the plot look to a more conservative “black & white” theme by adding the `theme_bw()` function

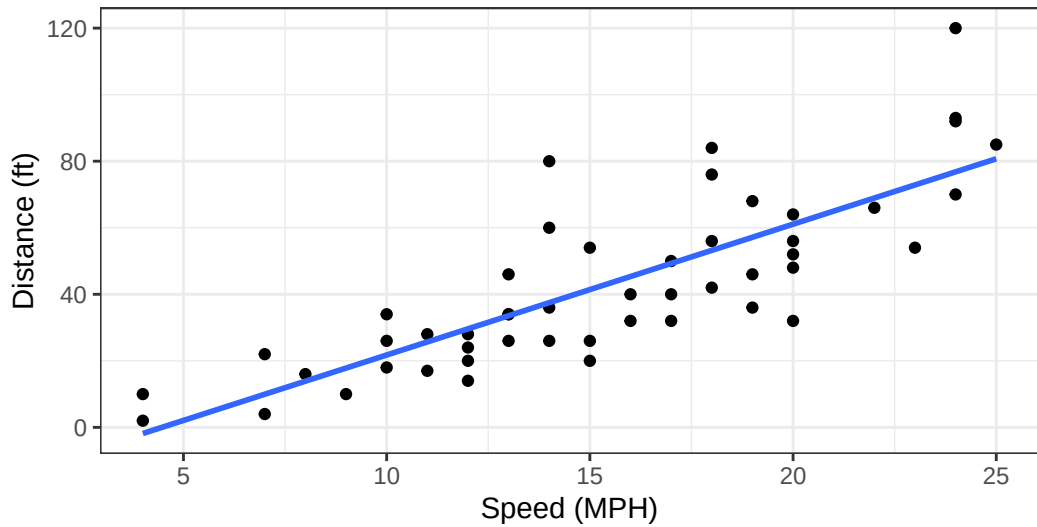
A4: to do this, I added labs at the end of the aesthetics. I labeled the graph “Speed vs. Distance to stop”, x axis as “Speed (MPH)”, y axis as “Distance (ft)”, as well as a caption indicating what dataset I used. And to add the theme, I added it at the end of the code using `theme_bw()`

```
ggplot(cars) +
  aes(x=speed, y=dist) +
  geom_point() +
  labs(title= "Speed Vs. Distance to stop",
        x= "Speed (MPH)",
        y= "Distance (ft)",
        subtitle = "How far do cars travel before stopping, depending on their speeds",
        caption = "Dataset: 'cars'")+
  geom_smooth(method="lm", se=FALSE)+
  theme_bw()
```

``geom_smooth()`` using formula = 'y ~ x'

Speed Vs. Distance to stop

How far do cars travel before stopping, depending on their speeds



Dataset: 'cars'

Adding more plot aesthetics through aes()

Adding data set to R

```
url <- "https://bioboot.github.io/bimm143_S20/class-material/up_down_expression.txt"
genes <- read.delim(url)
head(genes)
```

	Gene	Condition1	Condition2	State
1	A4GNT	-3.6808610	-3.4401355	unchanging
2	AAAS	4.5479580	4.3864126	unchanging
3	AASDH	3.7190695	3.4787276	unchanging
4	AATF	5.0784720	5.0151916	unchanging
5	AATK	0.4711421	0.5598642	unchanging
6	AB015752.4	-3.6808610	-3.5921390	unchanging

Q5: Use the `nrow()` function to find out how many genes are in this data set. What is your answer?

```
# Answer 5: number of rows: 5196
nrow(genes)
```

```
[1] 5196
```

Q6: Use the `colnames()` function and the `ncol()` function on the `genes` data frame to find out what the column names are (we will need these later) and how many columns there are. How many columns did you find?

```
# Answer 6: column name: gene, number of columns: 4
colnames(genes)
```

```
[1] "Gene"          "Condition1" "Condition2" "State"
```

```
ncol(genes)
```

```
[1] 4
```

Q7: Use the `table()` function on the `State` column of this data.frame to find out how many 'up' regulated genes there are. What is your answer?

```
# Answer 7: There are 127 unregulated genes
table(genes$State)
```

down	unchanging	up
72	4997	127

Q8: Using your values above and 2 significant figures. What fraction of total genes is up-regulated in this dataset?

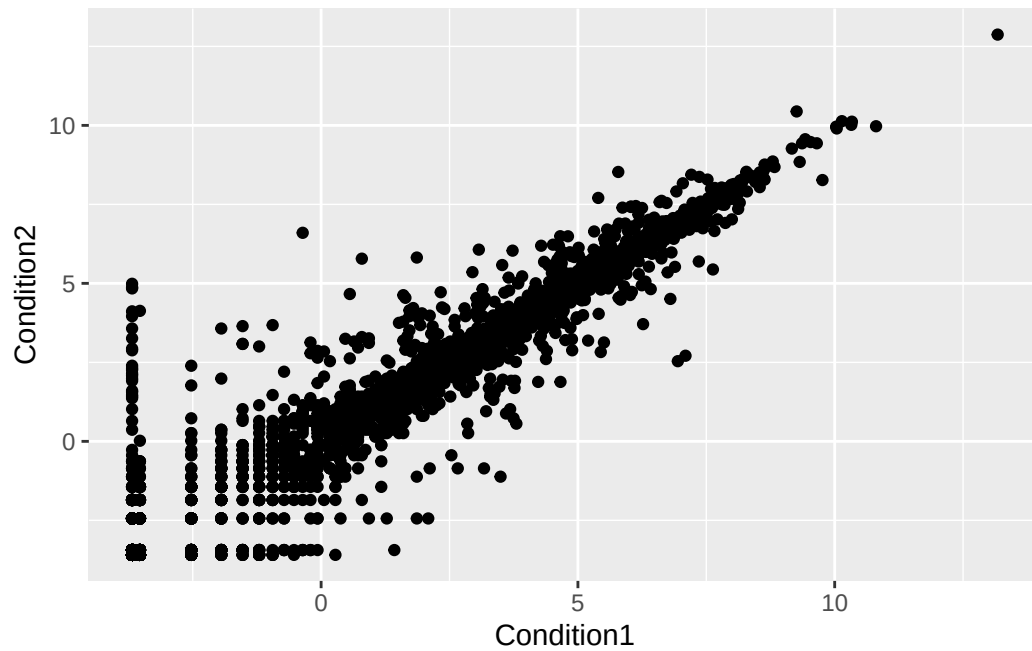
```
# Answer 8: There are 2.44% unregulated genes
round( table(genes$State)/nrow(genes)*100,2)
```

down	unchanging	up
1.39	96.17	2.44

Q9: Complete the code below to produce the following plot `ggplot() + aes(x=Condition1, y=)`

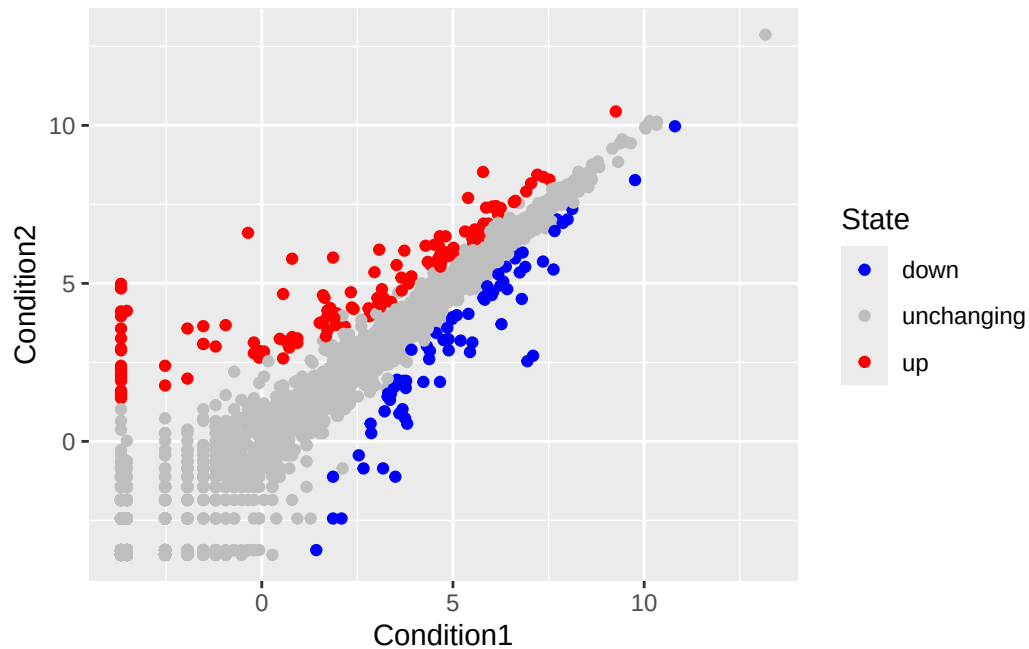
A9: filled in dataset as `genes`, y-axis as `Condition 2`, and told R to use scatter plot

```
ggplot(genes) +
  aes(x=Condition1, y=Condition2)+
  geom_point()
```



mapping plot with state

```
ggplot(genes) +
  aes(x=Condition1, y=Condition2, col=State) +
  geom_point() +
  scale_colour_manual( values=c("blue","gray","red") )
```

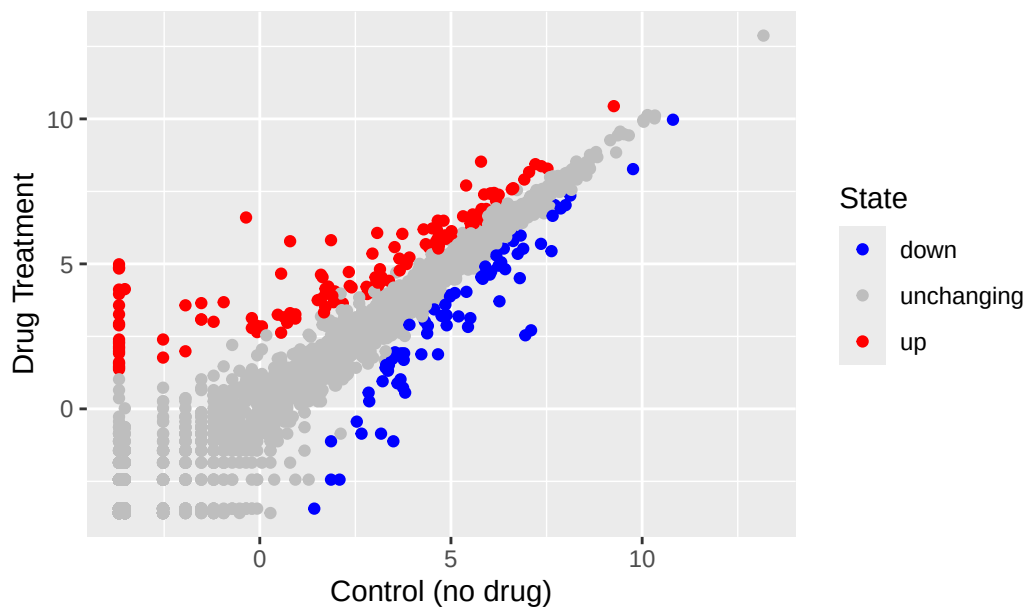


Q10: Nice, now add some plot annotations to the p object with the labs() function so your plot looks like the following:

A10: Added labels to the graph

```
ggplot(genes) +
  aes(x=Condition1, y=Condition2, col=State) +
  geom_point() +
  scale_colour_manual( values=c("blue","gray","red") )+
  labs(title= "Gene Expression Changes Upon Drug Treatment", x= "Control (no drug)", y= "Drug")
```

Gene Expression Changes Upon Drug Treatment



SECTION 7

installing package

```
url <- "https://raw.githubusercontent.com/jennybc/gapminder/master/inst/extdata/gapminder.tsv"

gapminder <- read.delim(url)

# install.packages("dplyr") ## un-comment to install if needed
library(dplyr)
```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

filter, lag

The following objects are masked from 'package:base':

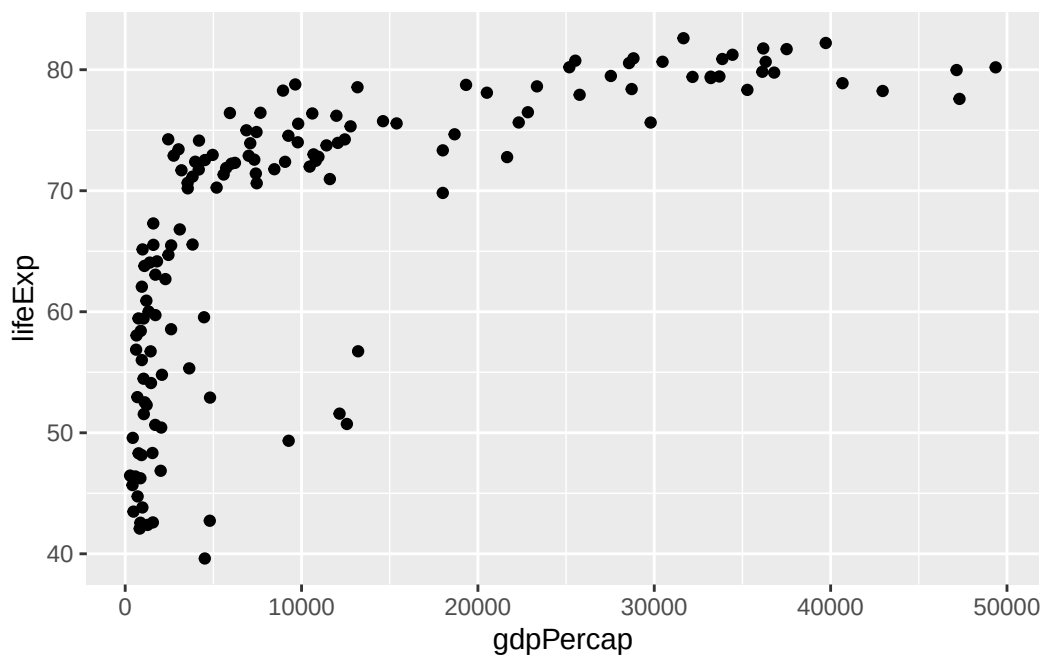
intersect, setdiff, setequal, union

```
gapminder_2007 <- gapminder %>% filter(year==2007)
```

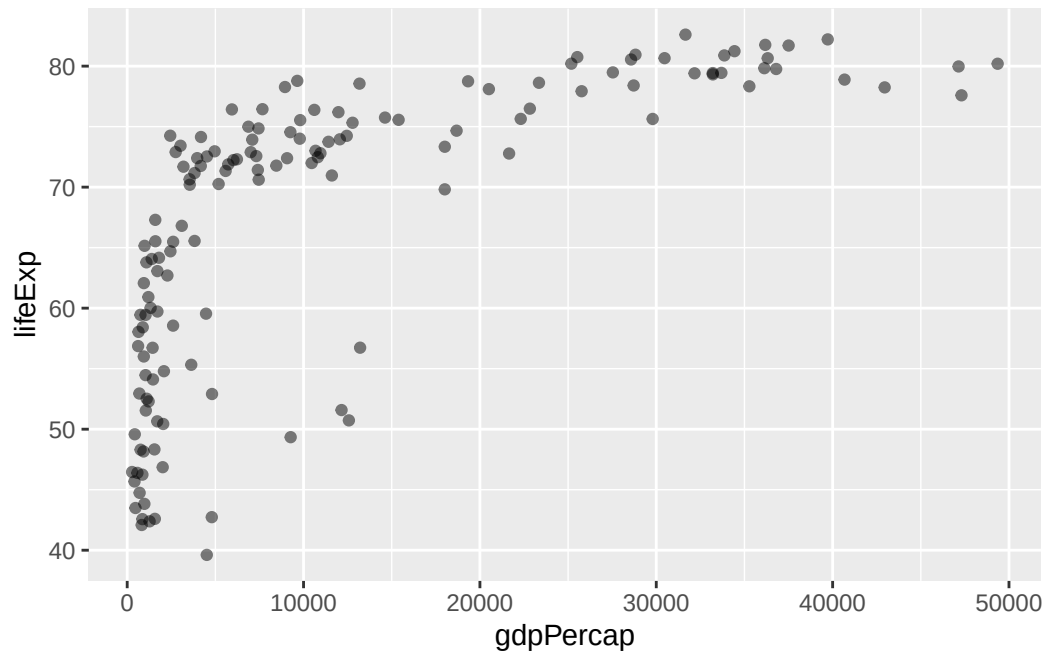
Q1:Q. Complete the code below to produce a first basic scatter plot of this `gapminder_2007` dataset

A1: I added the x variable as gdp per capita (`gdpPercap`), and the y variable as life expectancy (`lifeExp`), and also told R to make it a scatter plot by assigning `geom_point` to the plot

```
ggplot(gapminder_2007) +  
  aes(x=gdpPercap, y=lifeExp)+  
  geom_point()
```

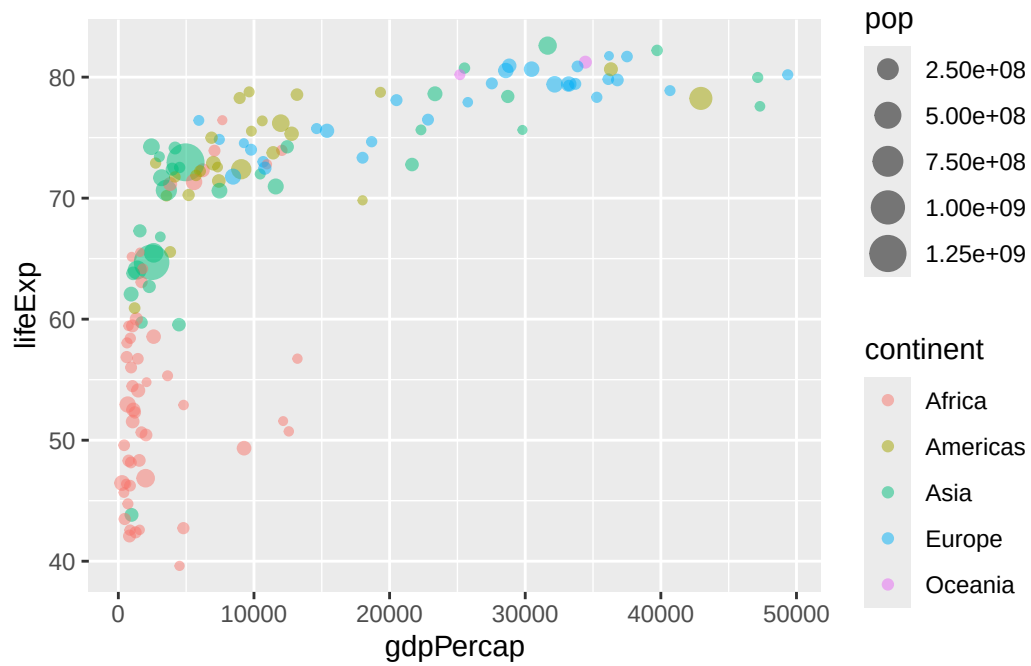


```
ggplot(gapminder_2007) +  
  aes(x=gdpPercap, y=lifeExp) +  
  geom_point(alpha=0.5)
```

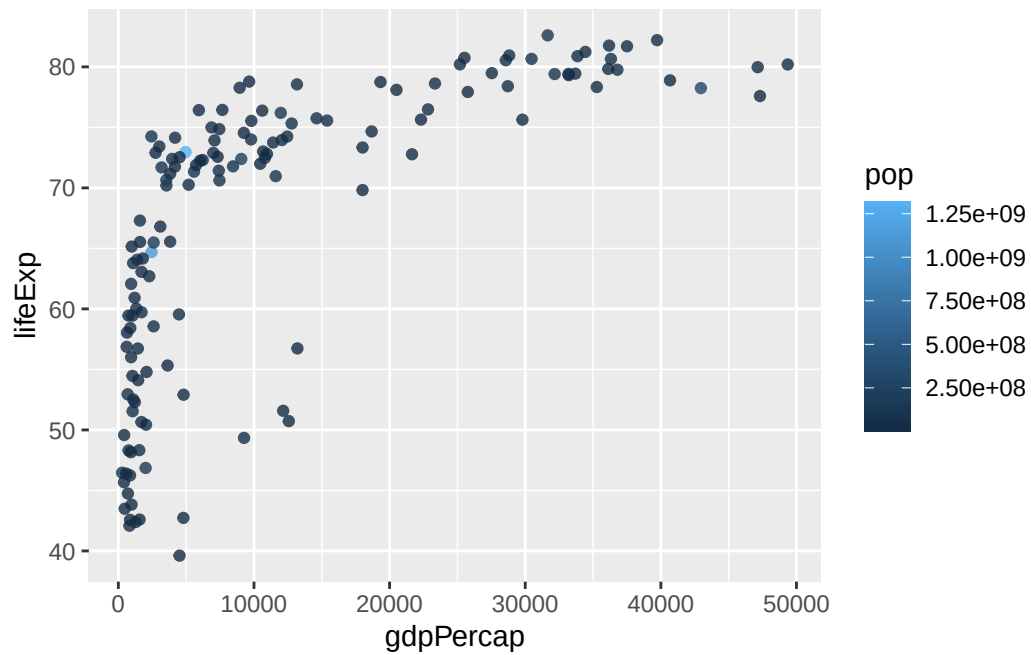



Adding more variables to aes()

```
ggplot(gapminder_2007) +  
  aes(x=gdpPerCap, y=lifeExp, color=continent, size=pop) +  
  geom_point(alpha=0.5)
```

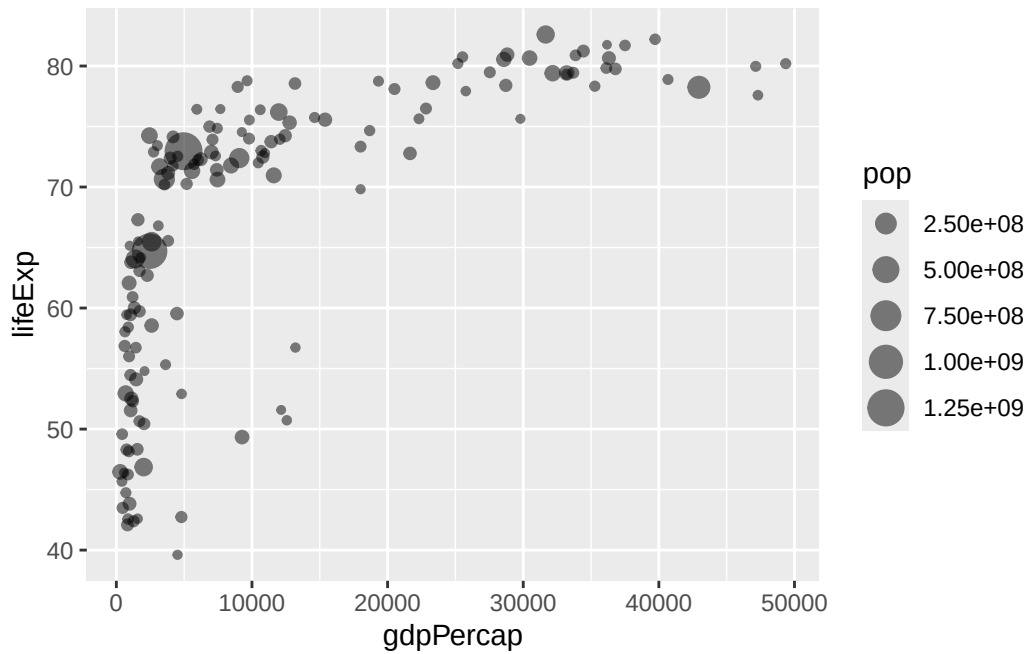


```
ggplot(gapminder_2007) +
  aes(x = gdpPerCap, y = lifeExp, color = pop) +
  geom_point(alpha=0.8)
```

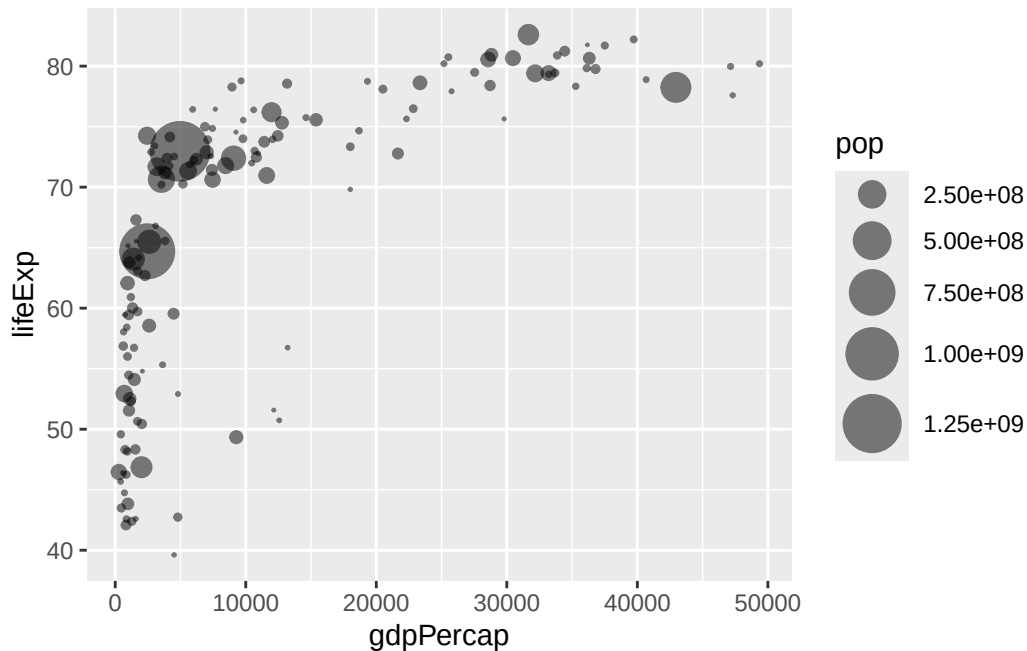


Adjusting point size

```
ggplot(gapminder_2007) +  
  aes(x = gdpPerCap, y = lifeExp, size = pop) +  
  geom_point(alpha=0.5)
```

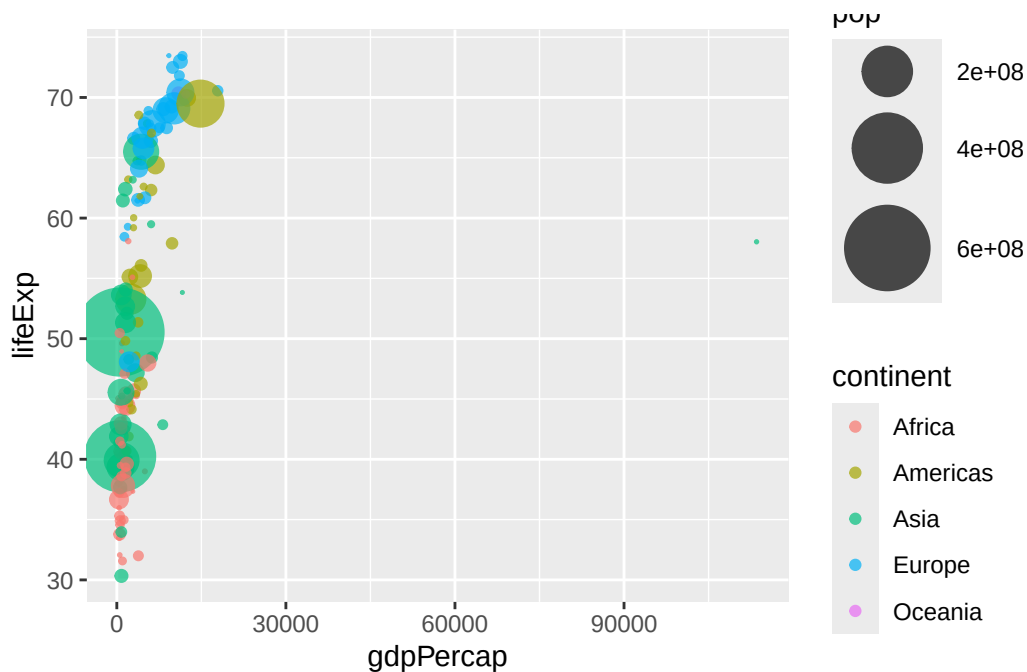


```
ggplot(gapminder_2007) +  
  geom_point(aes(x = gdpPerCap, y = lifeExp,  
                 size = pop), alpha=0.5) +  
  scale_size_area(max_size = 10)
```



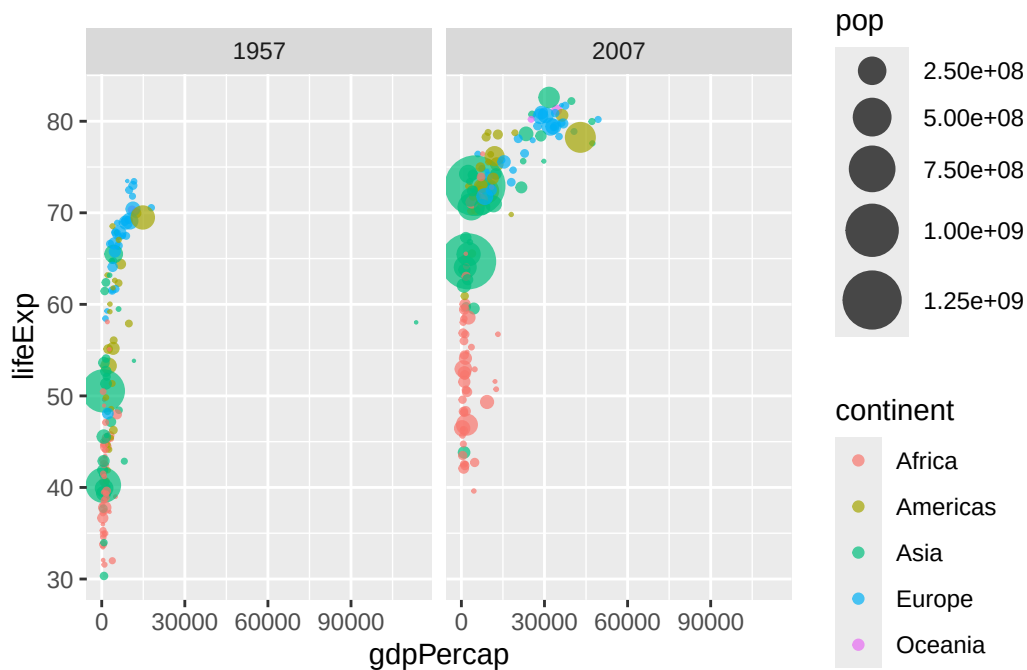
Q2: Can you adapt the code you have learned thus far to reproduce our gapminder scatter plot for the year 1957? What do you notice about this plot is it easy to compare with the one for 2007?

```
#create object gapminder_1957 by filtering the data of gapminder dataset
gapminder_1957 <- gapminder %>% filter(year==1957)
#used geom_point to make a scatter plot of the data, x variable was the gdp per capita, y variable was life expectancy
ggplot(gapminder_1957)+
  aes(x=gdpPercap, y=lifeExp, color=continent, size = pop) +
  #used alpha point 0.7 so that the circles are more opaque and it is more obvious when circles overlap
  geom_point(alpha=0.7) +
  #used scale size set to max size 15 so that the point sizes reflect the actual population distribution
  scale_size_area(max_size = 15)
```



Q3: Do the same steps above but include 1957 and 2007 in your input dataset for `ggplot()`. You should now include the layer `facet_wrap(~year)` to produce the following plot:

```
#create object gapminder_1957 by filtering the data of gapminder dataset for 1957 and 2007
gapminder_1957 <- gapminder %>% filter(year==1957 | year==2007)
#used geom_point to make a scatter plot of the data, x variable was the gdp per capita, y variable was life expectancy
ggplot(gapminder_1957)+
  aes(x=gdpPercap, y=lifeExp, color=continent, size = pop) +
  #used alpha point 0.7 so that the circles are more opaque and it is more obvious when circles overlap
  geom_point(alpha=0.7) +
  #used scale_size set to max size 10 so that the point sizes reflect the actual population distribution
  scale_size_area(max_size = 10) +
  #used facet_wrap to make two plots differing by year.
  facet_wrap(~year)
```



Section 8: Bar charts

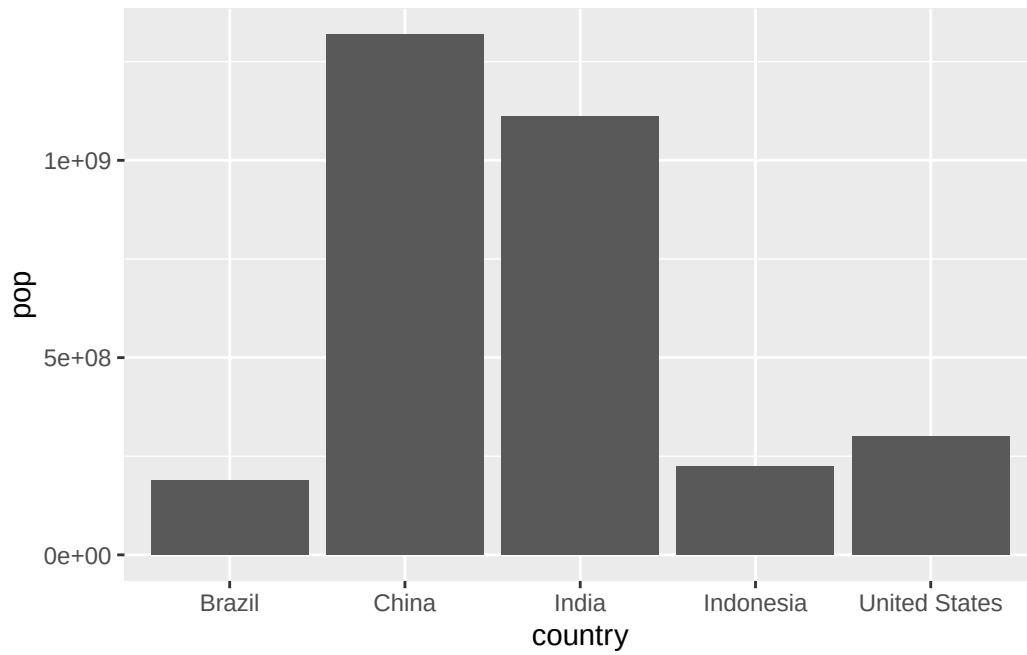
```
gapminder_top5 <- gapminder %>%
  filter(year==2007) %>%
  arrange(desc(pop)) %>%
  top_n(5, pop)
```

```
gapminder_top5
```

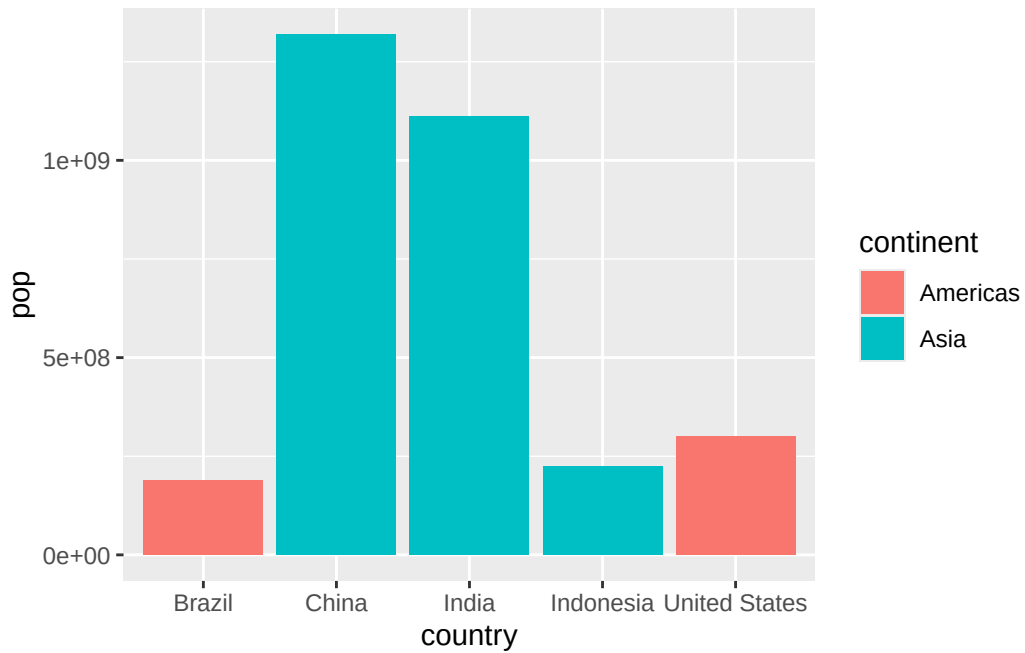
	country	continent	year	lifeExp	pop	gdpPercap
1	China	Asia	2007	72.961	1318683096	4959.115
2	India	Asia	2007	64.698	1110396331	2452.210
3	United States	Americas	2007	78.242	301139947	42951.653
4	Indonesia	Asia	2007	70.650	223547000	3540.652
5	Brazil	Americas	2007	72.390	190010647	9065.801

```
#country vs pop col plot
```

```
ggplot(gapminder_top5) +
  geom_col(aes(x = country, y = pop))
```



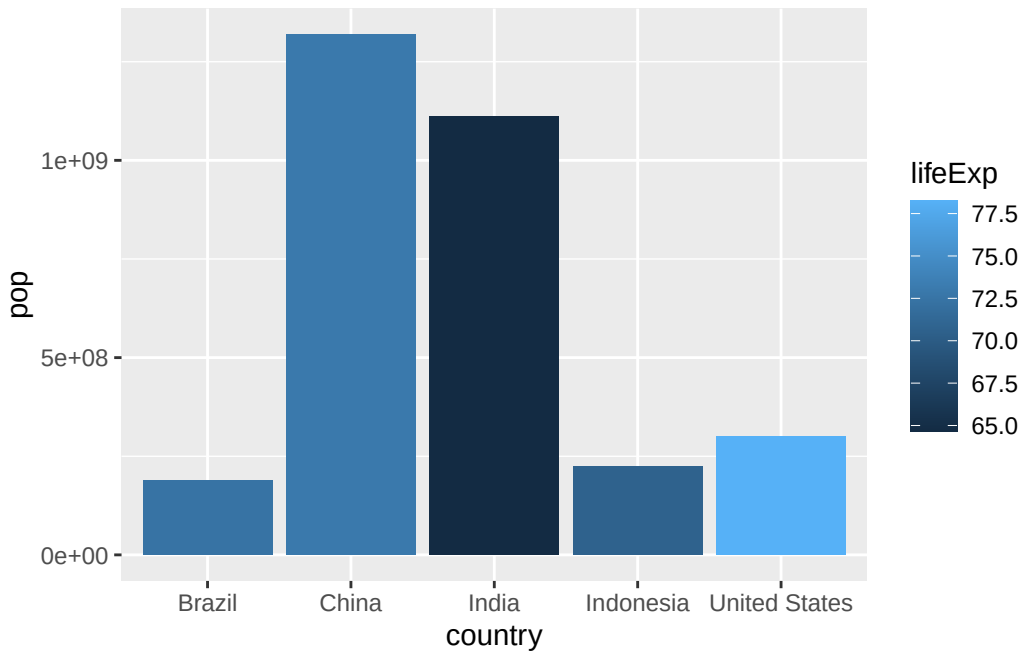
```
#country vs population col plot colored by continent  
  
ggplot(gapminder_top5) +  
  geom_col(aes(x = country, y = pop, fill = continent))
```



```
#used fill=country to color the bars according to the continent
```

```
#country vs pop col plot colored by continent
```

```
ggplot(gapminder_top5) +  
  geom_col(aes(x = country, y = pop, fill = lifeExp))
```

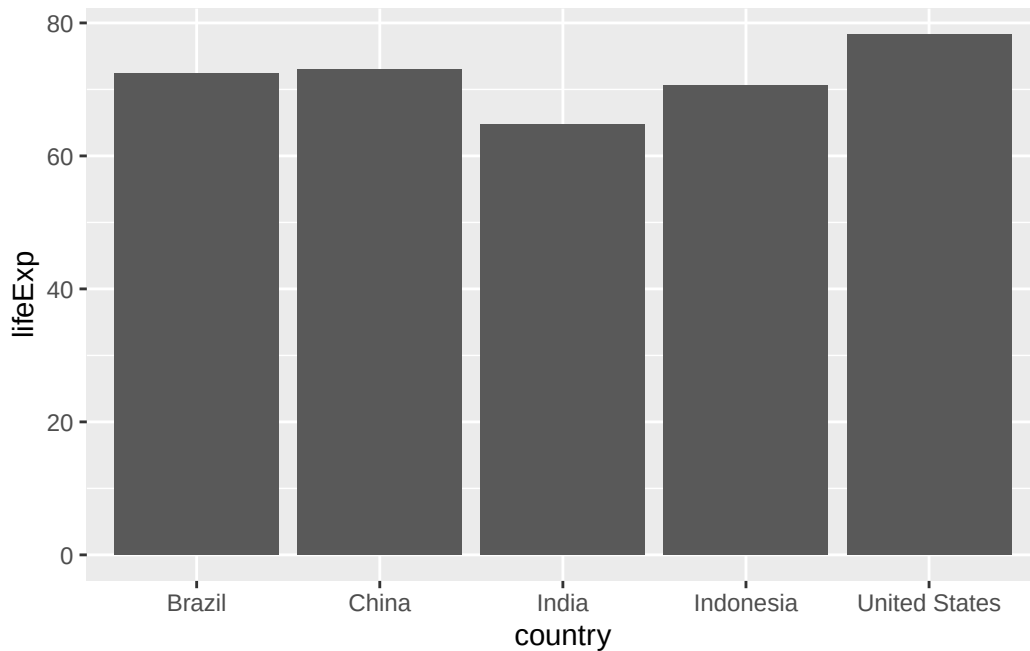
#used life expectatncy as what the coloring should be coded with

Q1: Create a bar chart showing the life expectancy of the five biggest countries by population in 2007

```
# Answer 1:
gapminder_top5 <- gapminder %>%
  filter (year==2007) %>%      #filtering 2007 out of the dataset
  arrange(desc(pop)) %>%      #arranging in descending order of population
  top_n(5, pop)                #getting the top 5 in the row
gapminder_top5
```

	country	continent	year	lifeExp	pop	gdpPercap
1	China	Asia	2007	72.961	1318683096	4959.115
2	India	Asia	2007	64.698	1110396331	2452.210
3	United States	Americas	2007	78.242	301139947	42951.653
4	Indonesia	Asia	2007	70.650	223547000	3540.652
5	Brazil	Americas	2007	72.390	190010647	9065.801

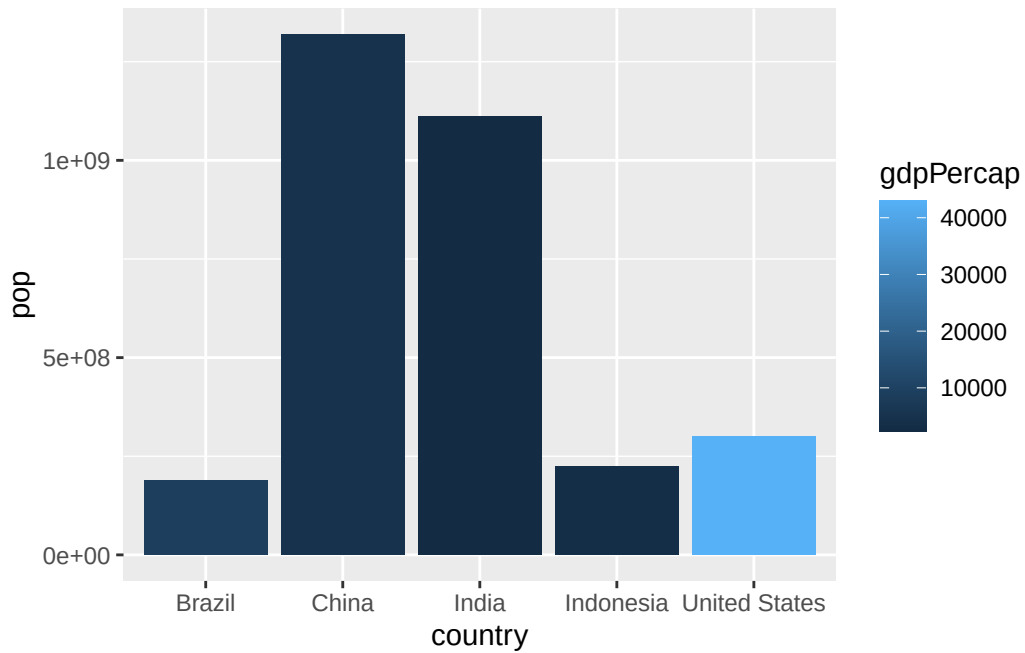
```
ggplot(gapminder_top5)+
  geom_col(aes(x=country, y=lifeExp))      #plotting bar graph with x variable being the c
```



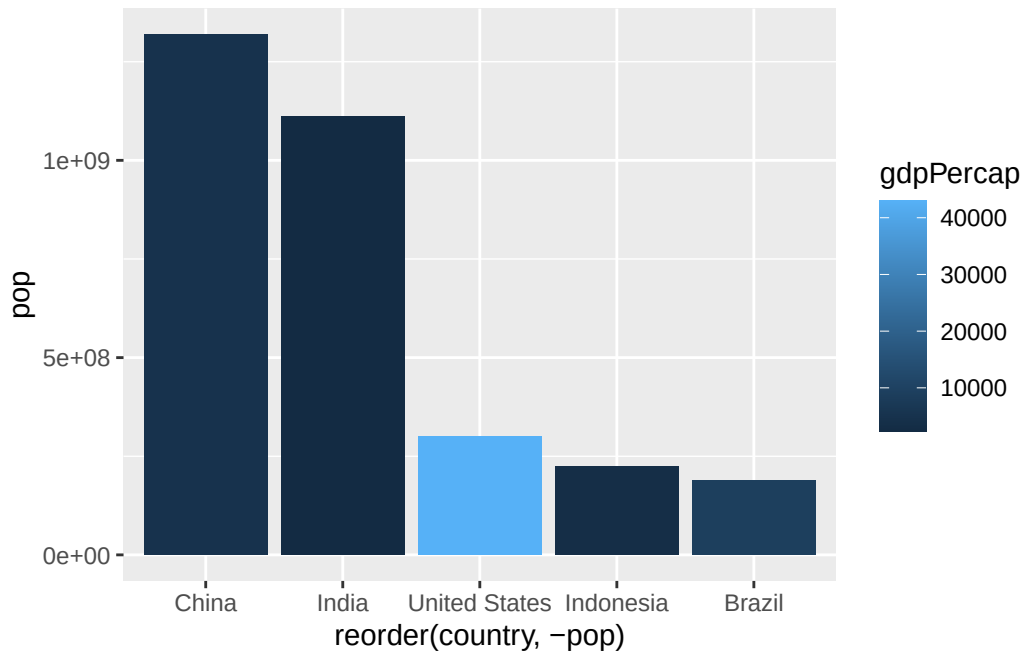
Q2: Plot population size by country. Create a bar chart showing the population (in millions) of the five biggest countries by population in 2007. Use the `ggplot()` function and specify the `gapminder_top5` dataset as input

Add a `geom_col()` layer to the plot Plot one bar for each country (x aesthetic) Use population `pop` as bar height (y aesthetic) Use the GDP per capita `gdpPercap` as fill aesthetic

```
ggplot(gapminder_top5) +  
  aes(country,  
        pop, fill=gdpPercap) +  
  #used gdp per capita as fill so the bars could be color coded accoring to capita.  
  geom_col ()
```

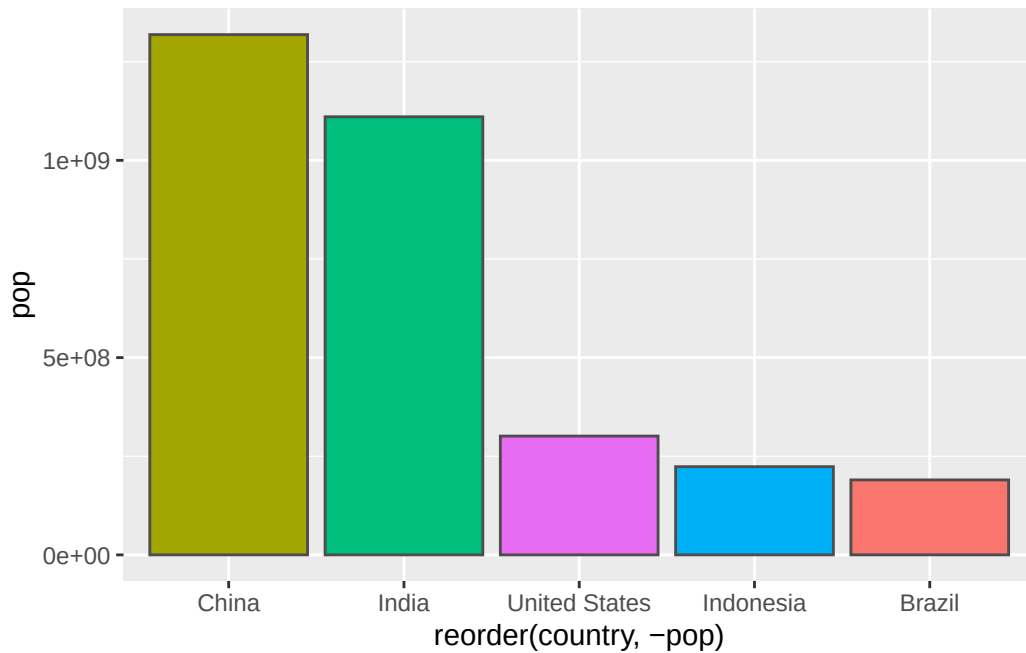


```
ggplot(gapminder_top5) +  
  aes(x=reorder(country, -pop),  
      # used x reorder to change the order of the bars  
      y=pop, fill=gdpPercap) +  
  #used gdp per capita as fill so the bars could be color coded accoring to capita.  
  geom_col()
```



```
#using fill by country so its color coded to countries

ggplot(gapminder_top5) +
  aes(x=reorder(country, -pop), y=pop, fill=country) +
  geom_col(col="gray30") +
  guides(fill="none")
```



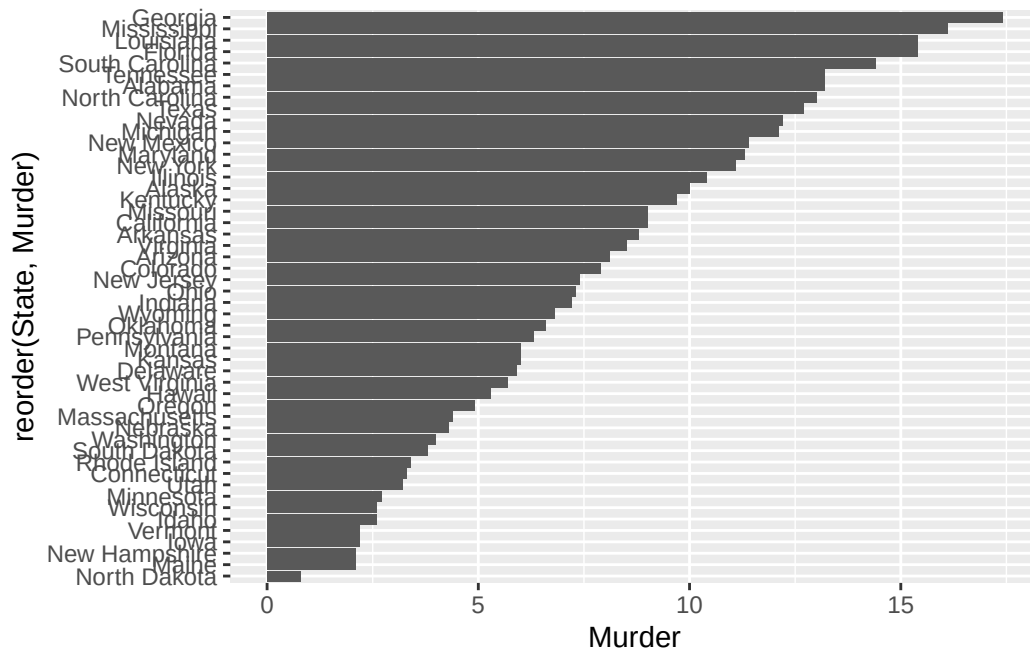
Flipping bar charts

```
#making flipped graph of US murders
#used coord_flip to flip the graph on to the side

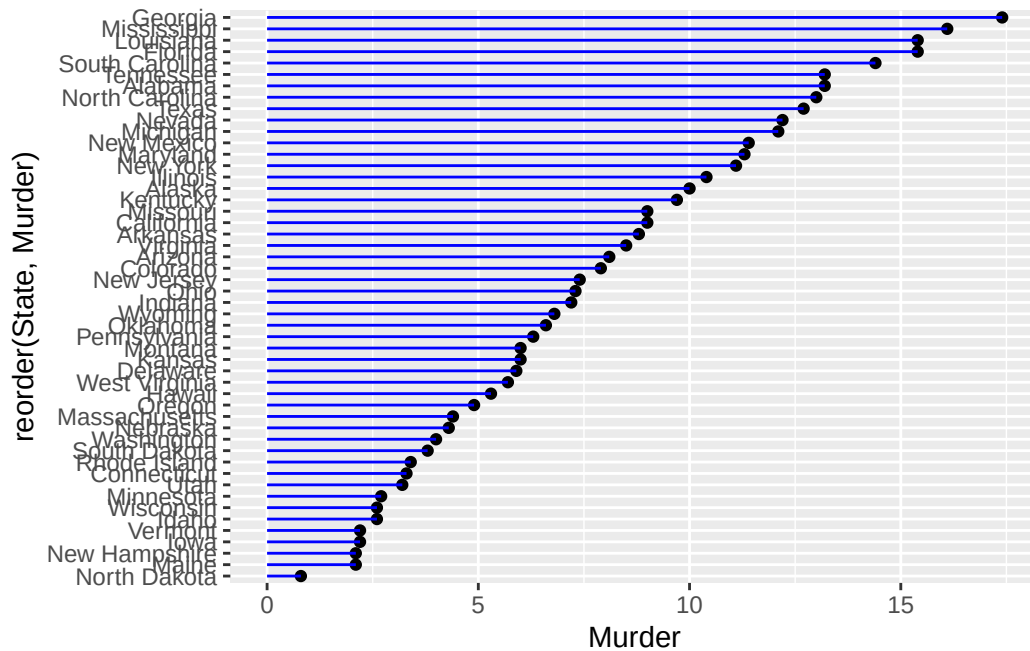
head(USArrests)
```

	Murder	Assault	UrbanPop	Rape
Alabama	13.2	236	58	21.2
Alaska	10.0	263	48	44.5
Arizona	8.1	294	80	31.0
Arkansas	8.8	190	50	19.5
California	9.0	276	91	40.6
Colorado	7.9	204	78	38.7

```
USArrests$State <- rownames(USArrests)
ggplot(USArrests) +
  aes(x=reorder(State,Murder), y=Murder) +
  geom_col() +
  coord_flip()
```



```
#changing the visualization using geom_point and geom_segment together
ggplot(USArrests) +
  aes(x=reorder(State,Murder), y=Murder) +
  geom_point() +
  geom_segment(aes(x=State,
                   xend=State,
                   y=0,
                   yend=Murder), color="blue") +
  coord_flip()
```



Section 9: Extensions: Animation

```
#installed packages in console
#loading packages in R

library(gapminder)
```

Attaching package: 'gapminder'

The following object is masked _by_ '.GlobalEnv':

gapminder

```
library(gganimate)

#setting up regular gapminder plot of life Exp vs. gdpPercap:

ggplot(gapminder, aes(gdpPercap, lifeExp, size = pop, colour = country)) +
```

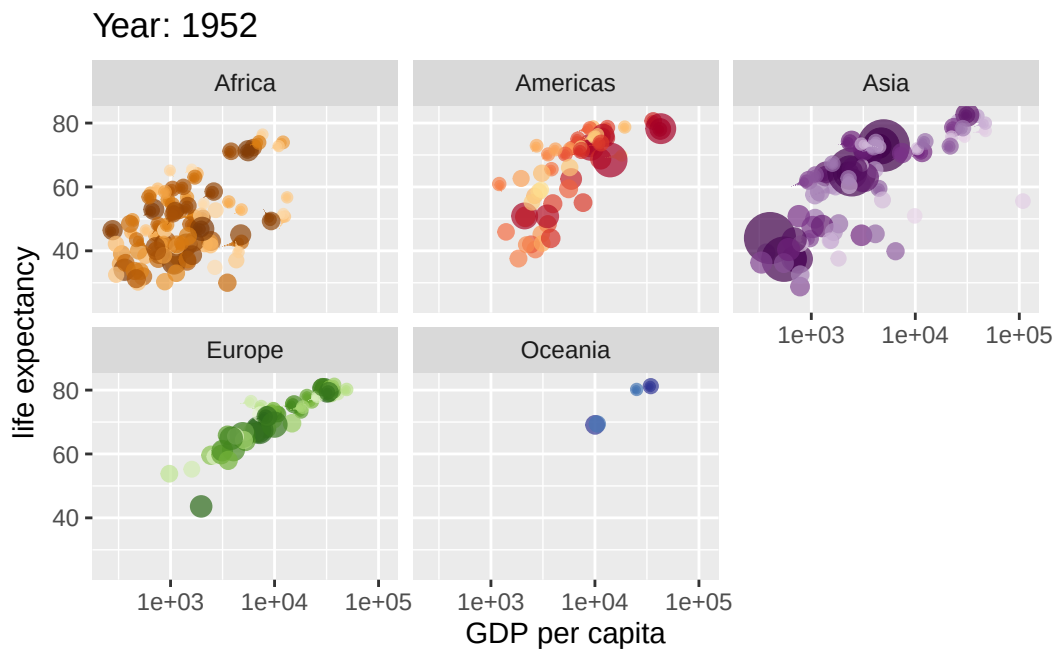
```

#color coded by country
geom_point(alpha = 0.7, show.legend = FALSE) +
#making a point scatter plot with transparency set to 70%, and no legend showing
scale_colour_manual(values = country_colors) +
#changing color scale
scale_size(range = c(2, 12)) +
  scale_x_log10()+
#changing it to logarithmic scale

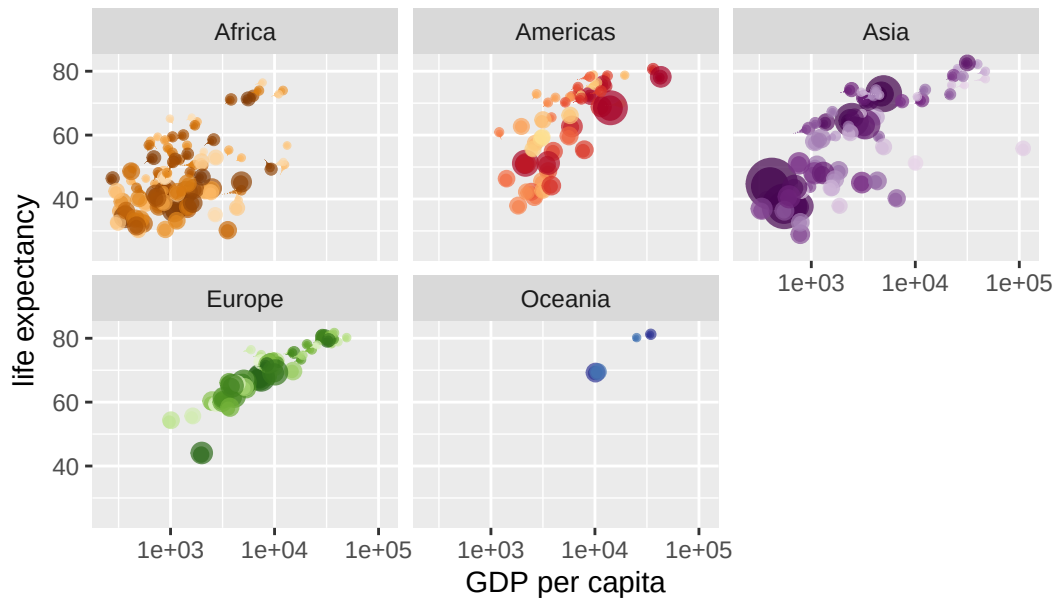
# Facet by continent so there are multiple graphs
facet_wrap(~continent) +
labs(title = 'Year: {frame_time}', x = 'GDP per capita', y = 'life expectancy') +
transition_time(year)+
#adding labels and adding in the transition and animation
shadow_wake(wake_length = 0.1, alpha = FALSE)

```

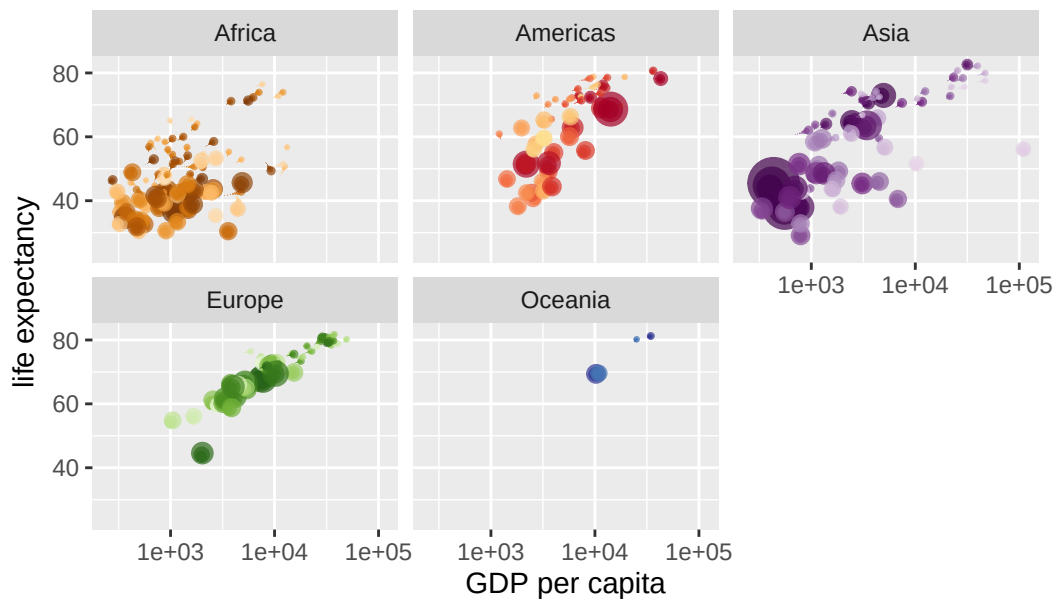
Warning in formalis(fun): argument is not a function



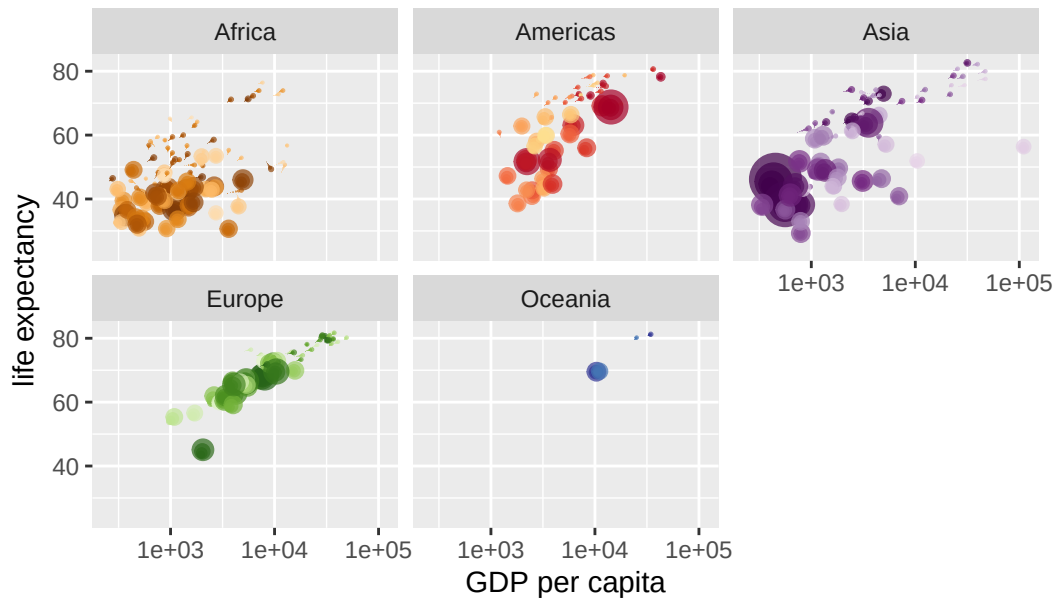
Year: 1953



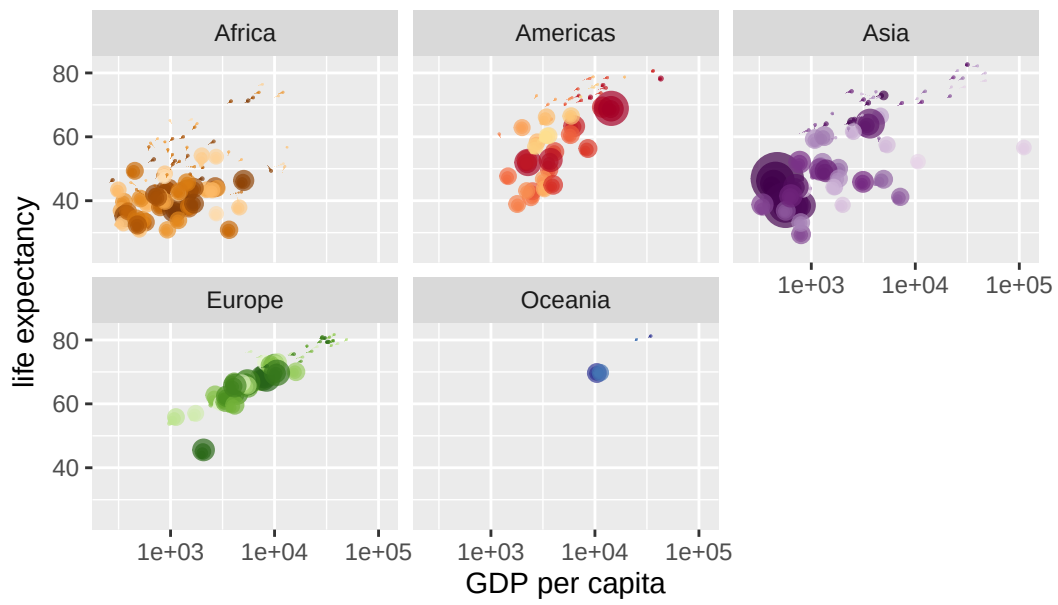
Year: 1953



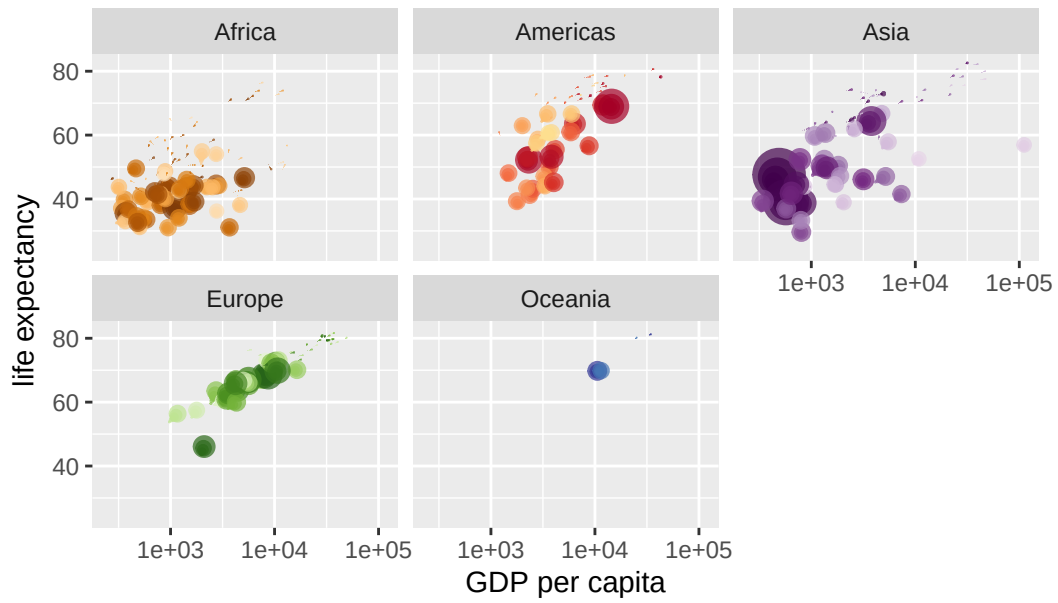
Year: 1954



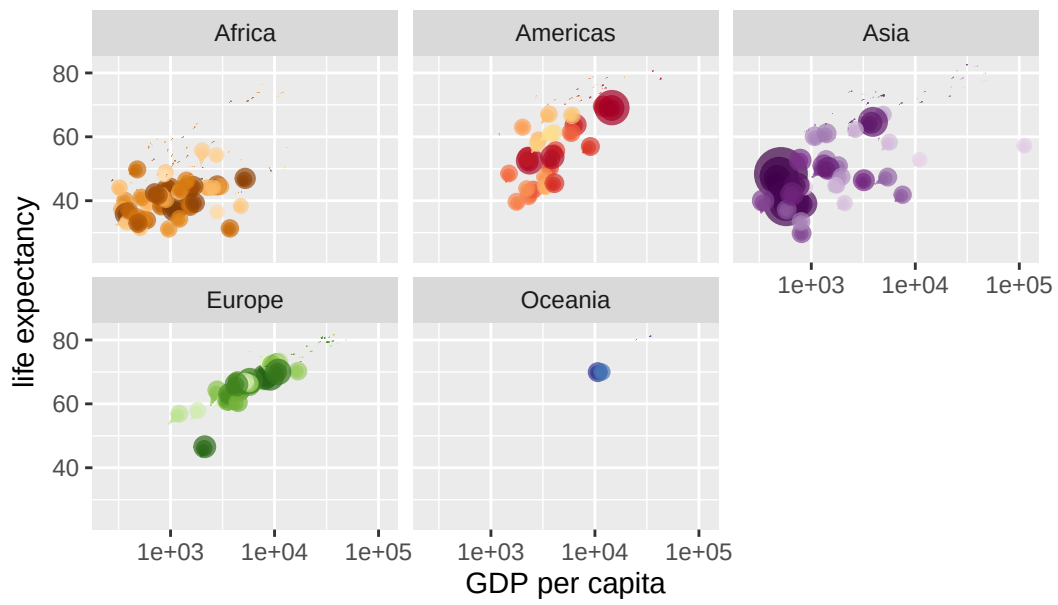
Year: 1954



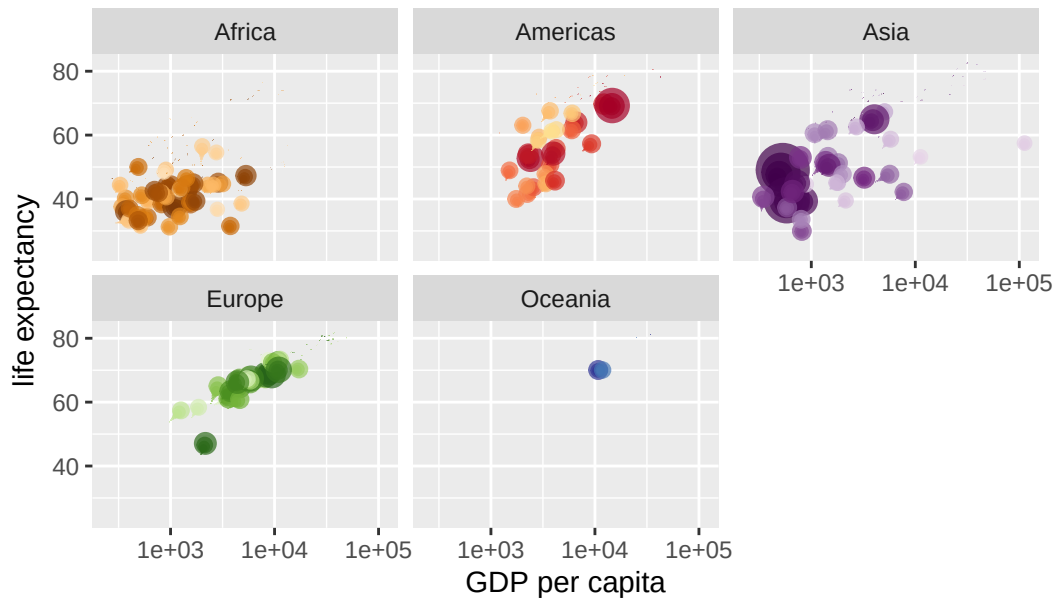
Year: 1955



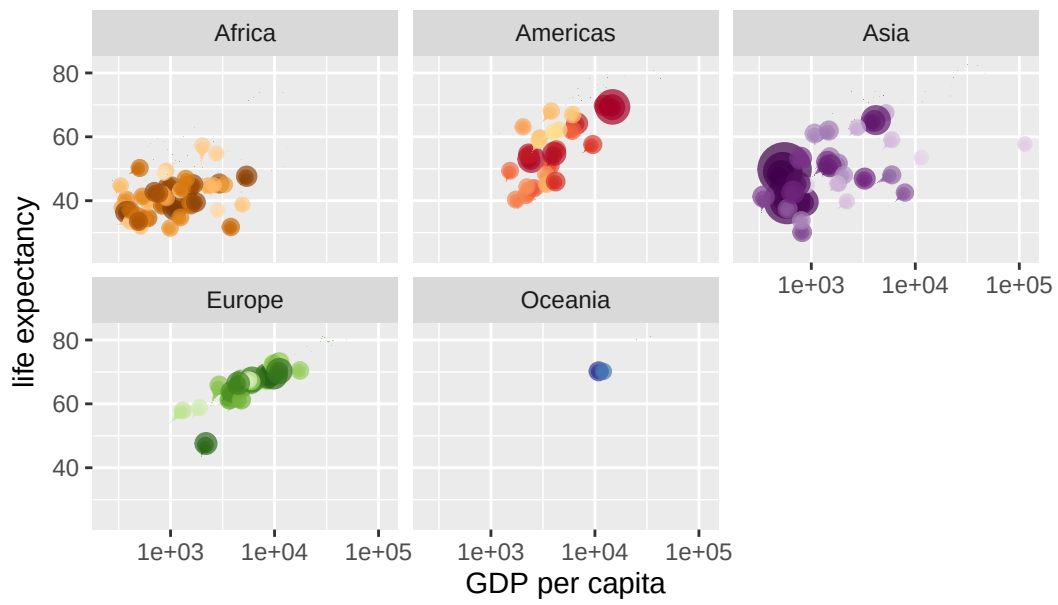
Year: 1955



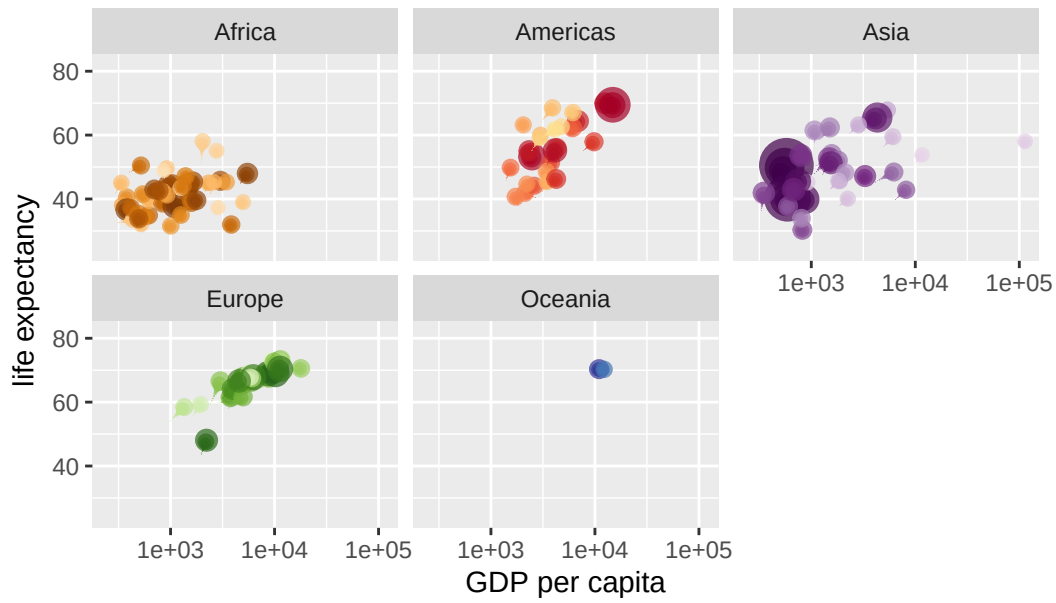
Year: 1956



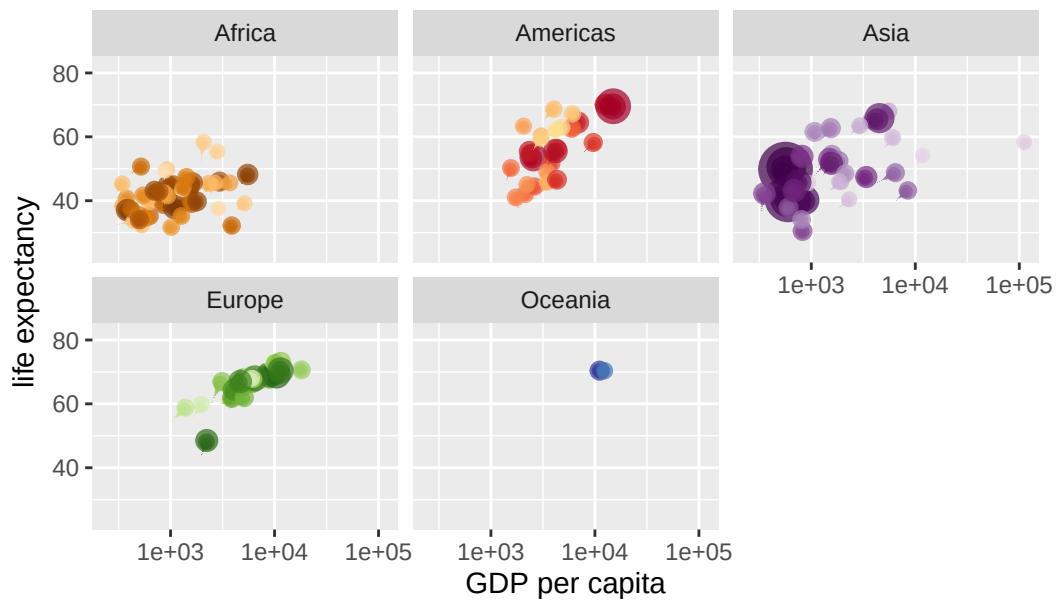
Year: 1956



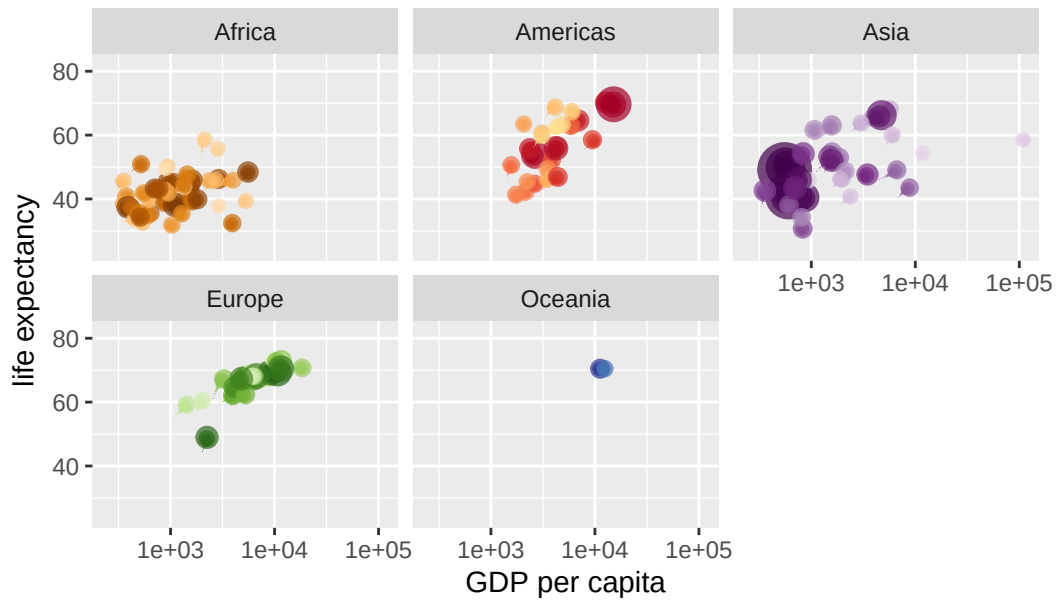
Year: 1957



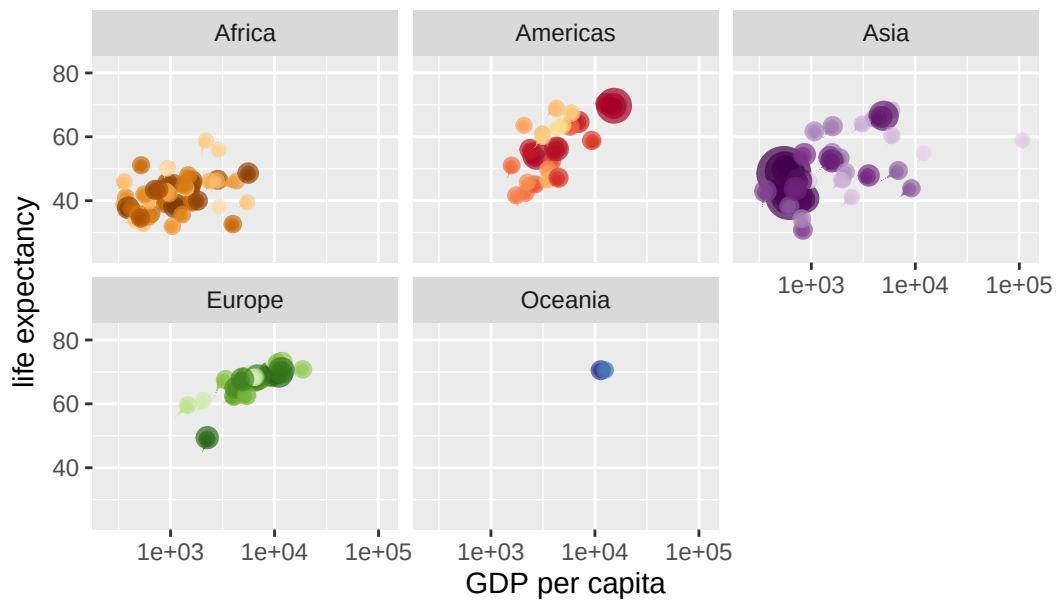
Year: 1958



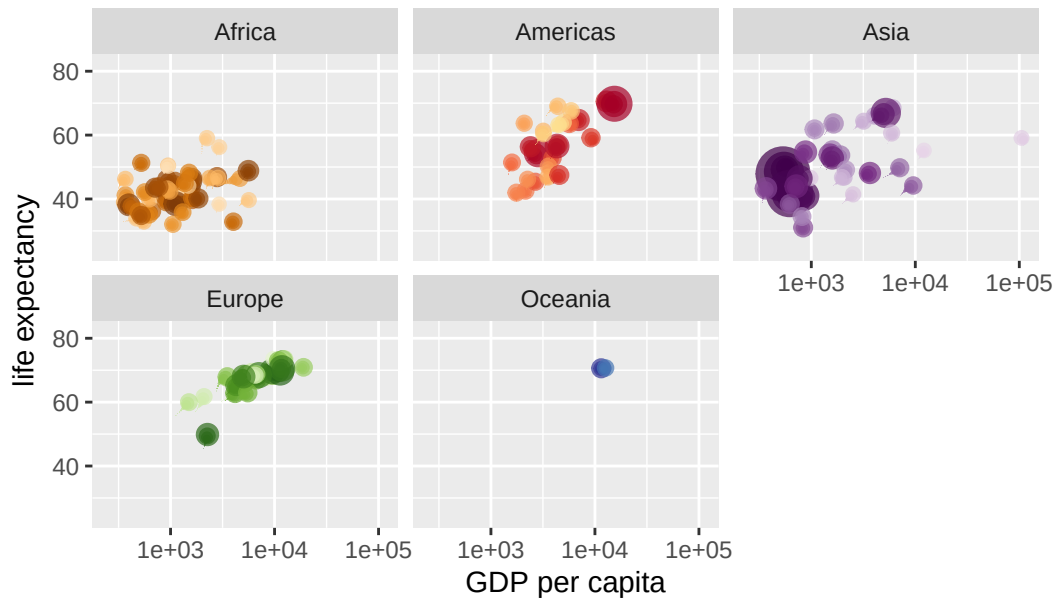
Year: 1958



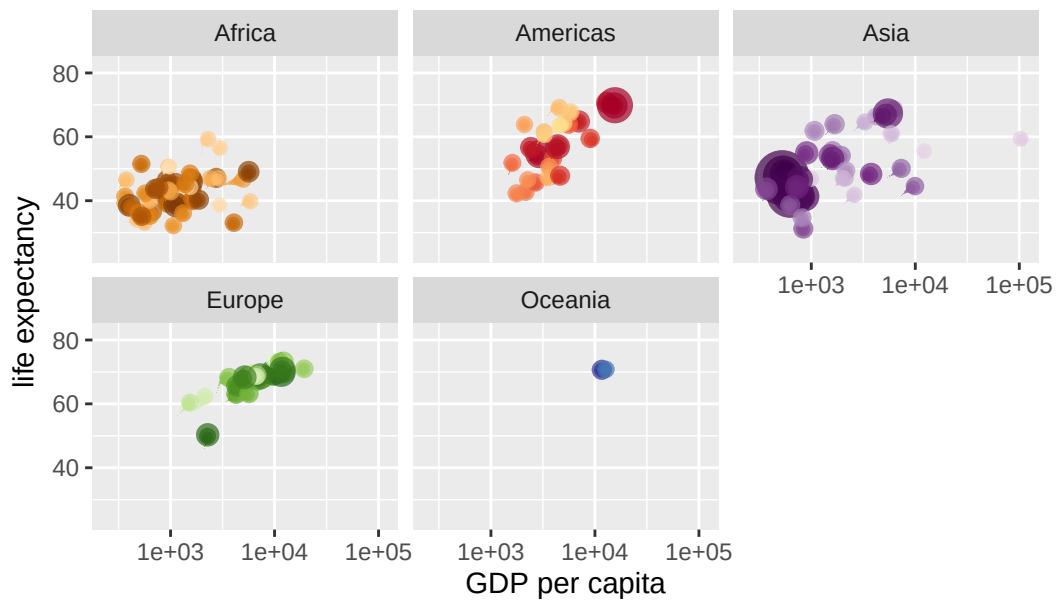
Year: 1959



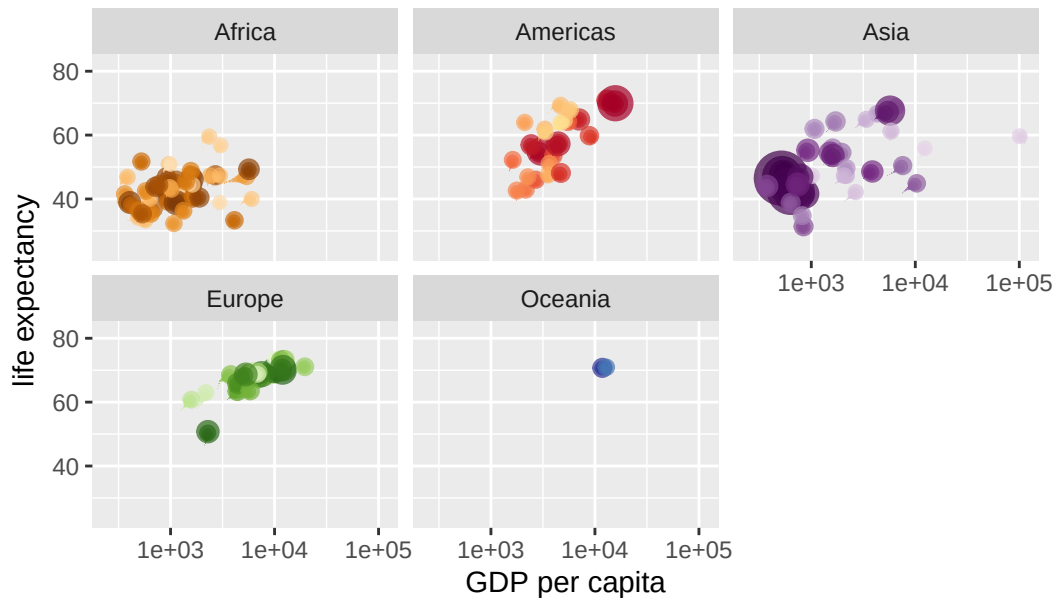
Year: 1959



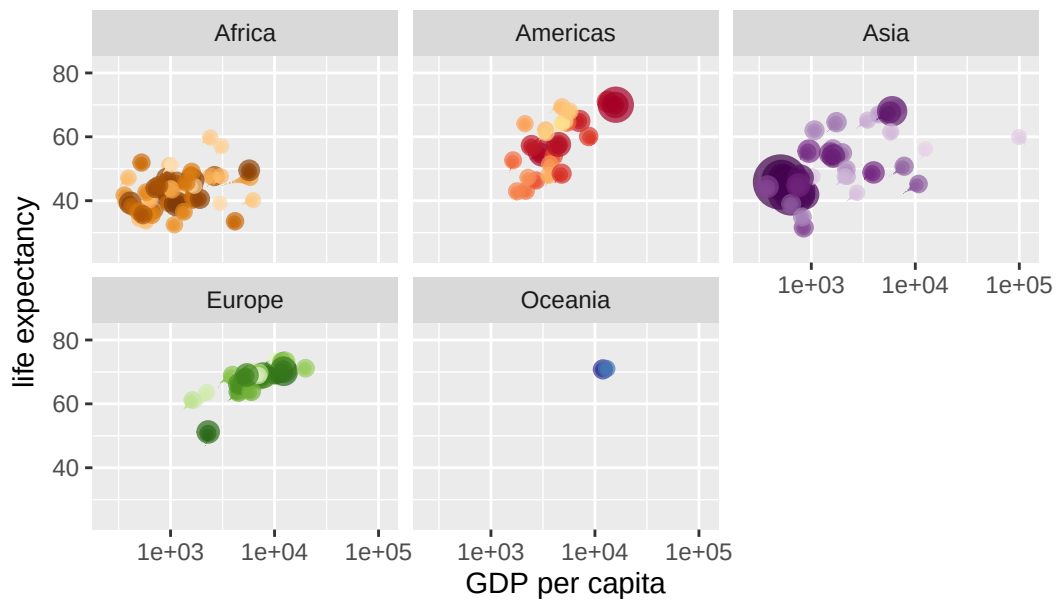
Year: 1960



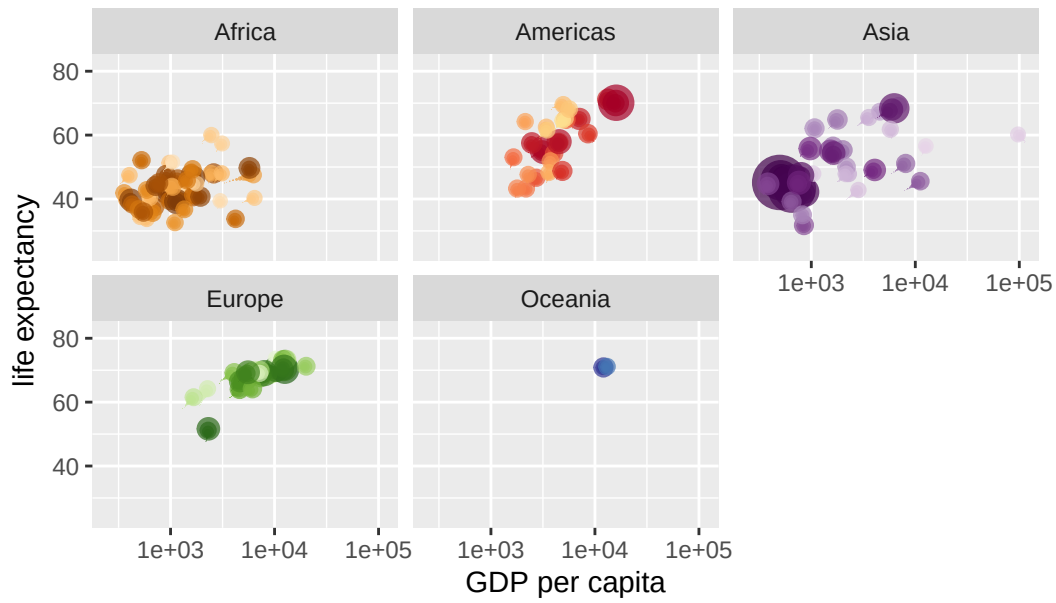
Year: 1960



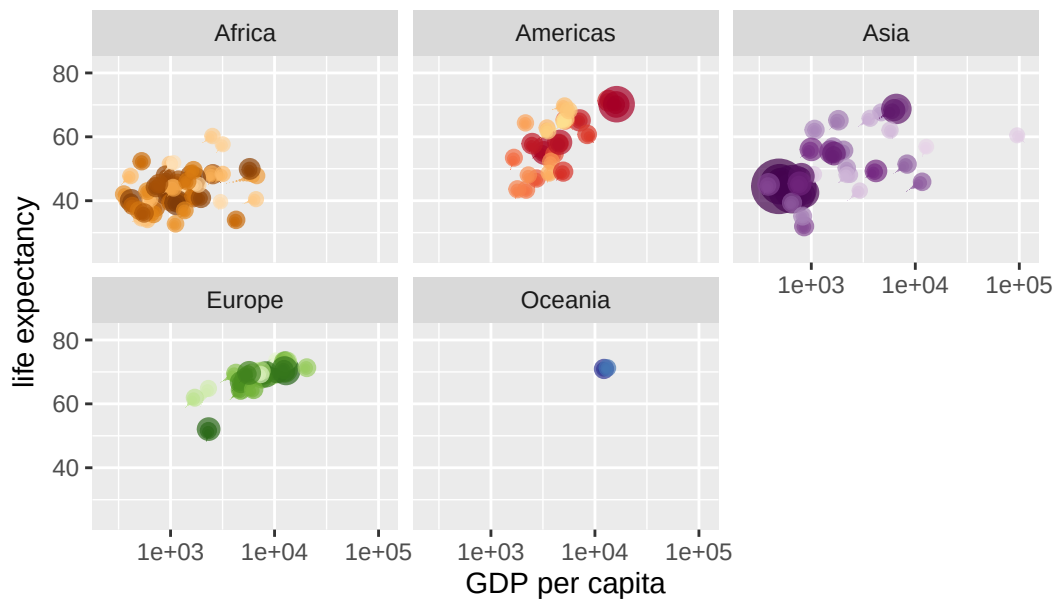
Year: 1961



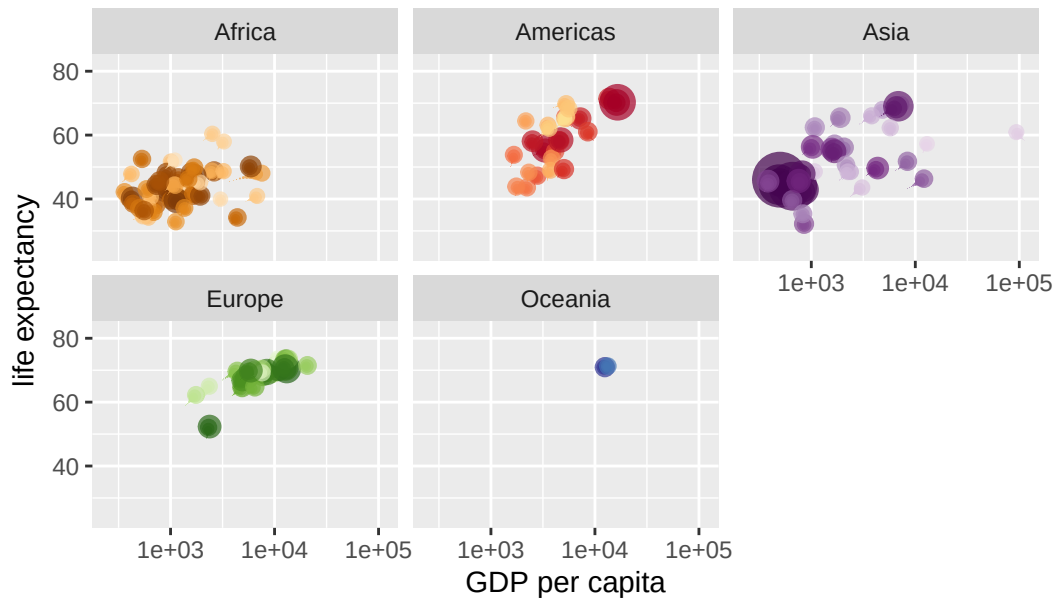
Year: 1961



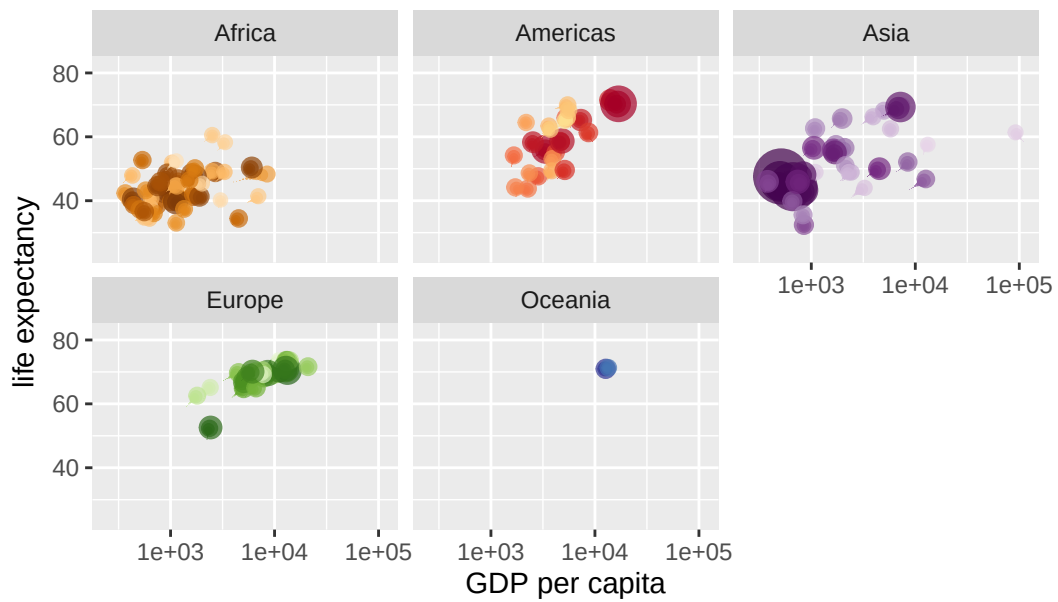
Year: 1962



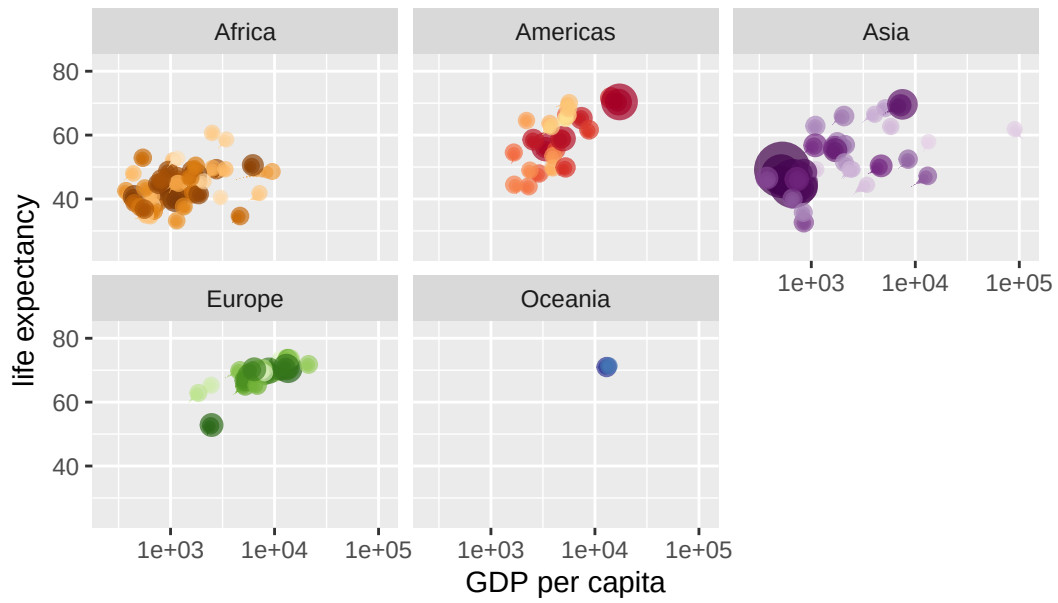
Year: 1963



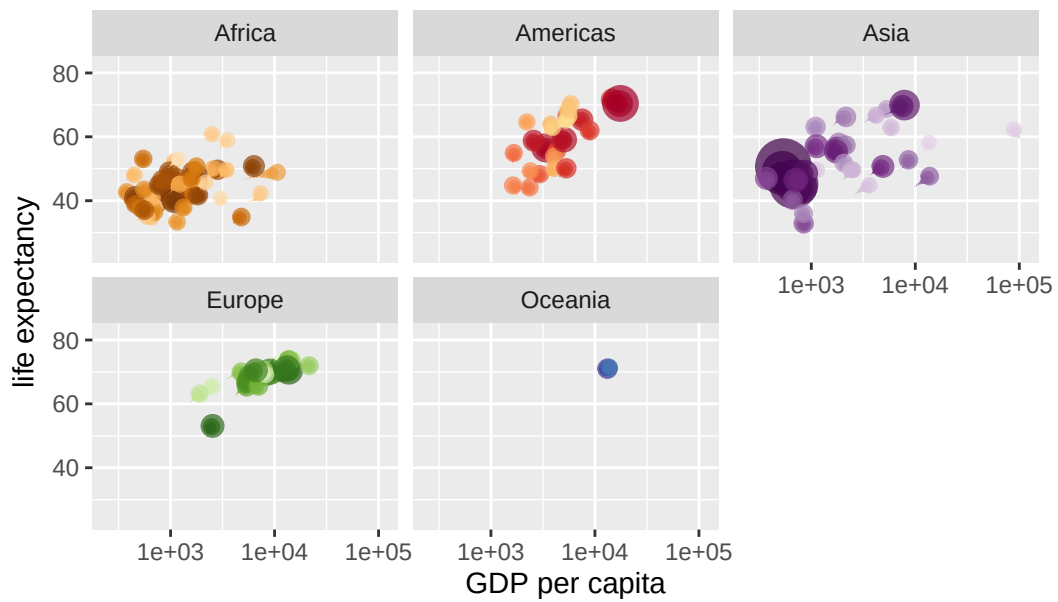
Year: 1963



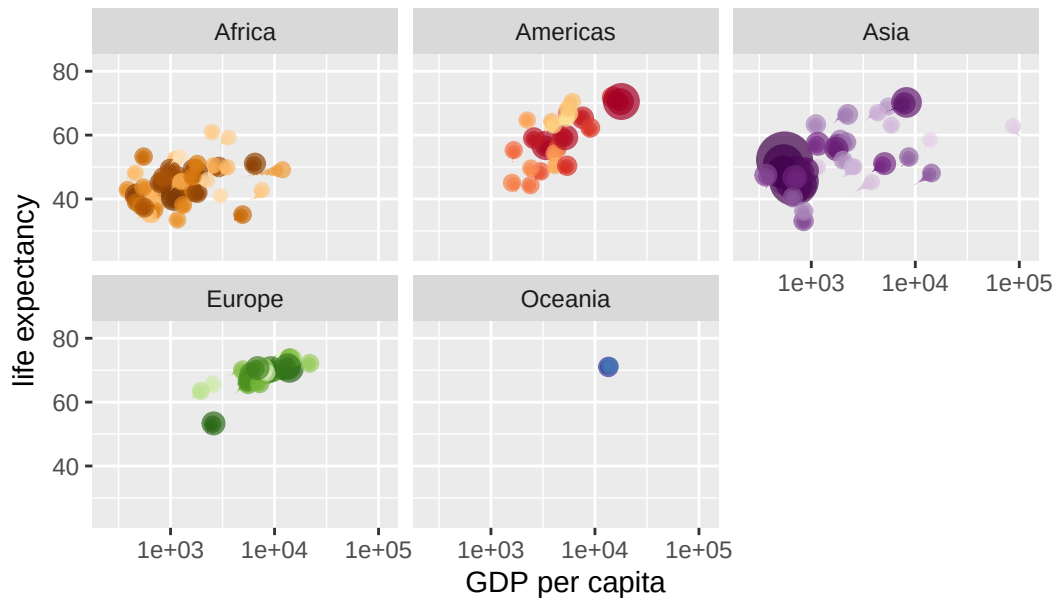
Year: 1964



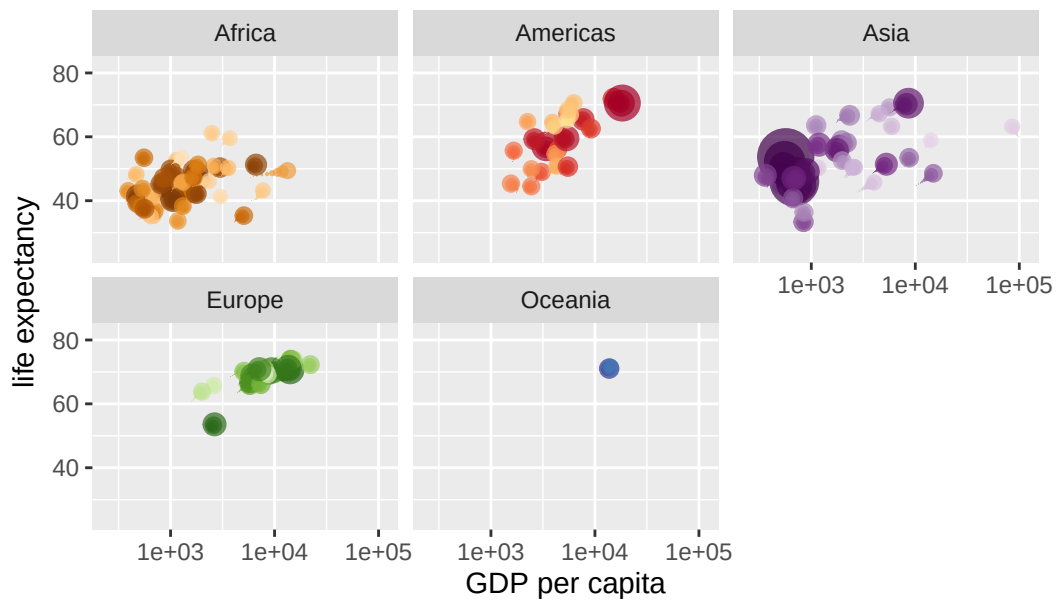
Year: 1964



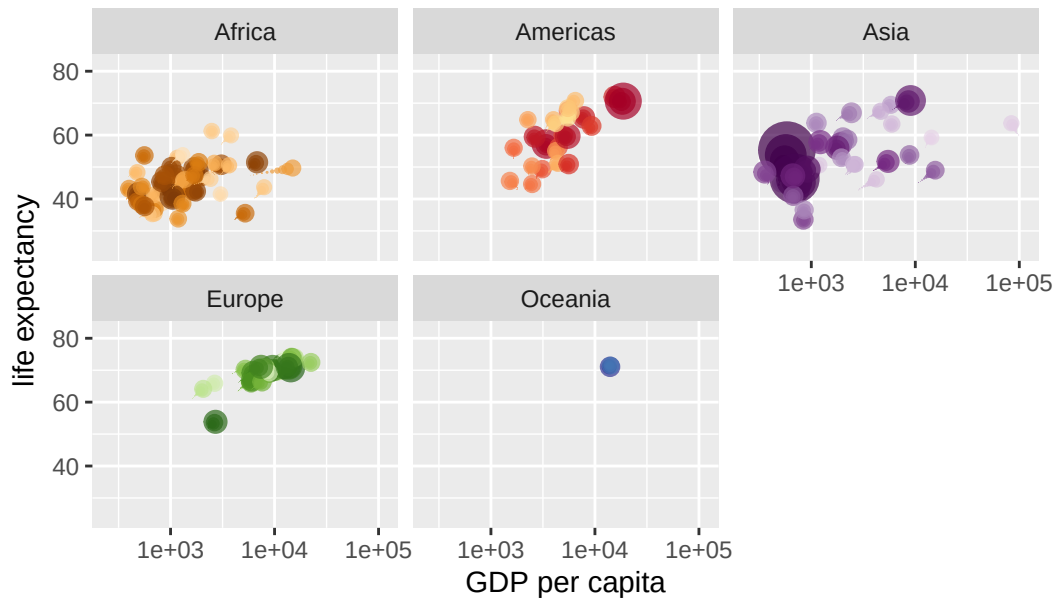
Year: 1965



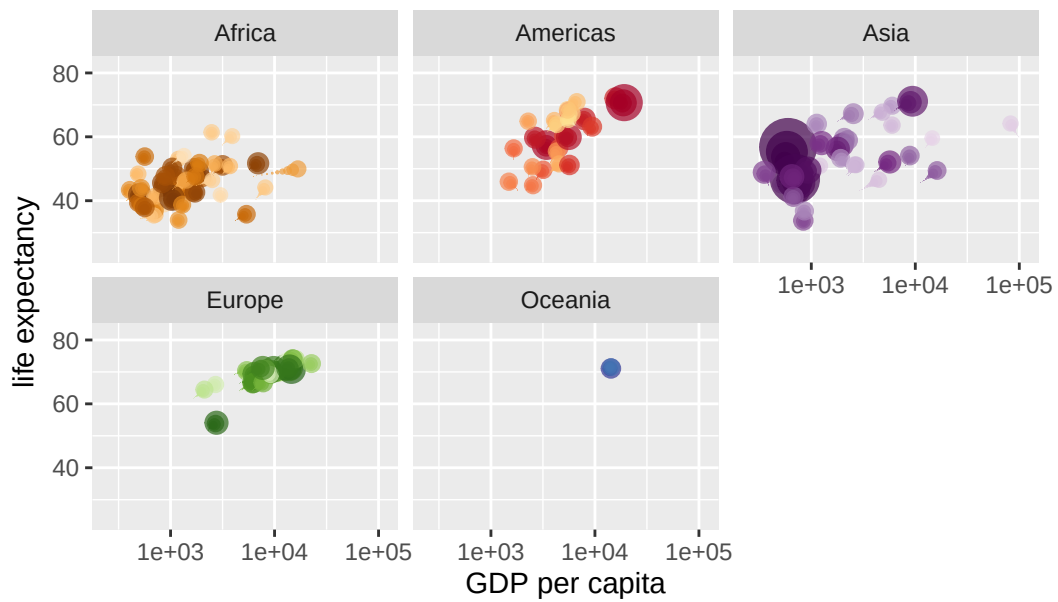
Year: 1965



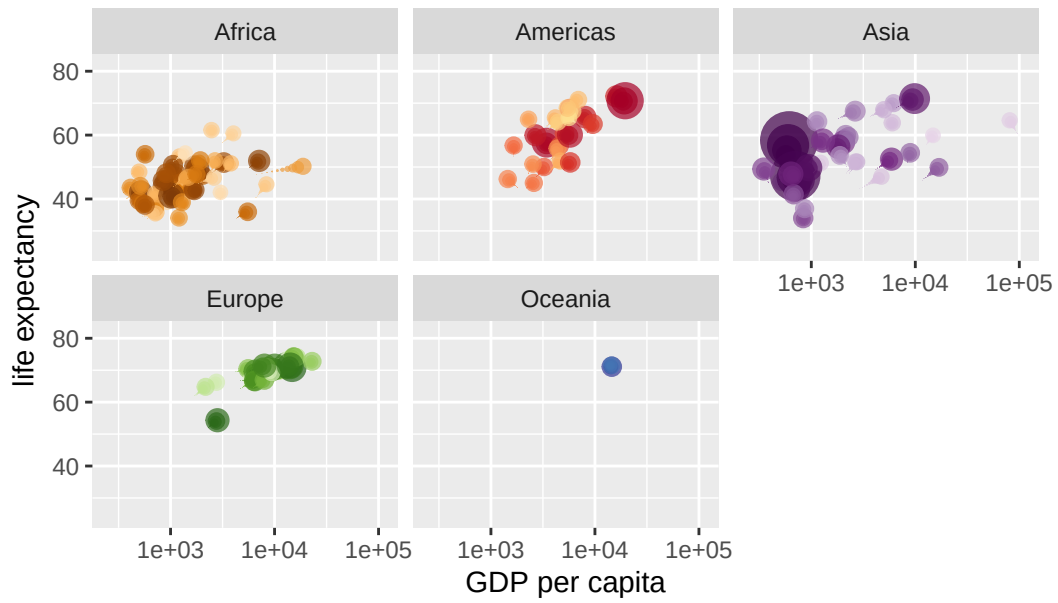
Year: 1966



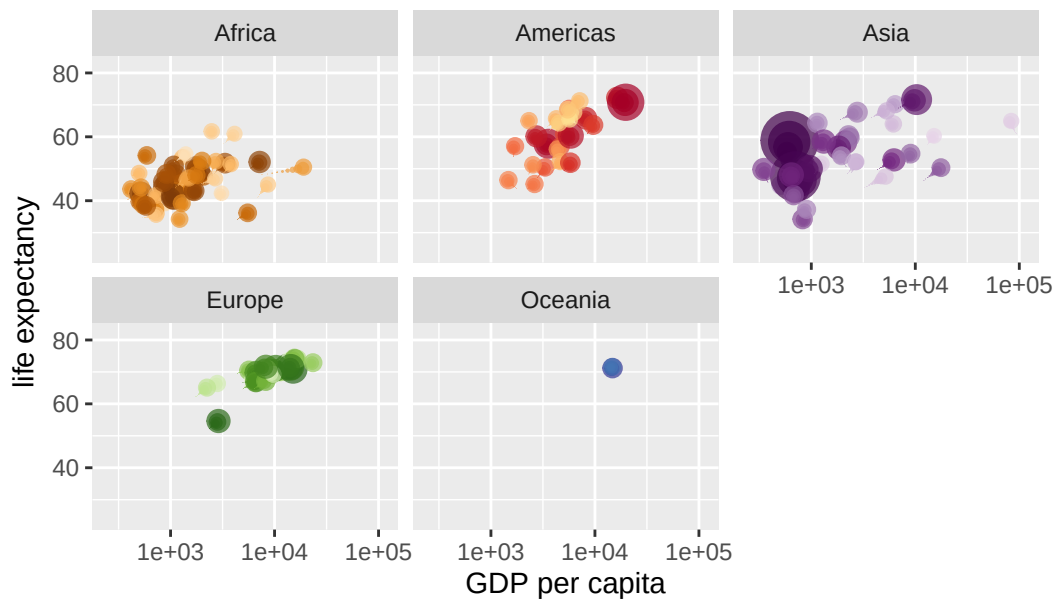
Year: 1966



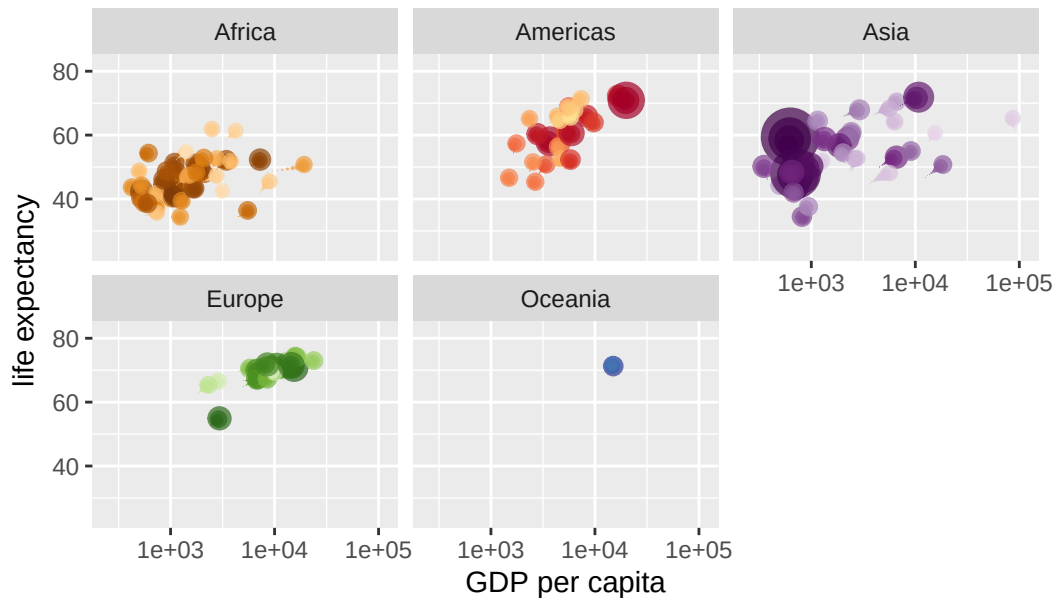
Year: 1967



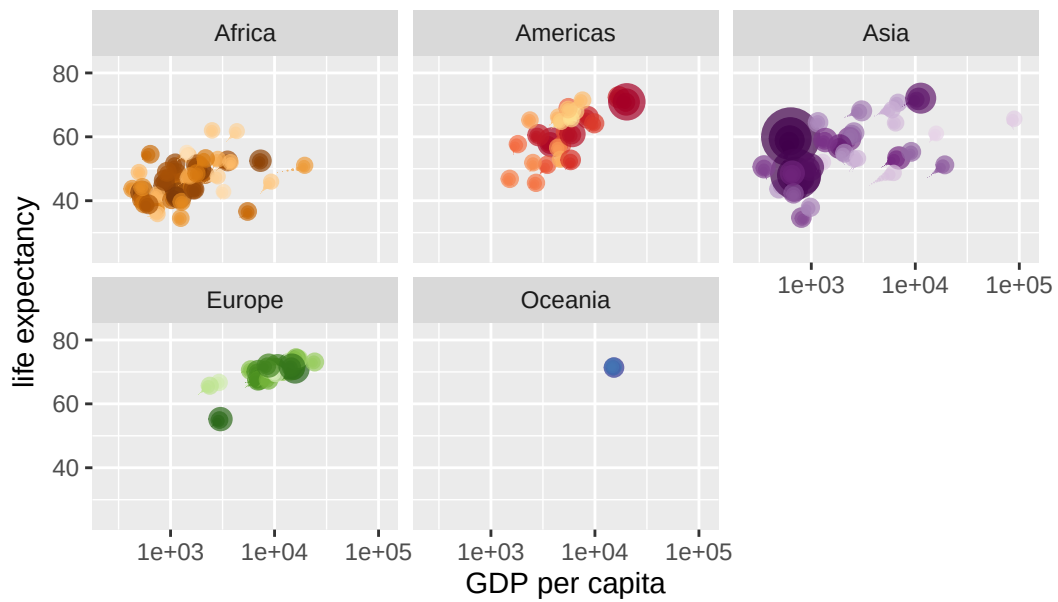
Year: 1968



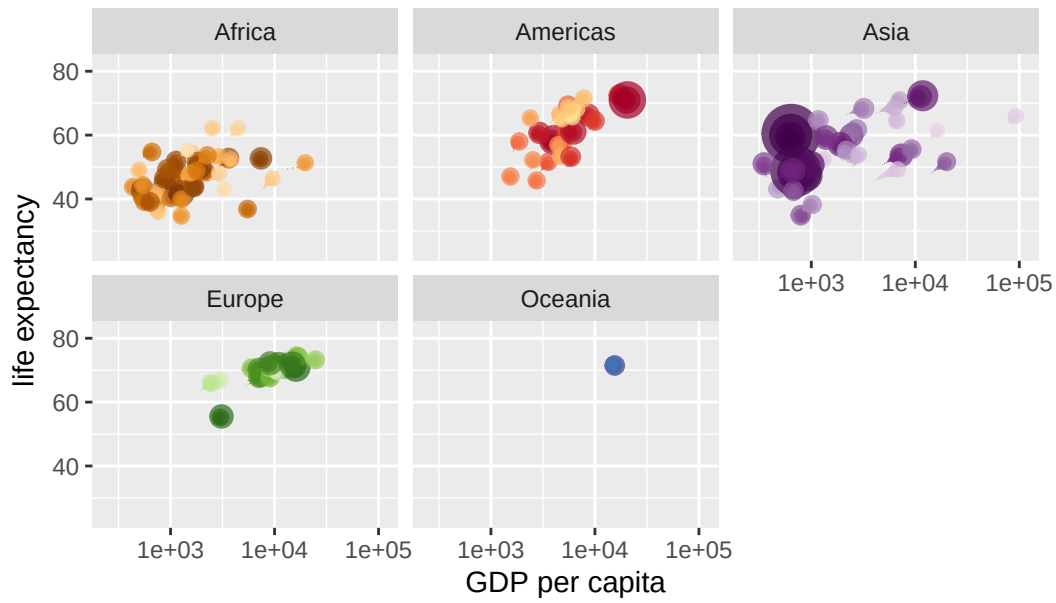
Year: 1968



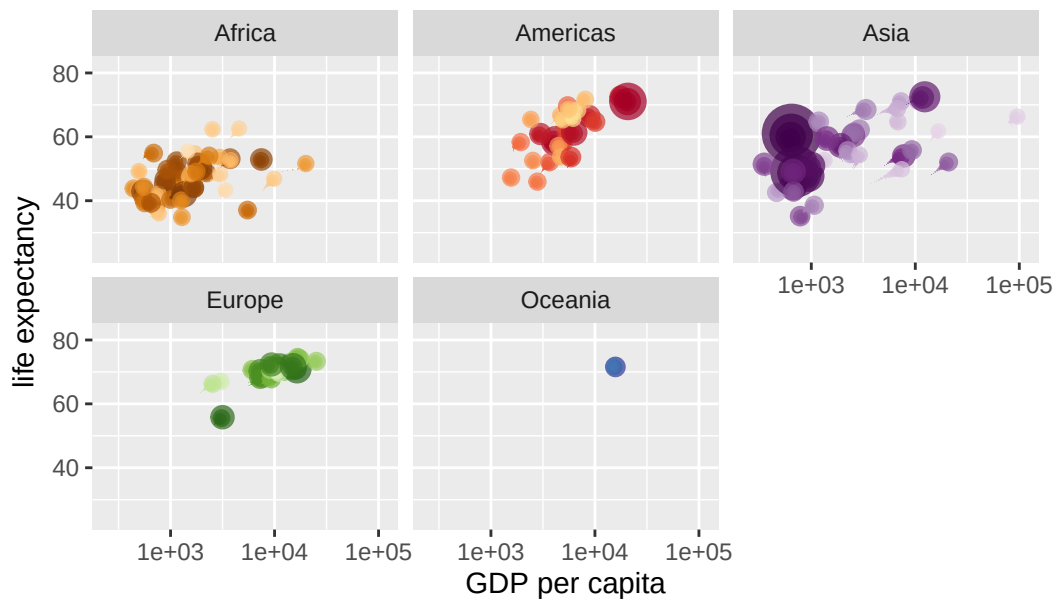
Year: 1969



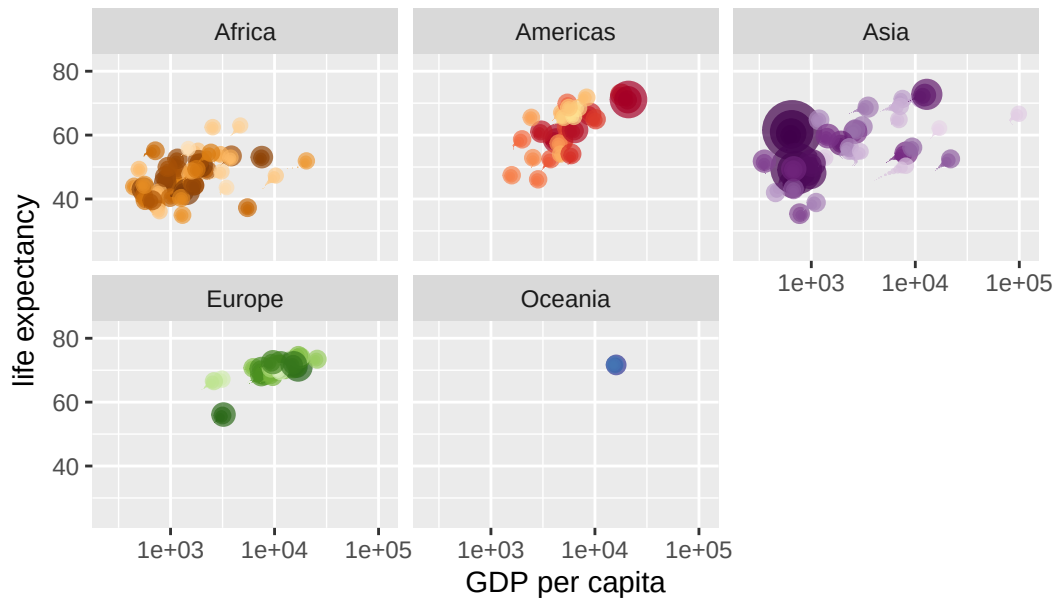
Year: 1969



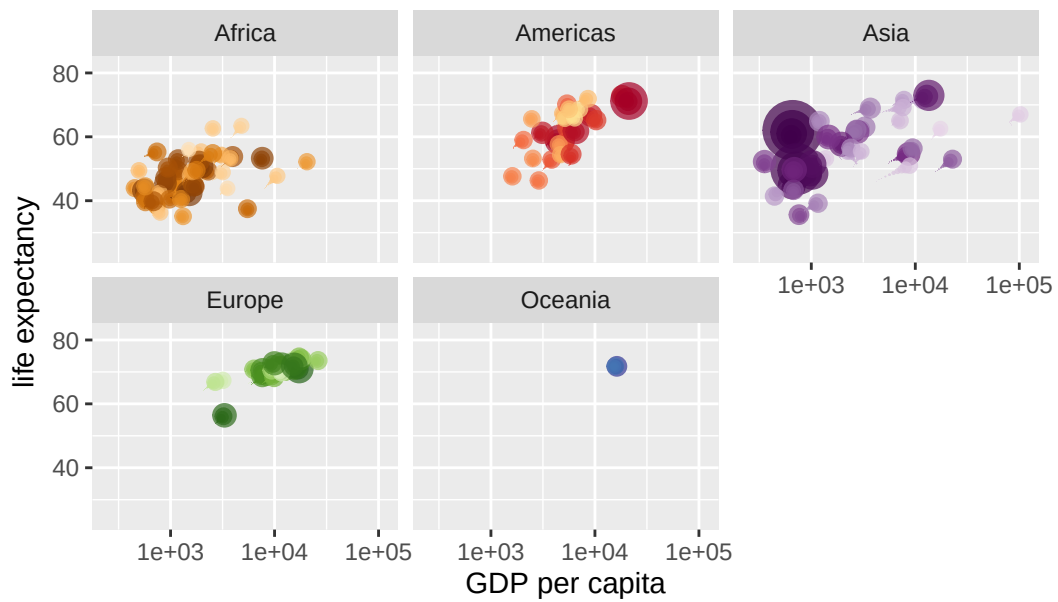
Year: 1970



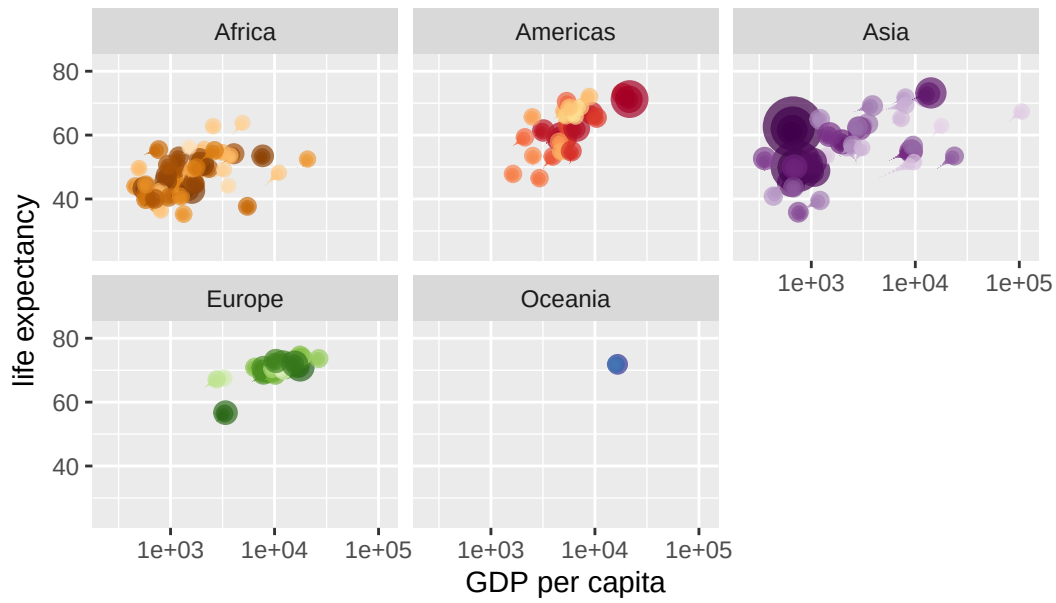
Year: 1970



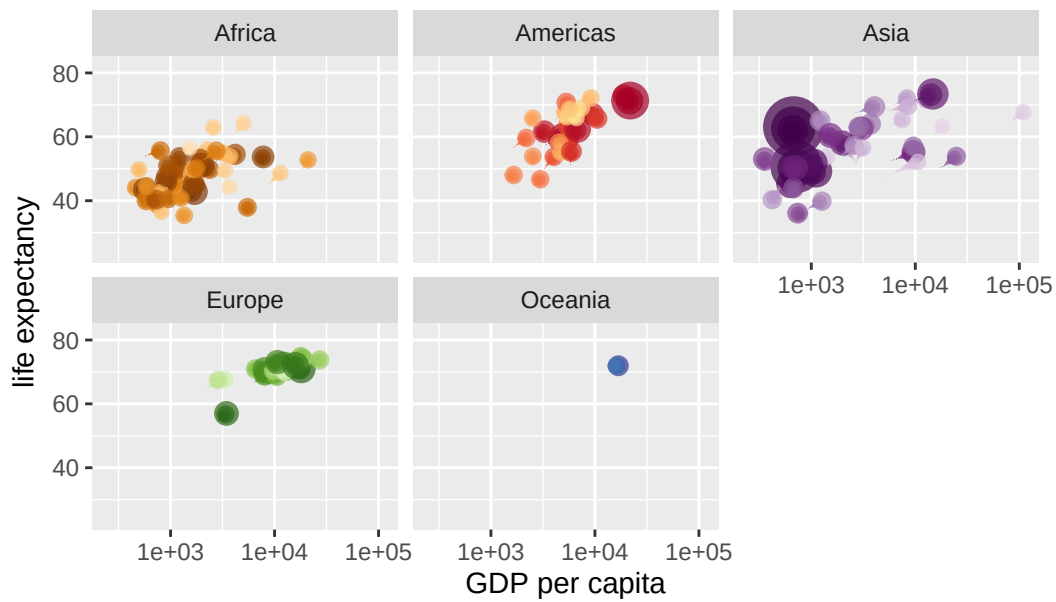
Year: 1971



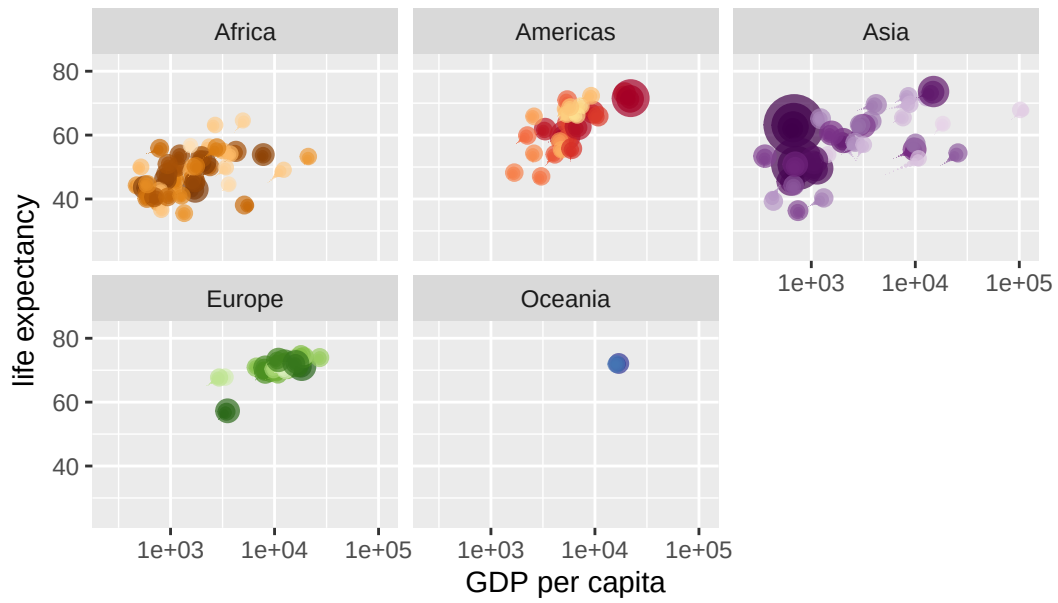
Year: 1971



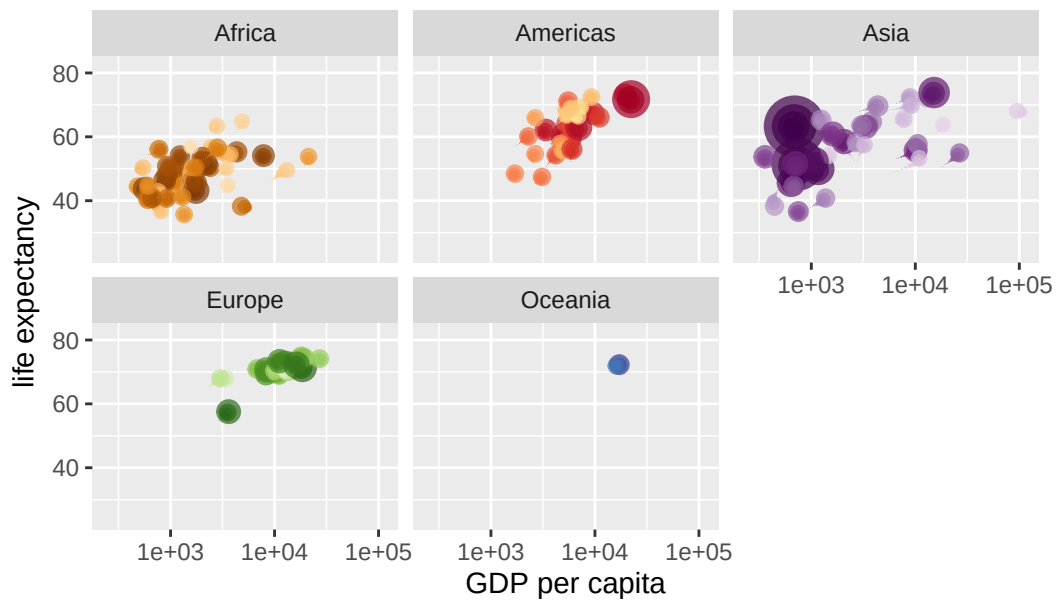
Year: 1972



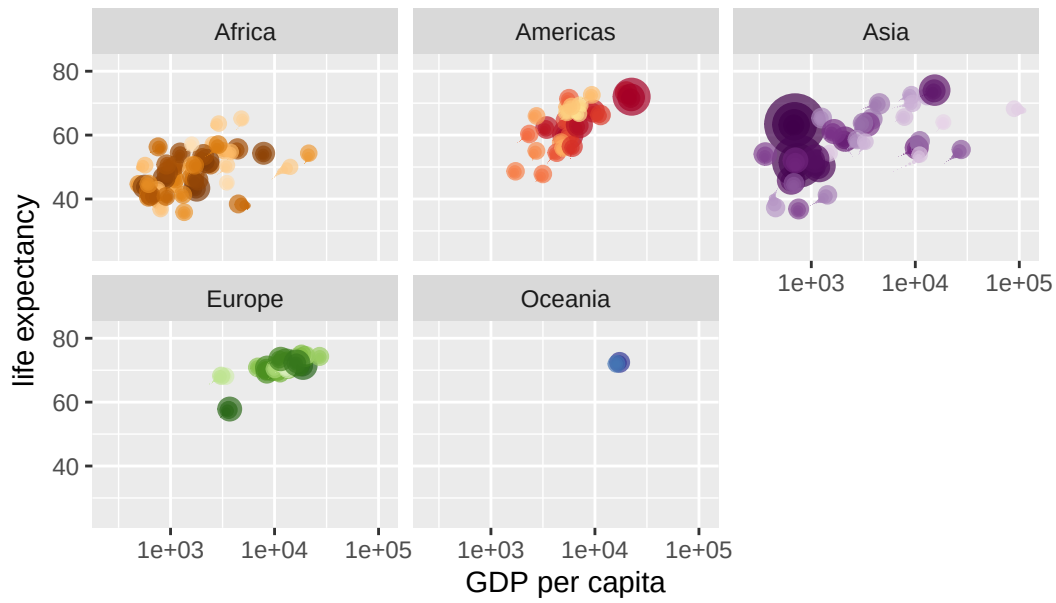
Year: 1973



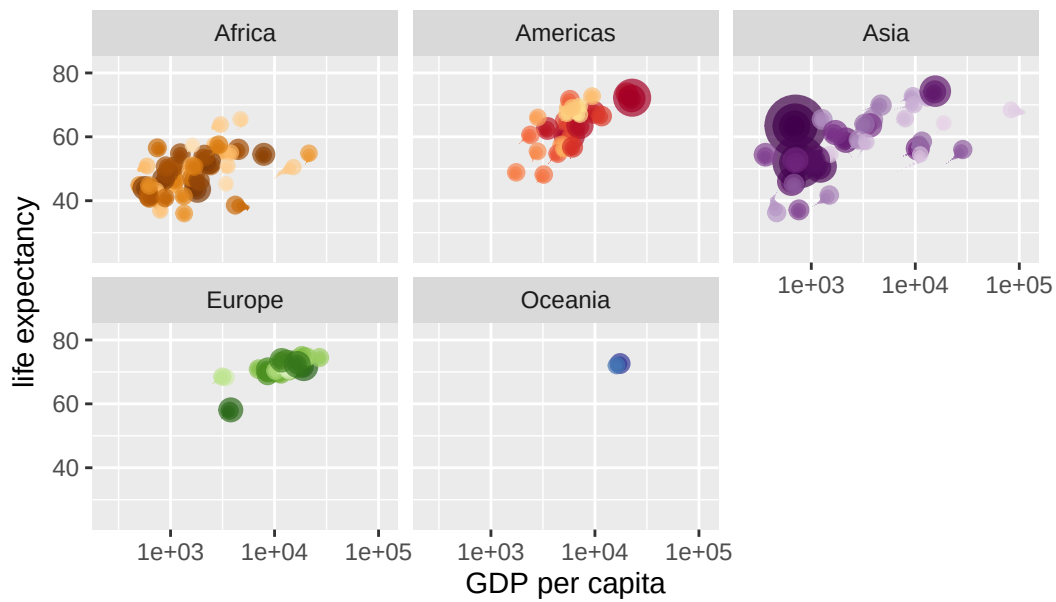
Year: 1973



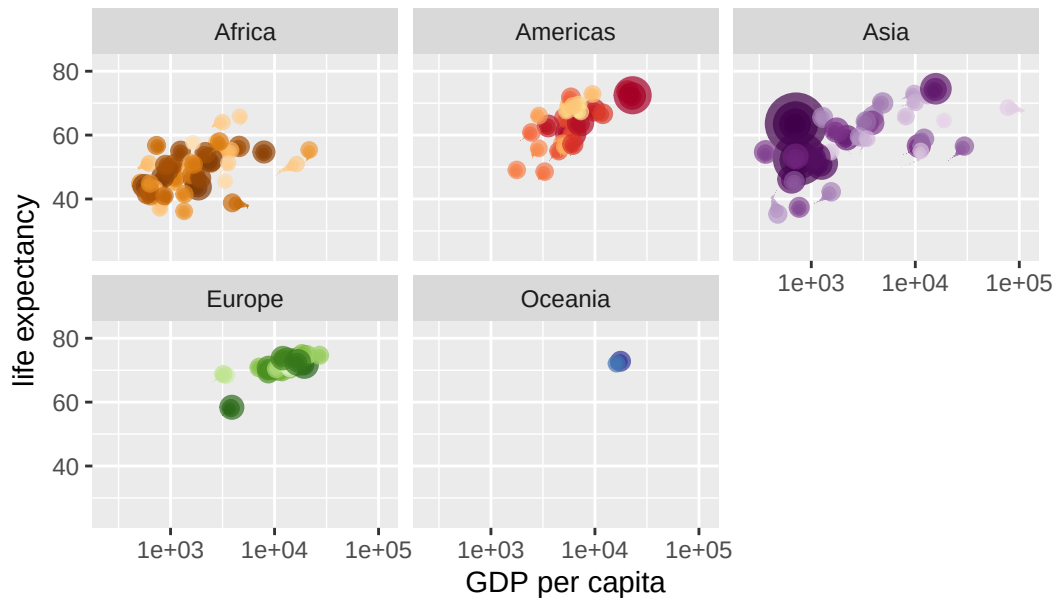
Year: 1974



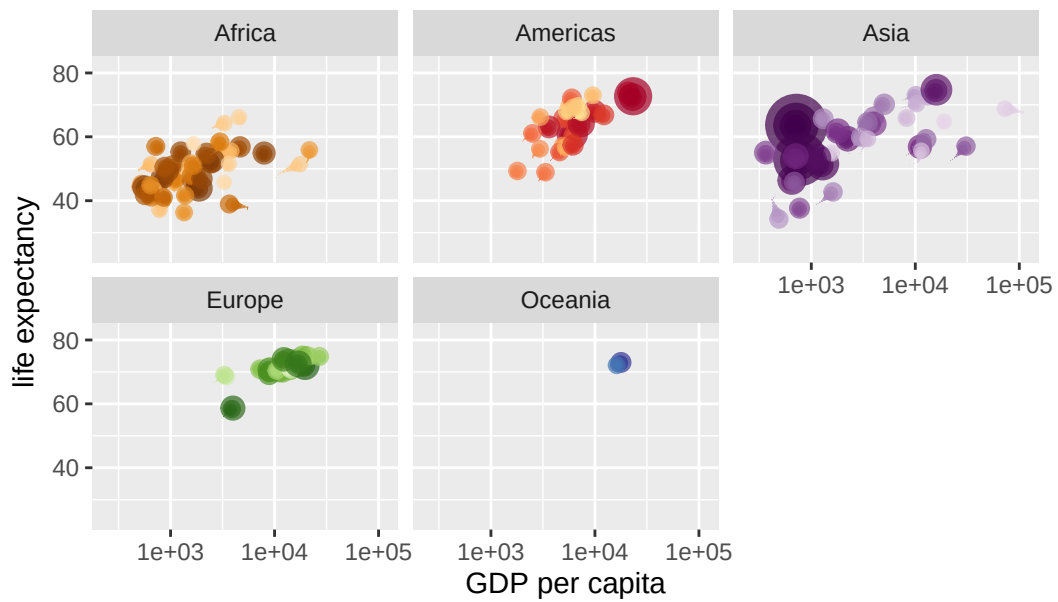
Year: 1974



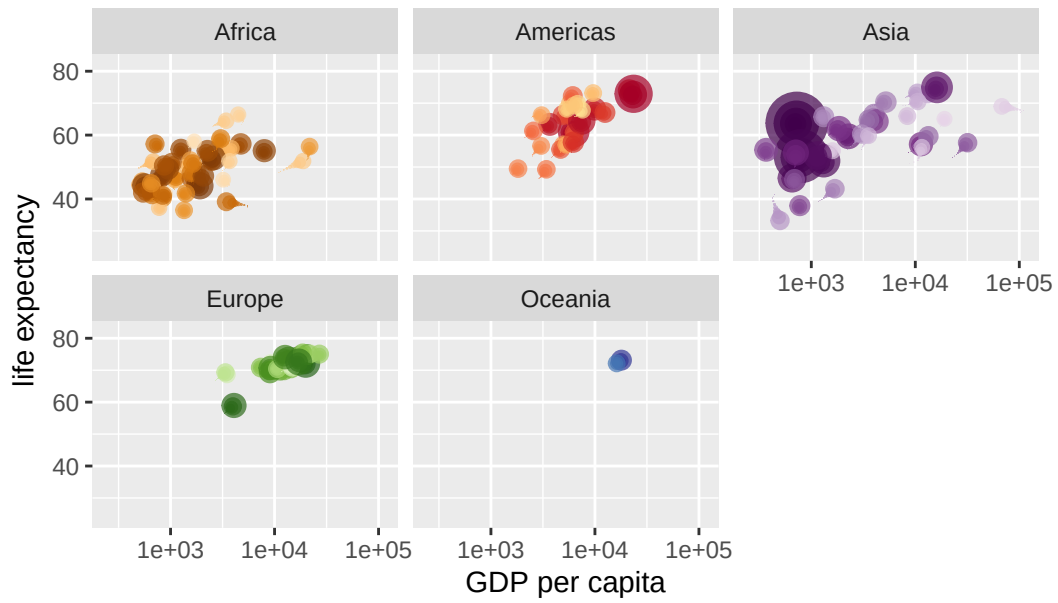
Year: 1975



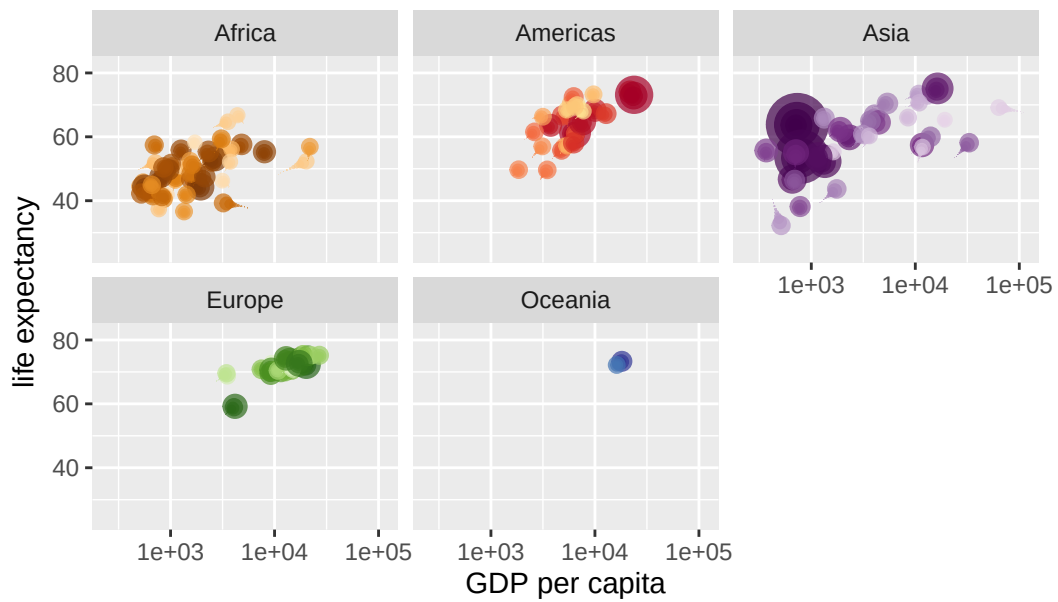
Year: 1975



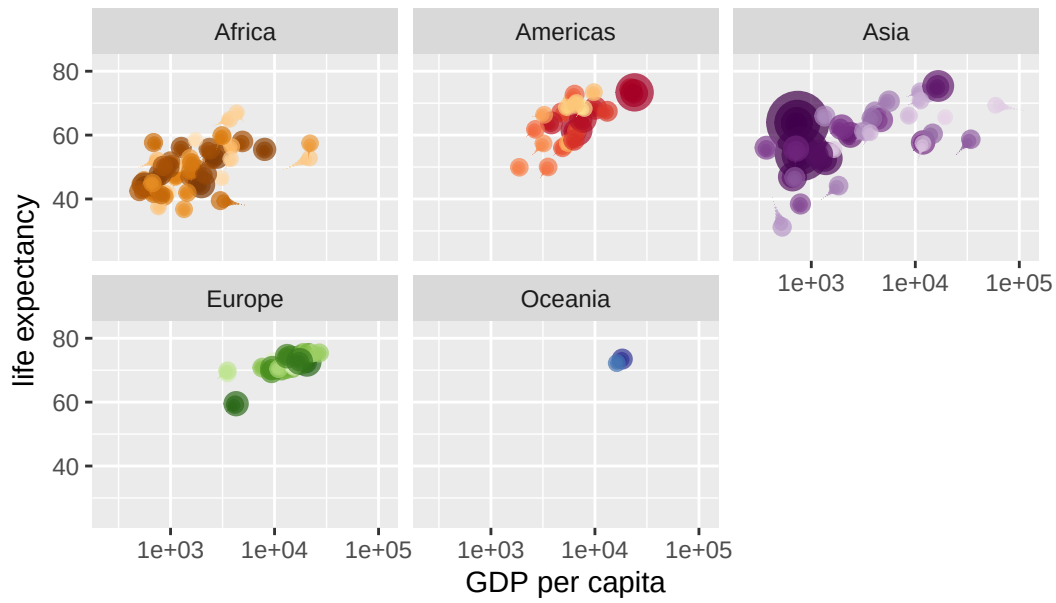
Year: 1976



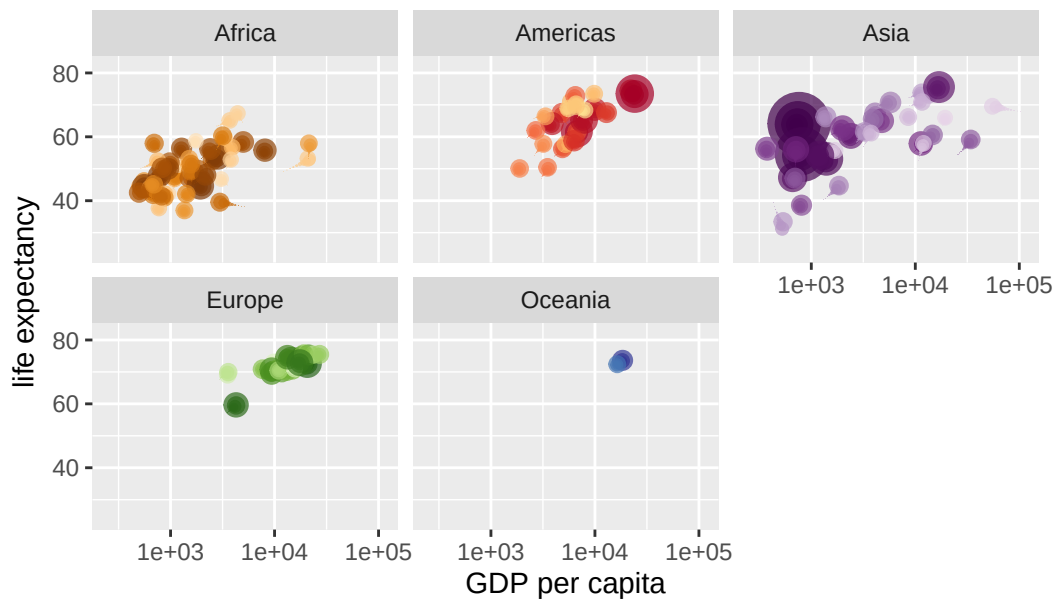
Year: 1976



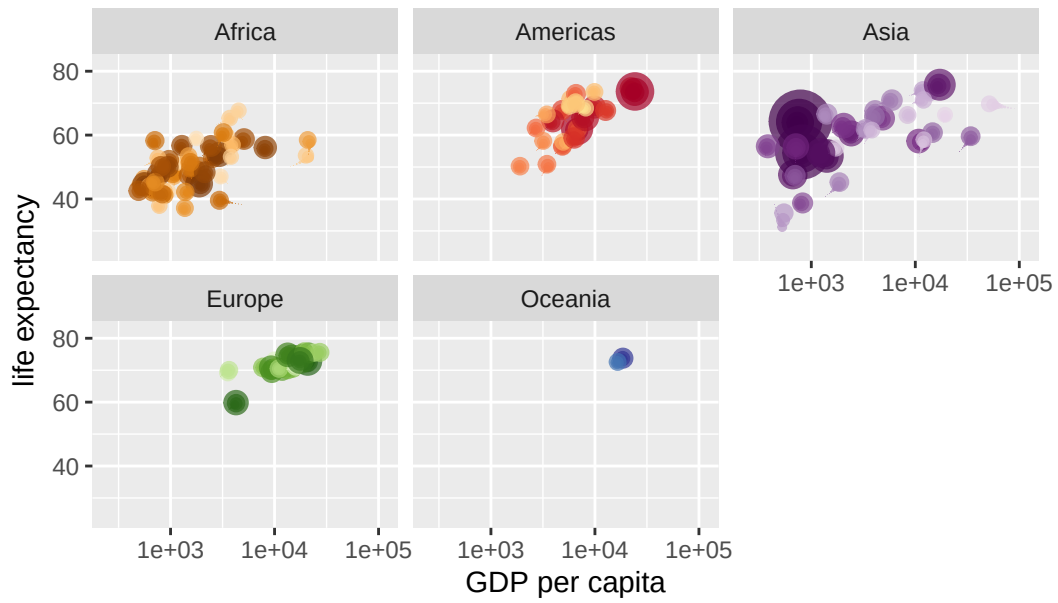
Year: 1977



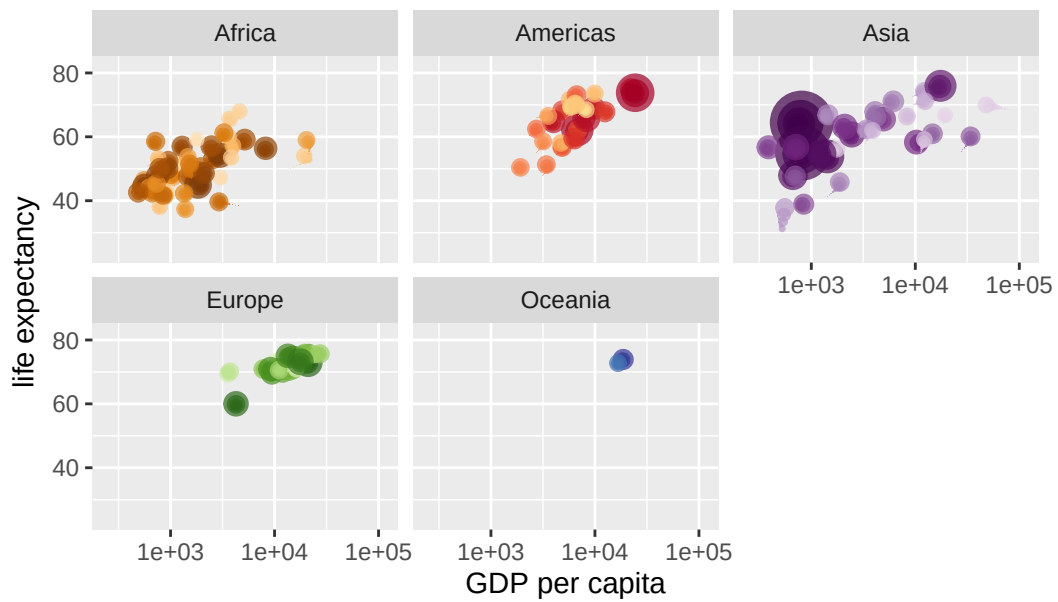
Year: 1978



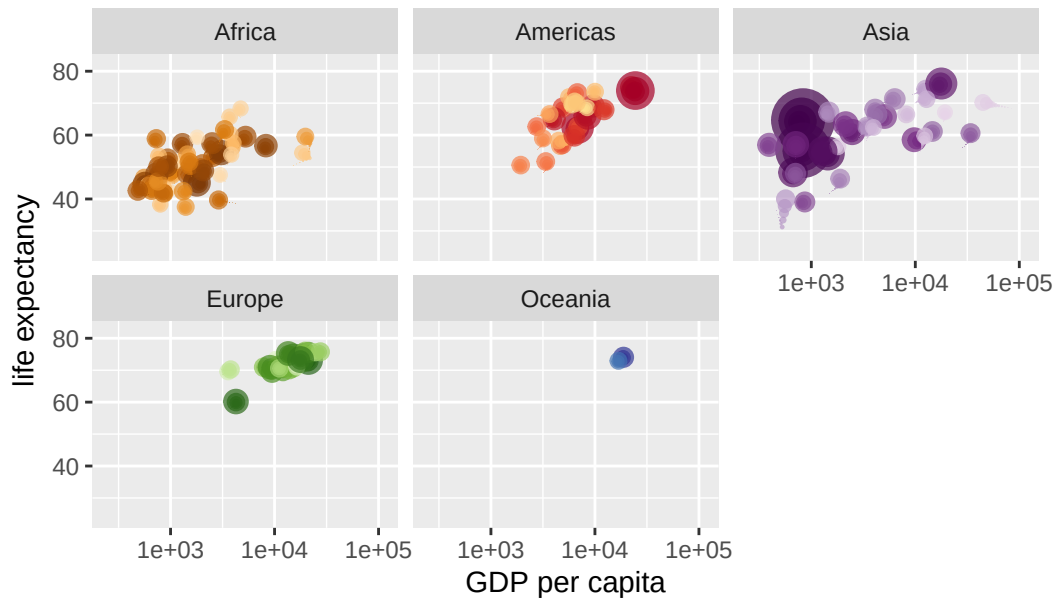
Year: 1978



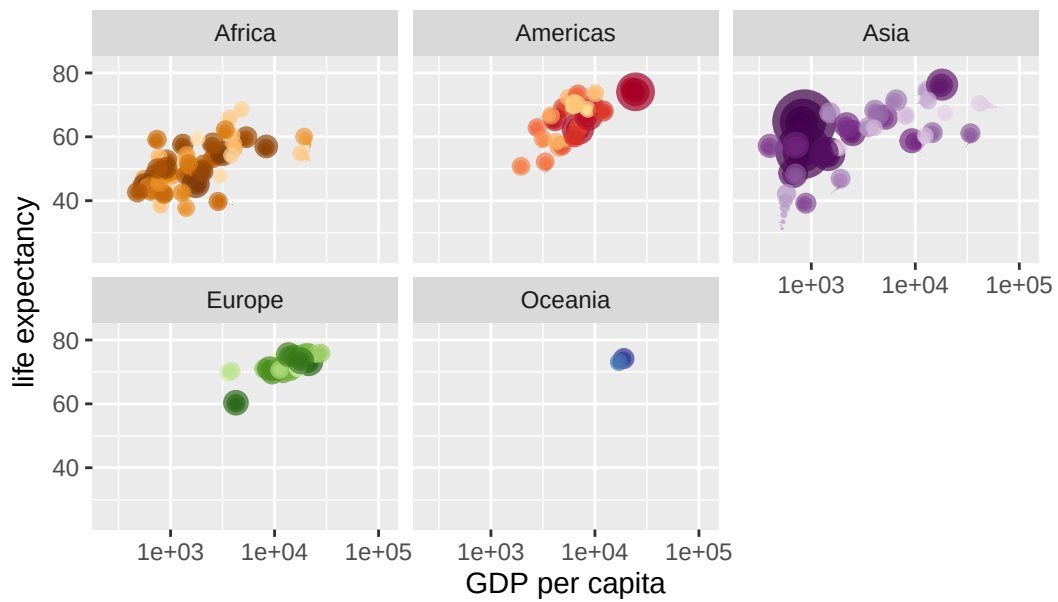
Year: 1979



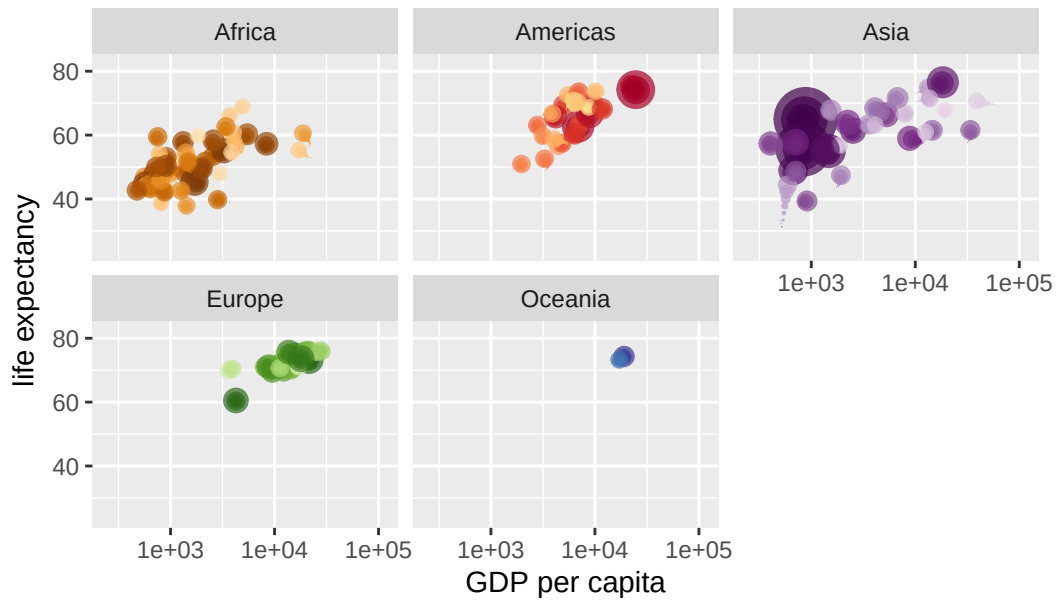
Year: 1979



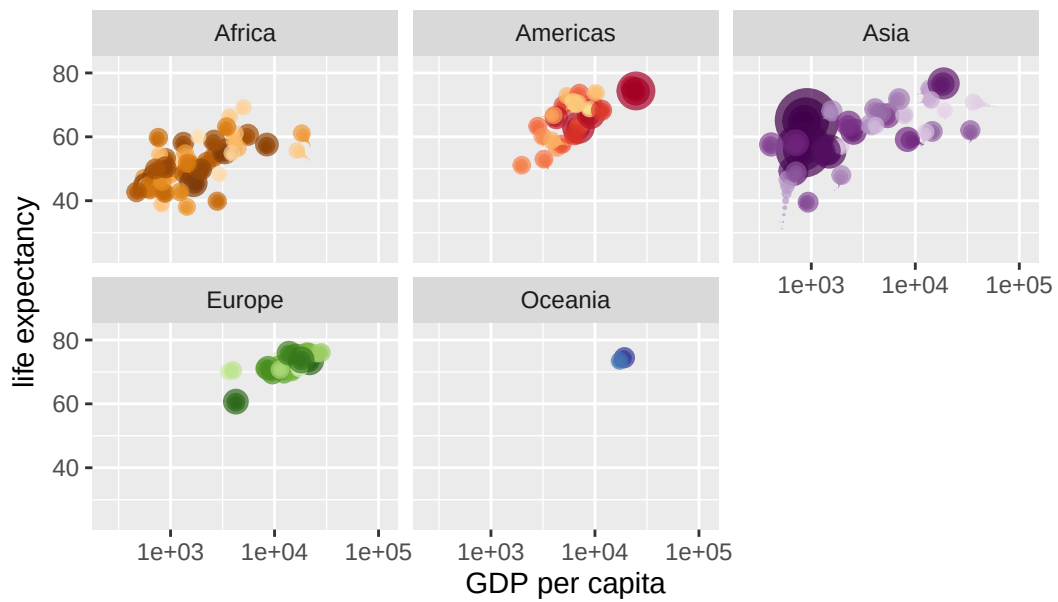
Year: 1980



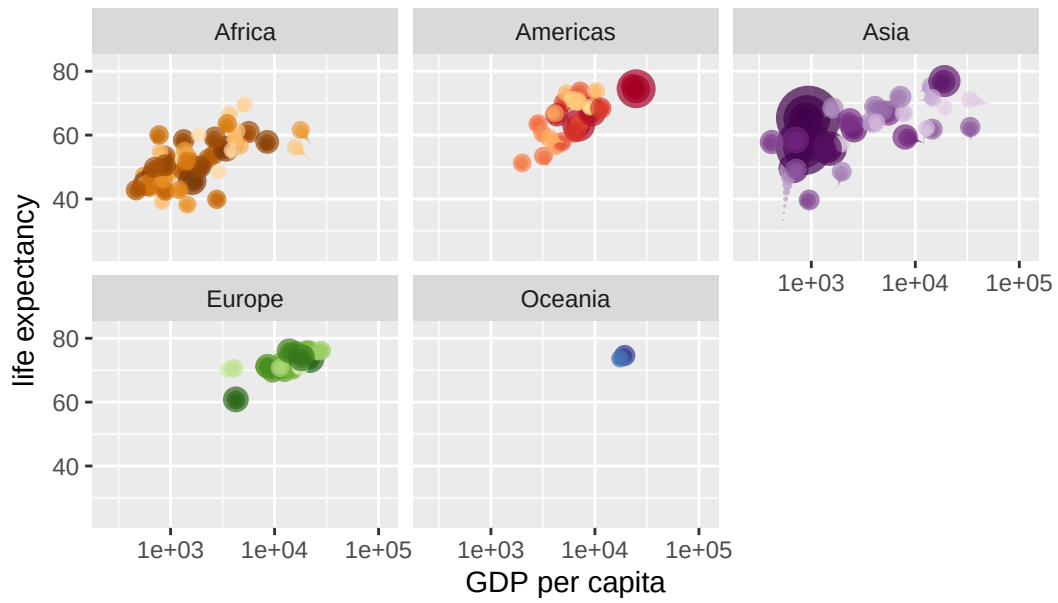
Year: 1980



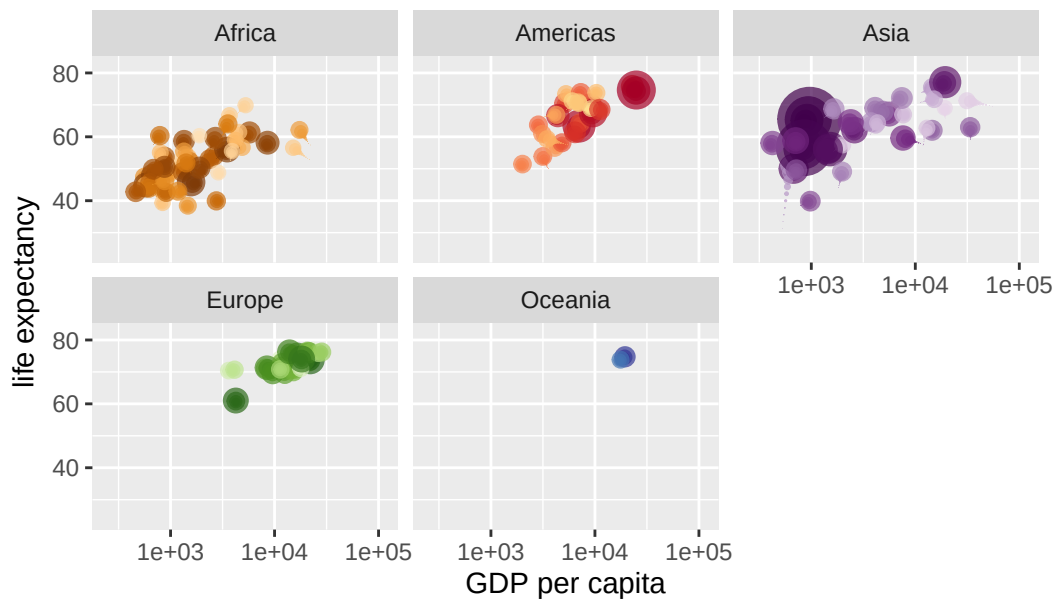
Year: 1981



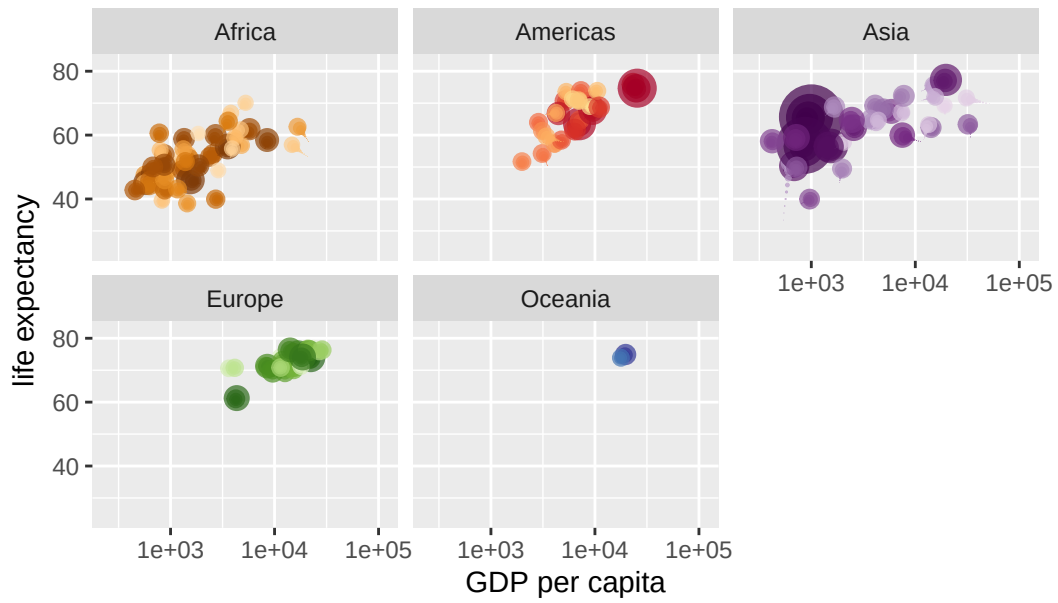
Year: 1981



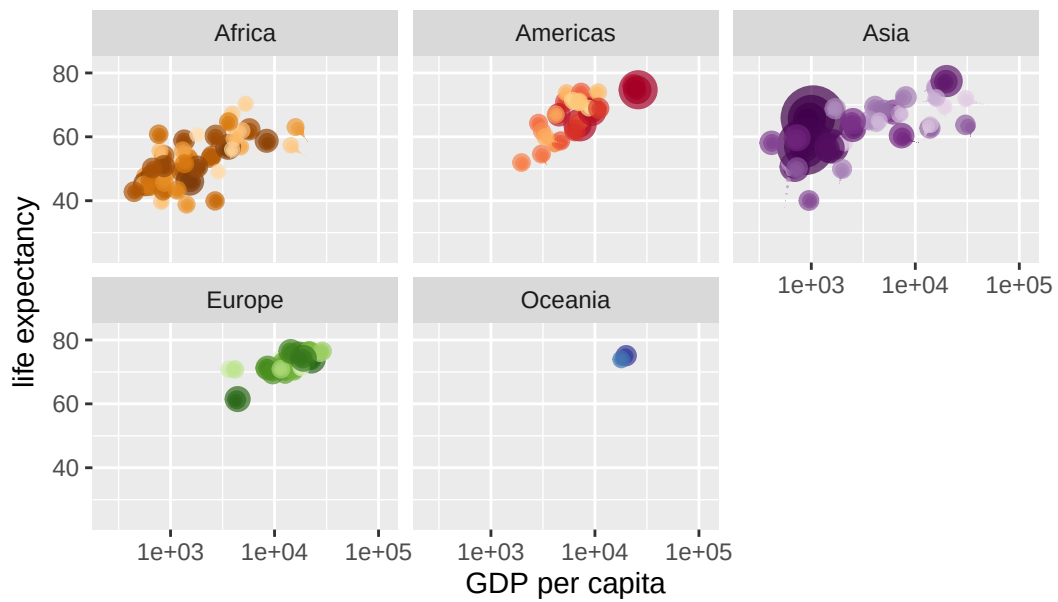
Year: 1982



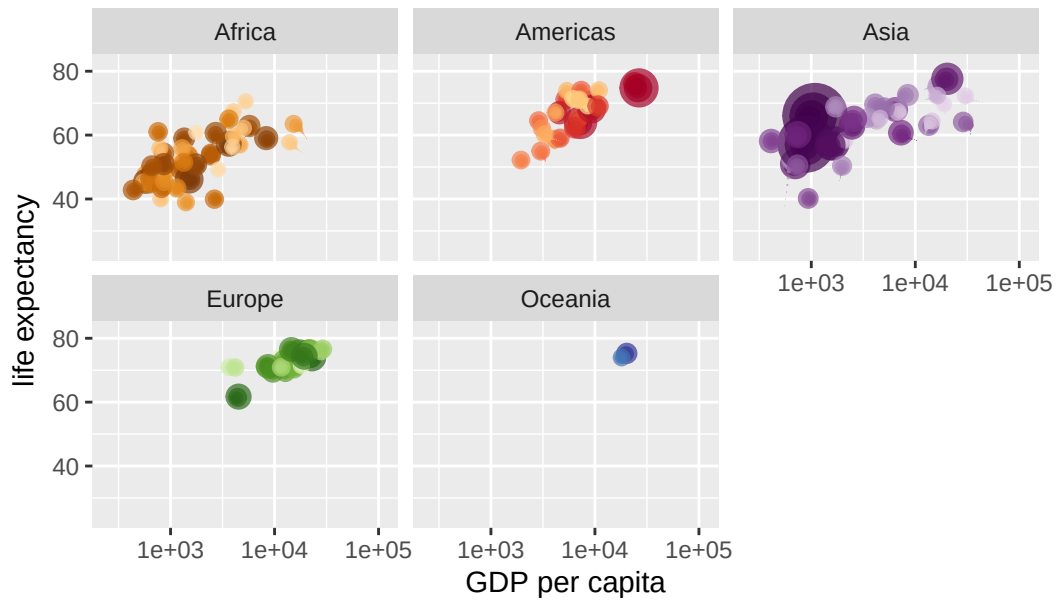
Year: 1983



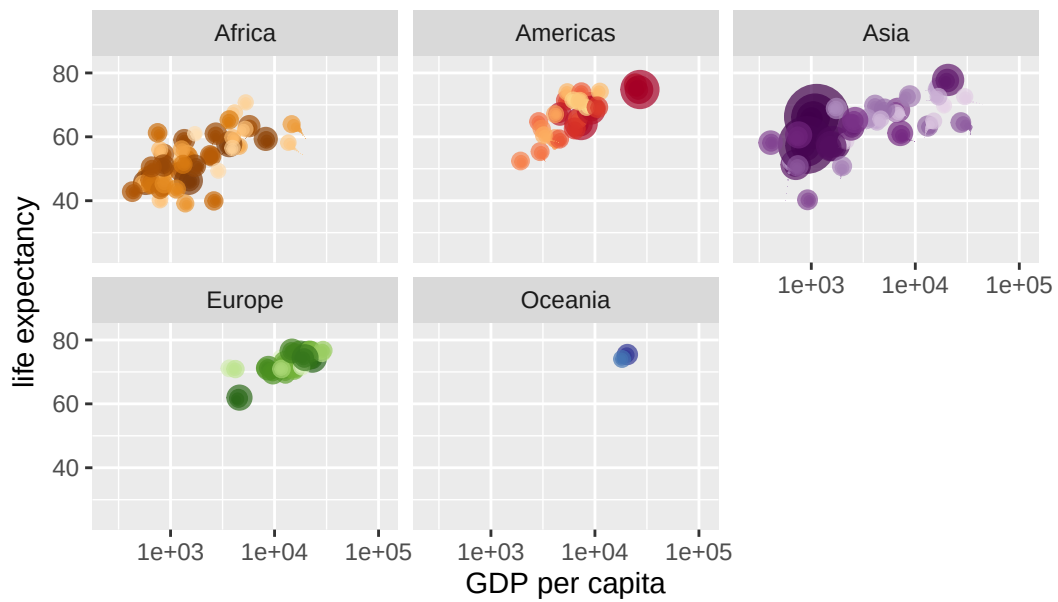
Year: 1983



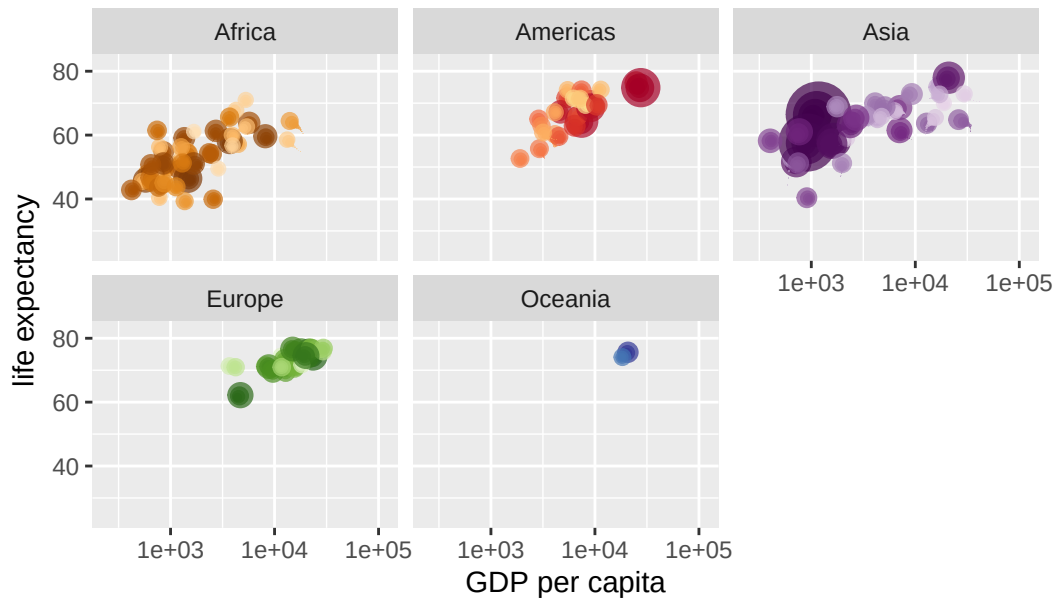
Year: 1984



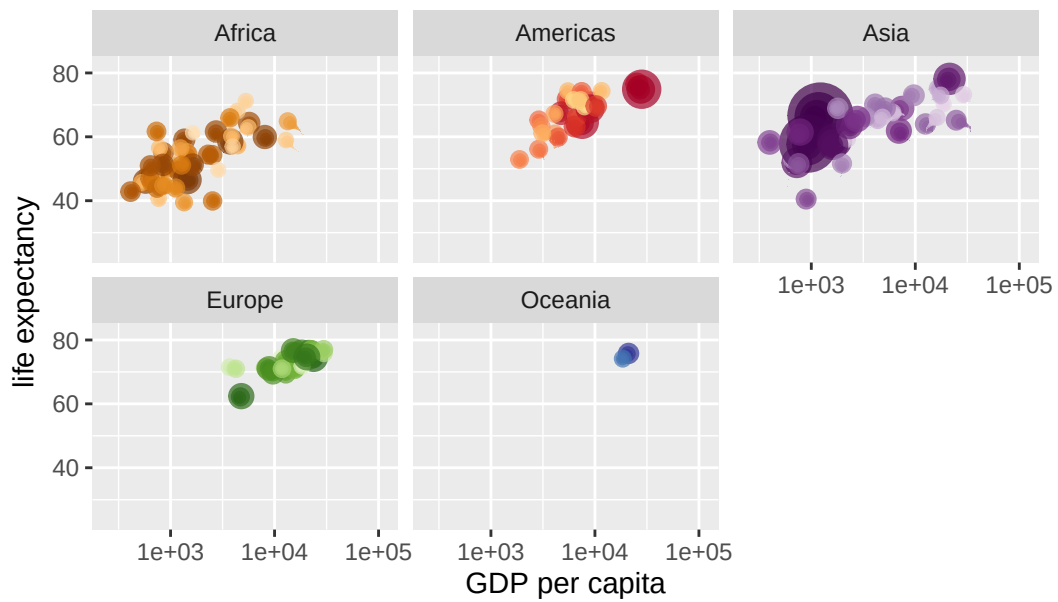
Year: 1984



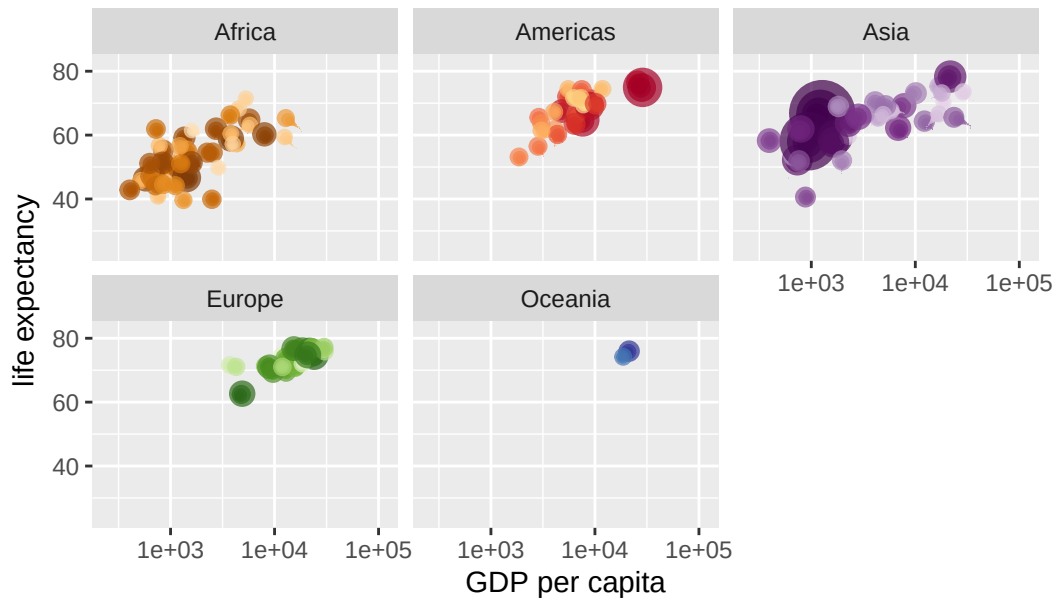
Year: 1985



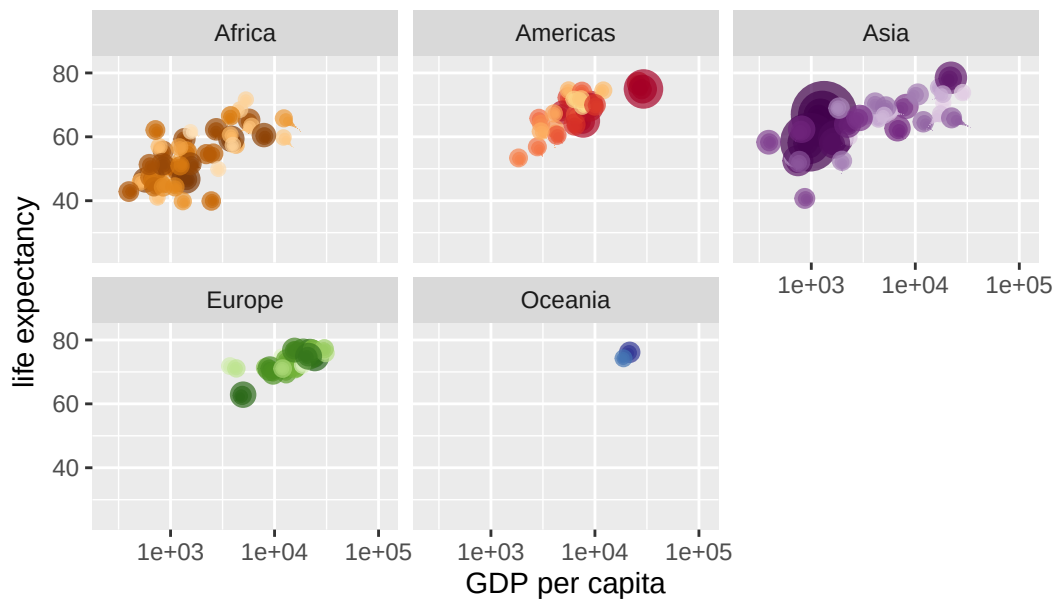
Year: 1985



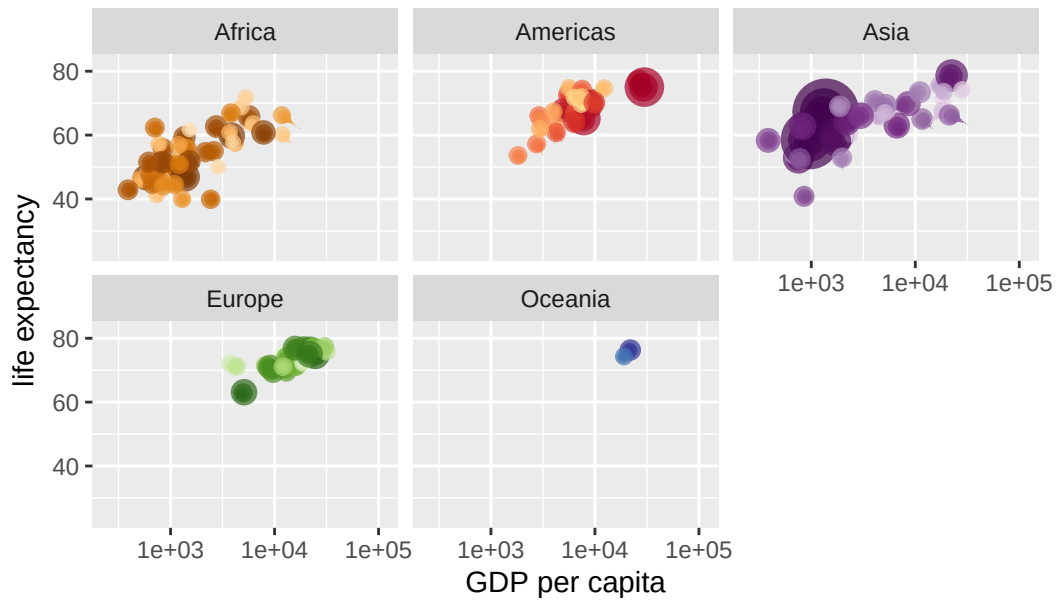
Year: 1986



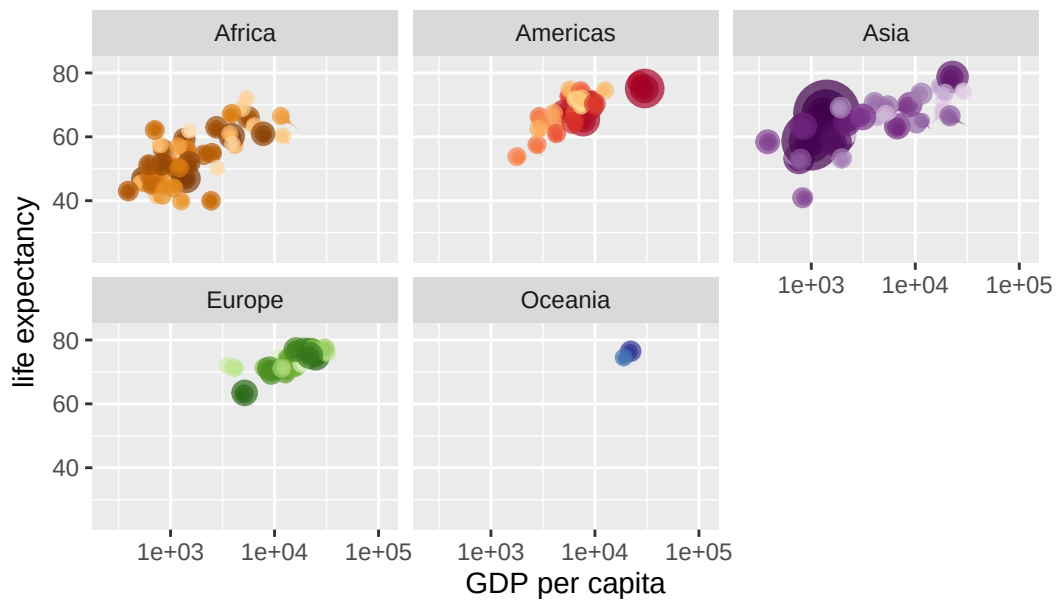
Year: 1986



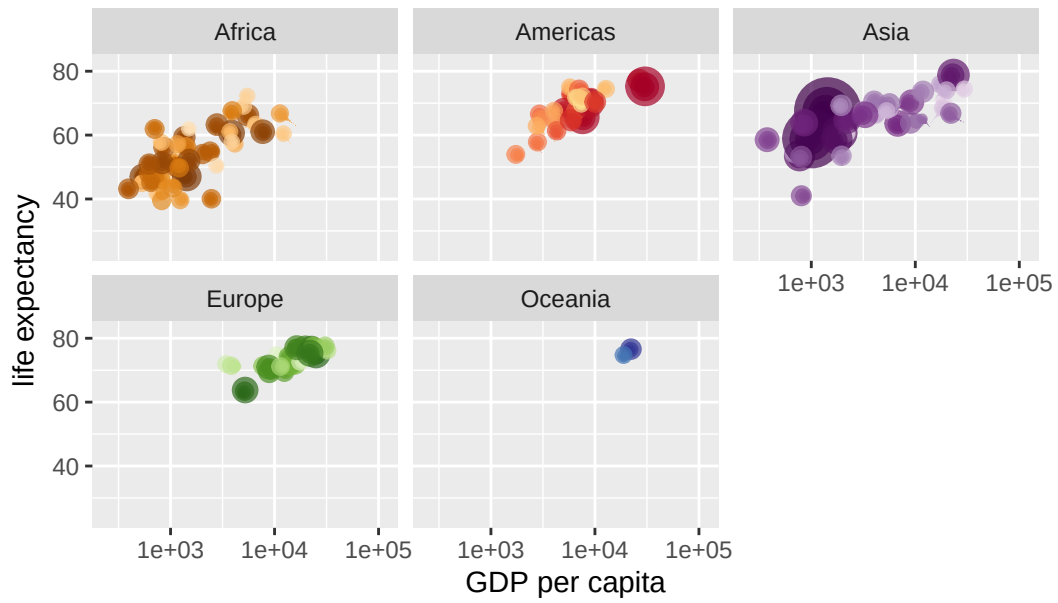
Year: 1987



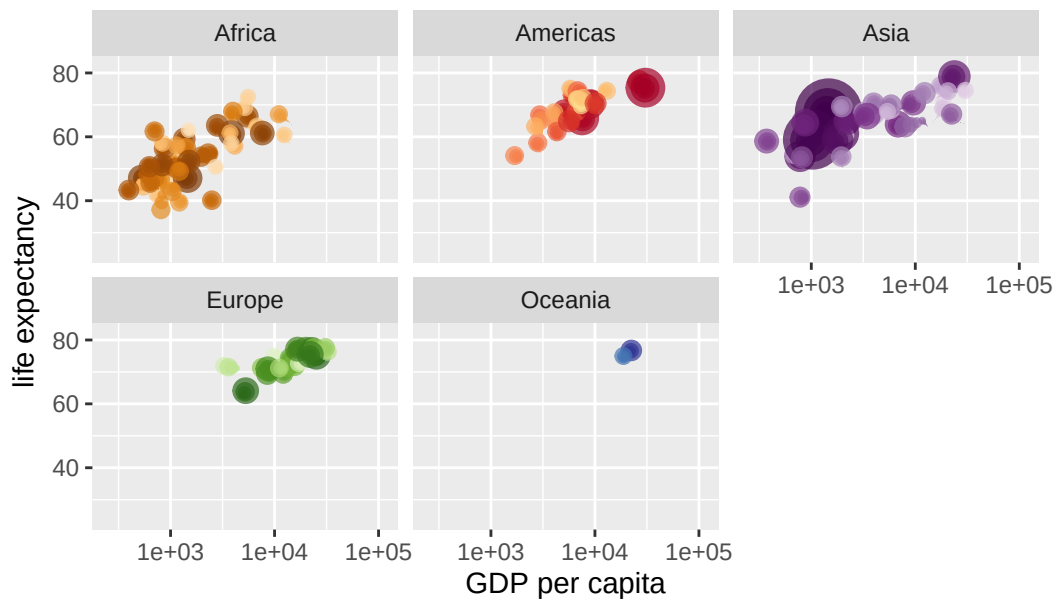
Year: 1988



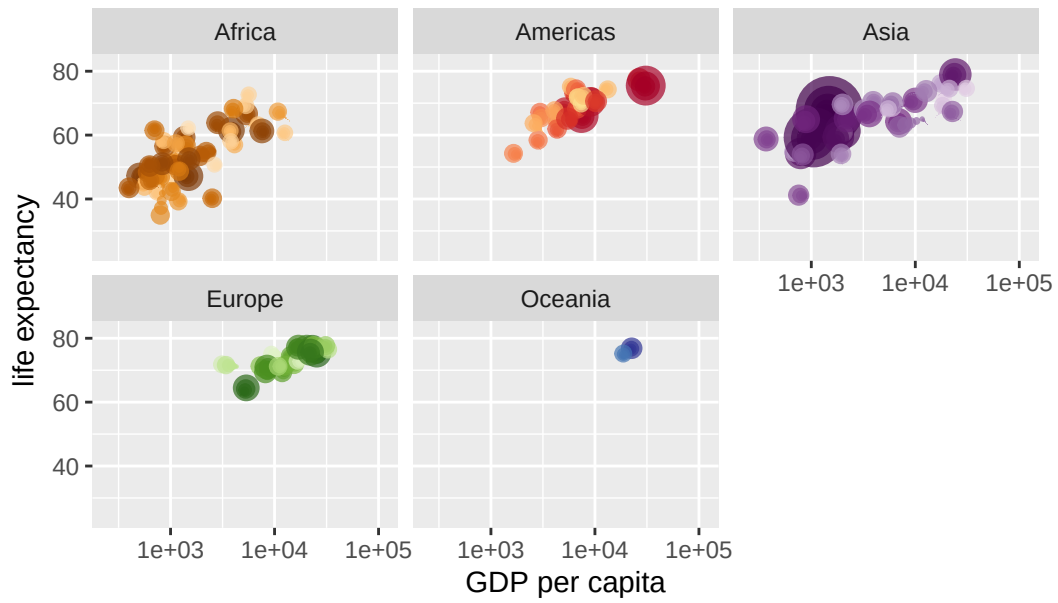
Year: 1988



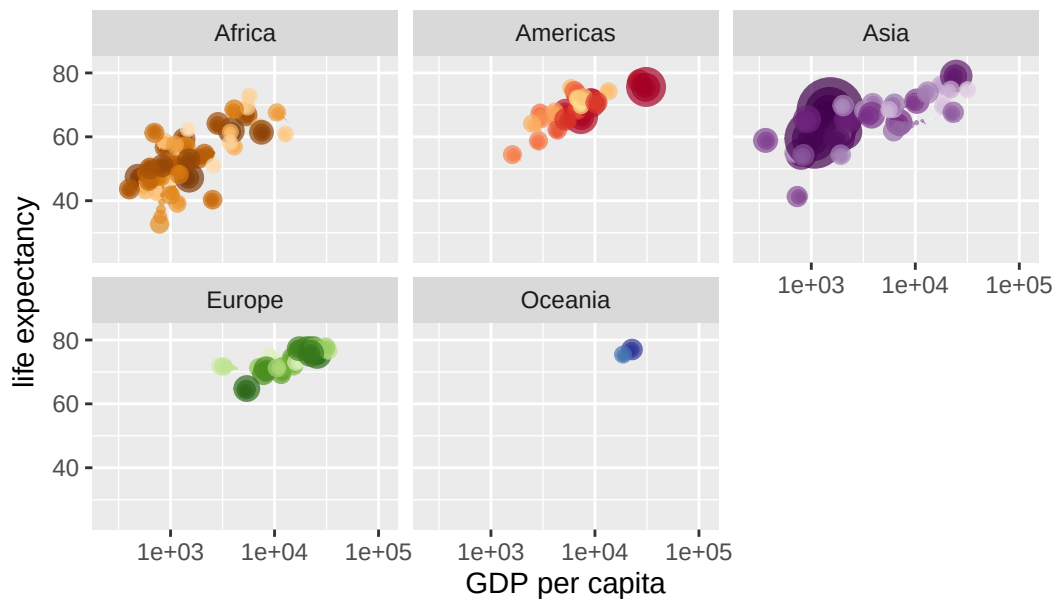
Year: 1989



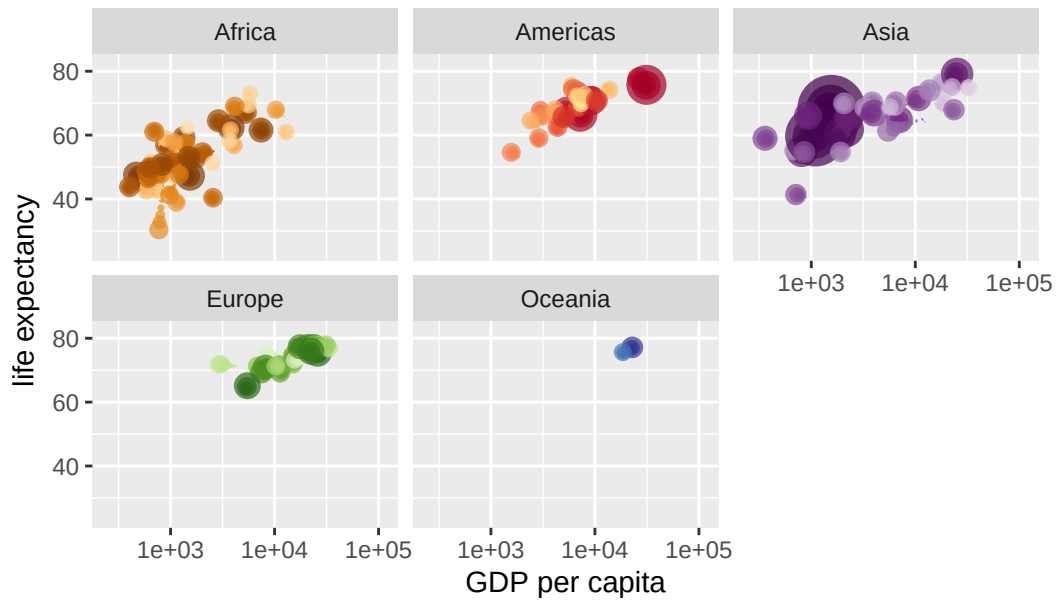
Year: 1989



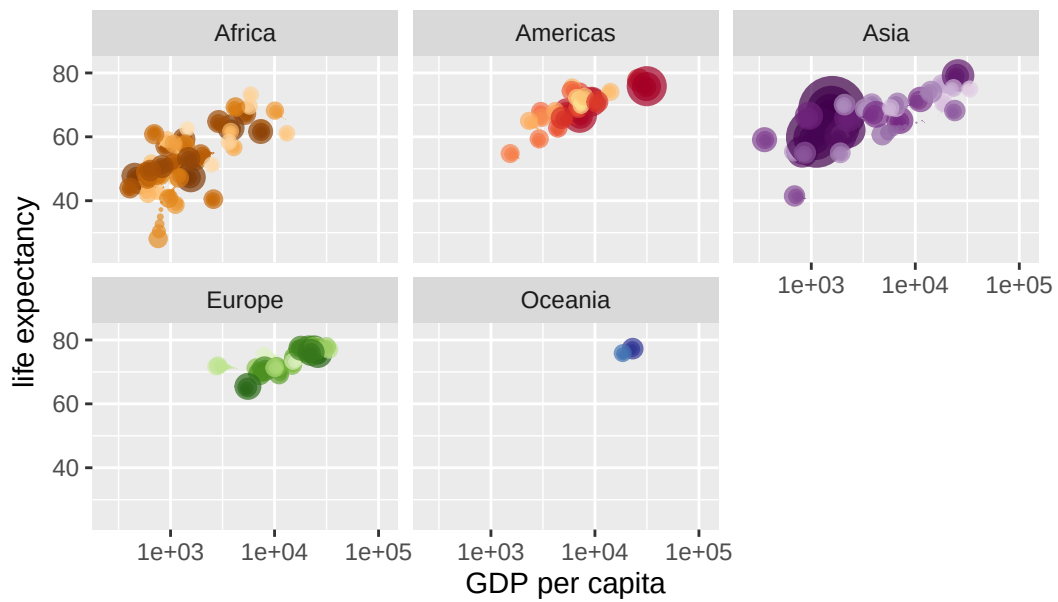
Year: 1990



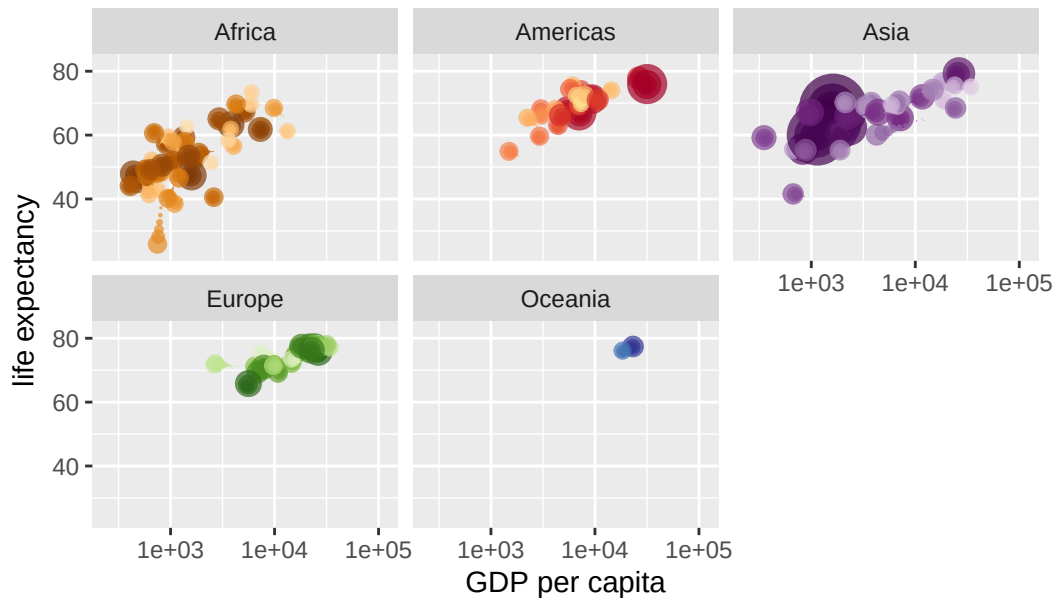
Year: 1990



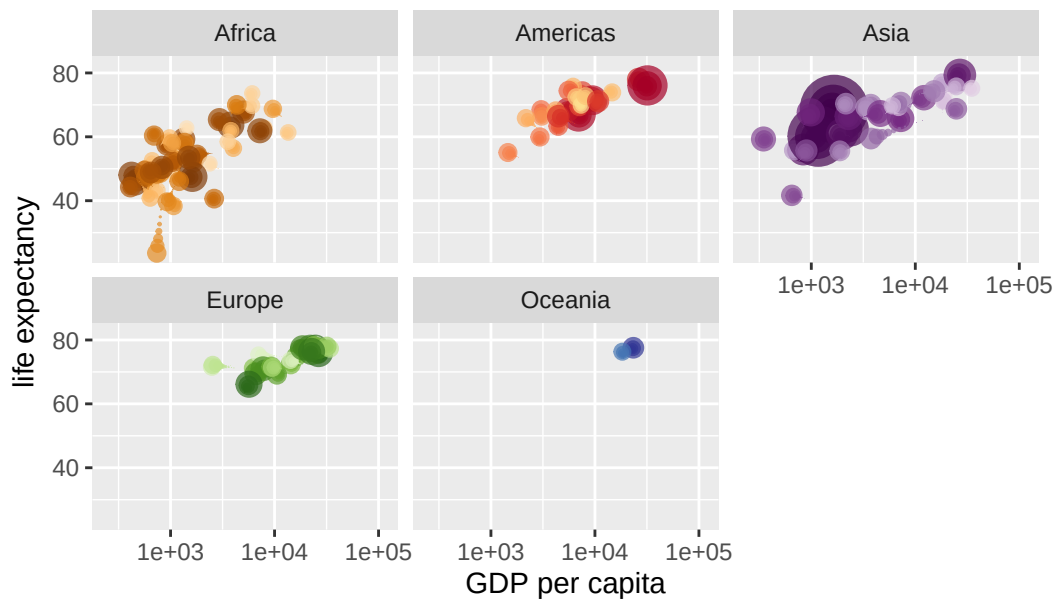
Year: 1991



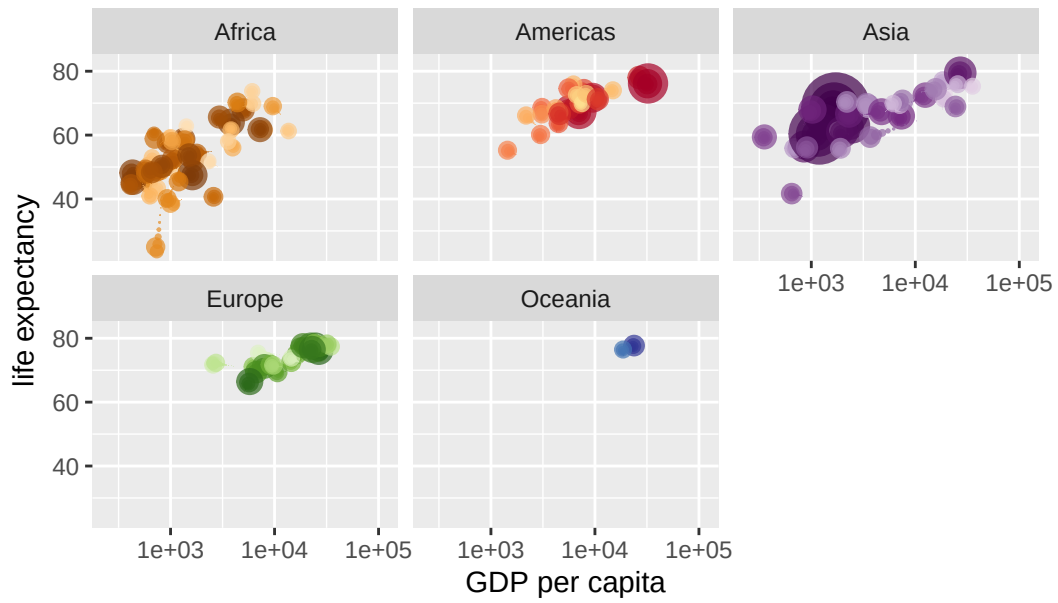
Year: 1991



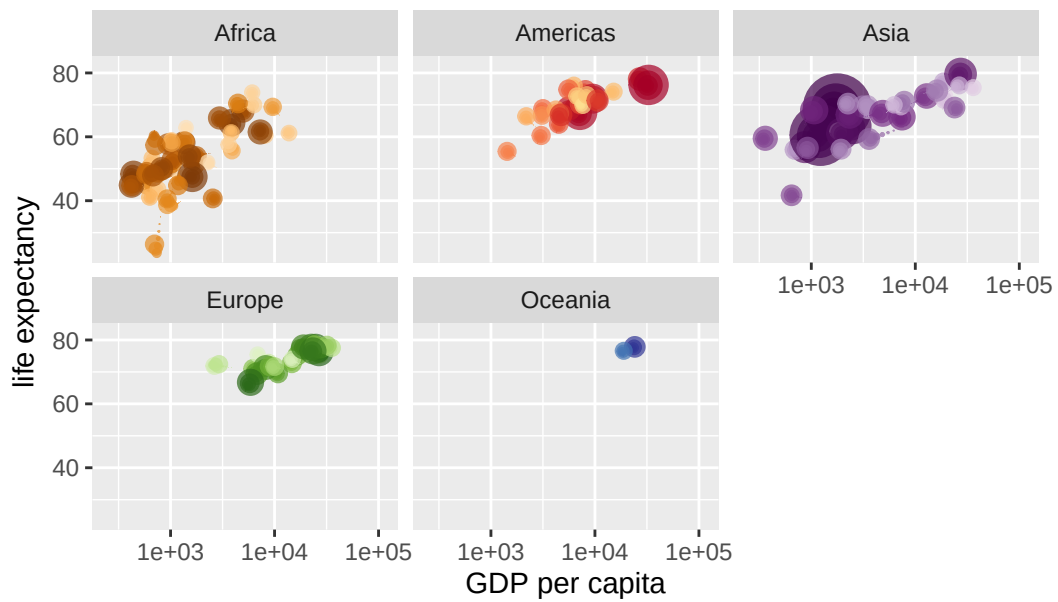
Year: 1992



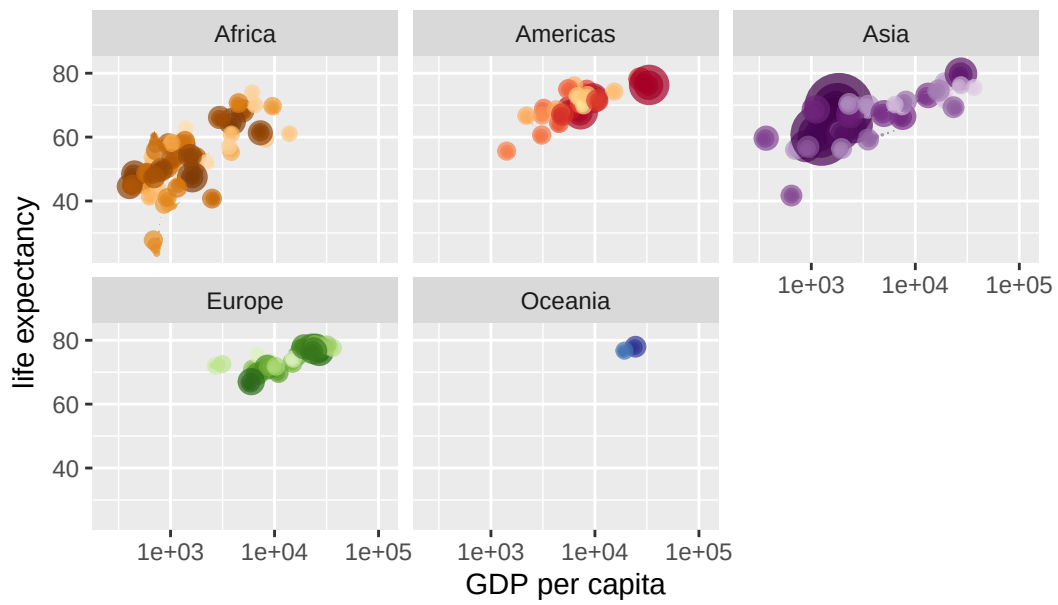
Year: 1993



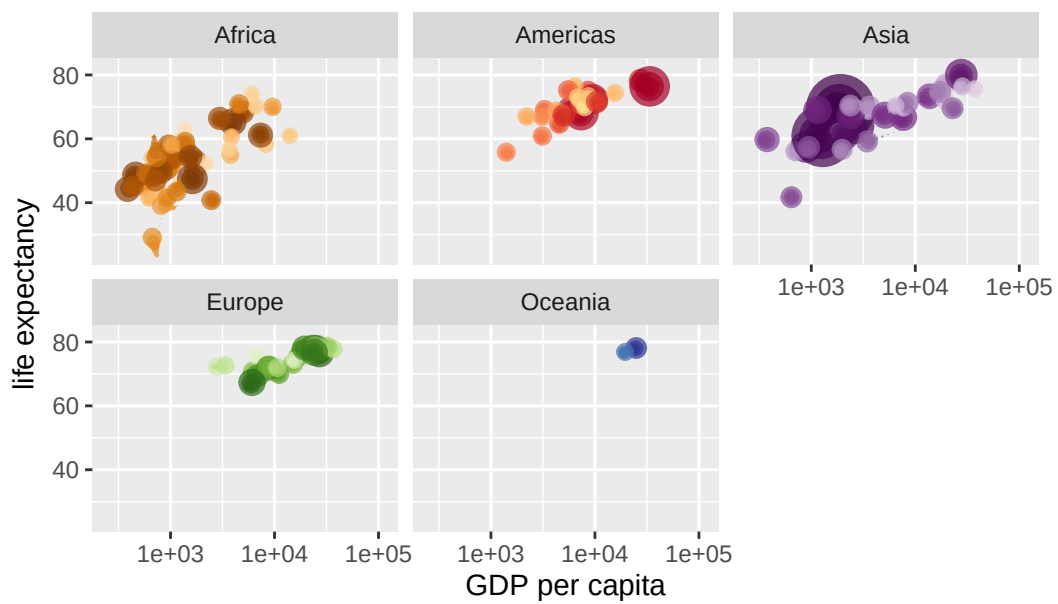
Year: 1993



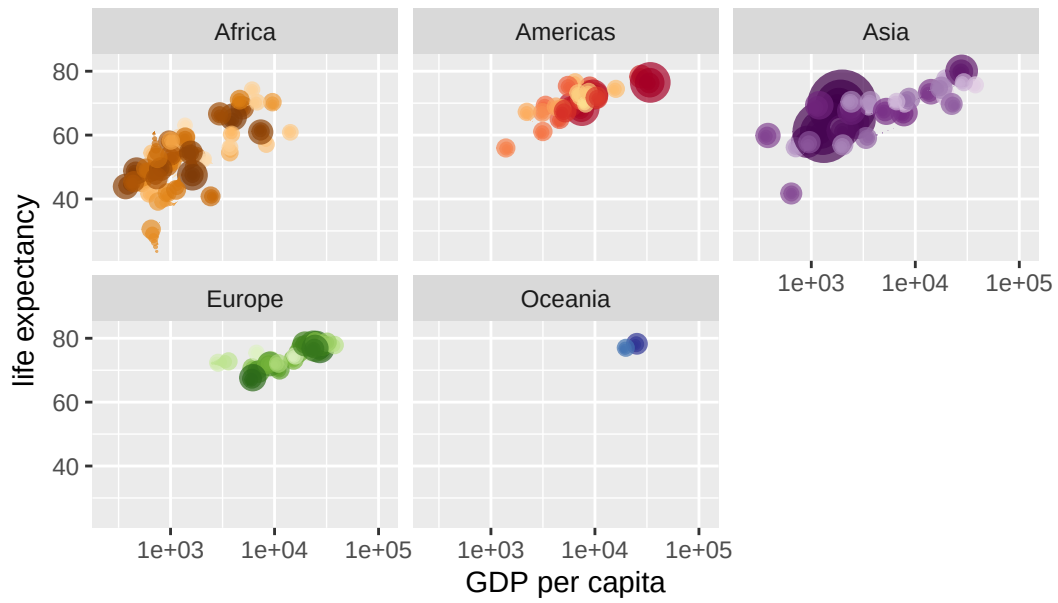
Year: 1994



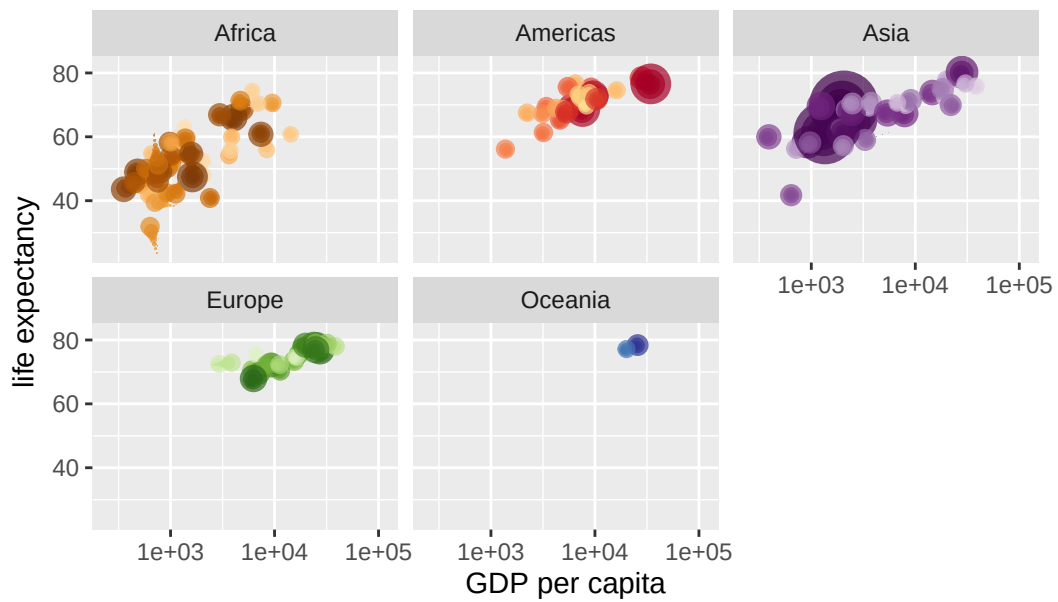
Year: 1994



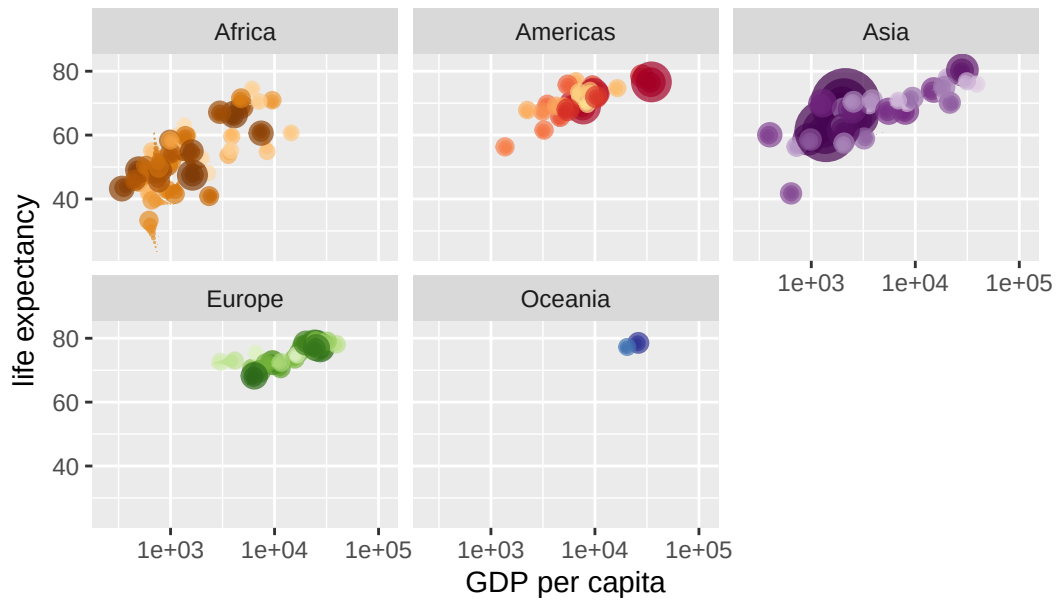
Year: 1995



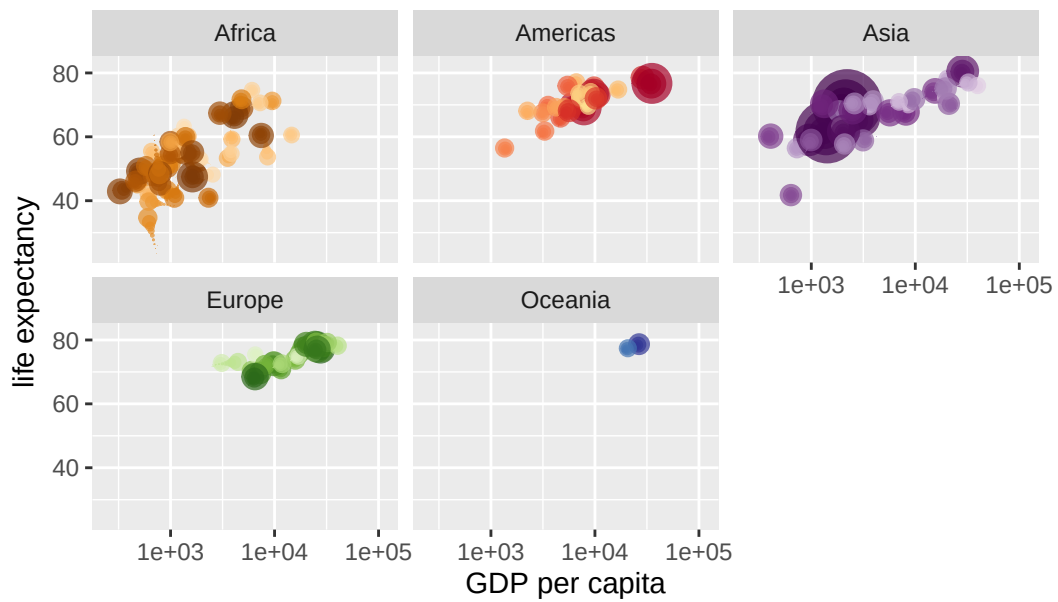
Year: 1995



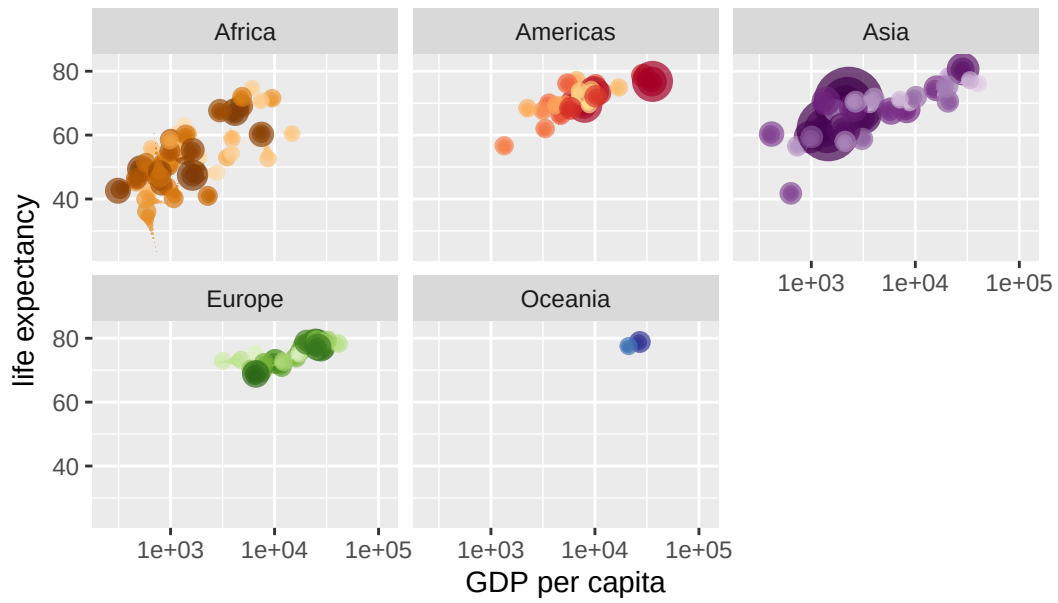
Year: 1996



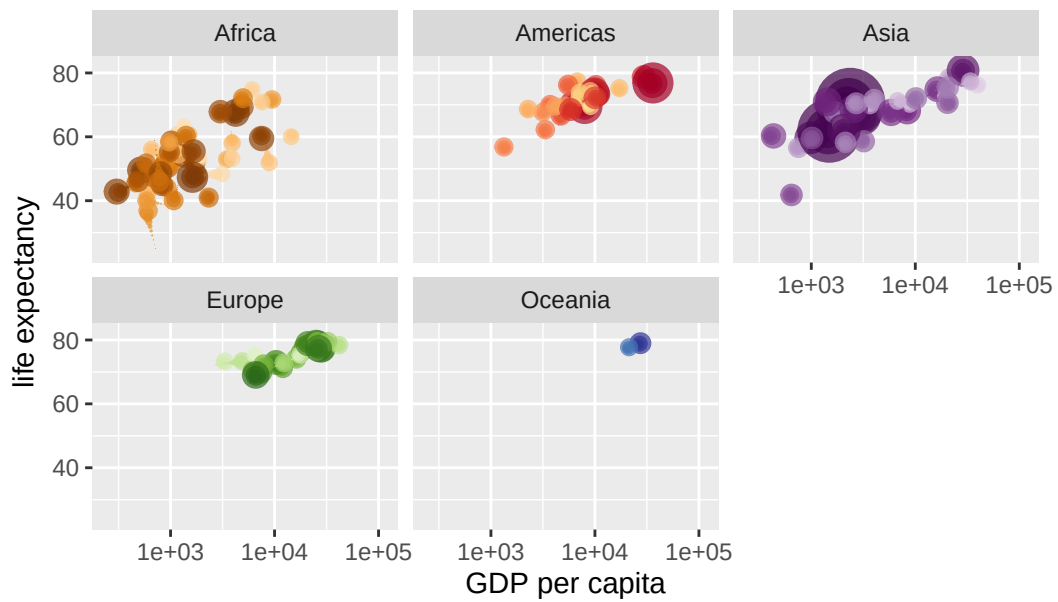
Year: 1996



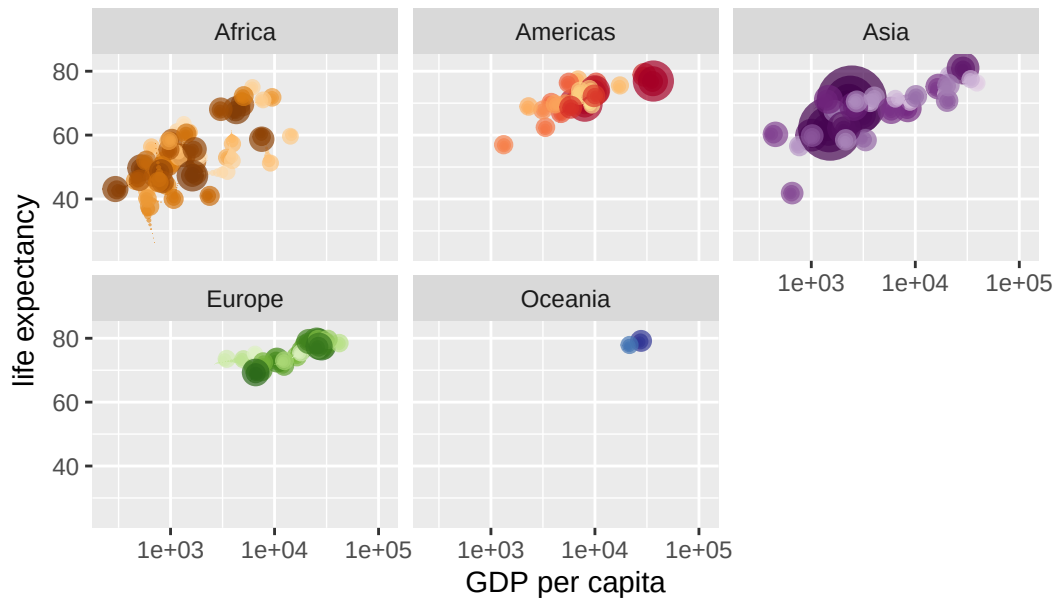
Year: 1997



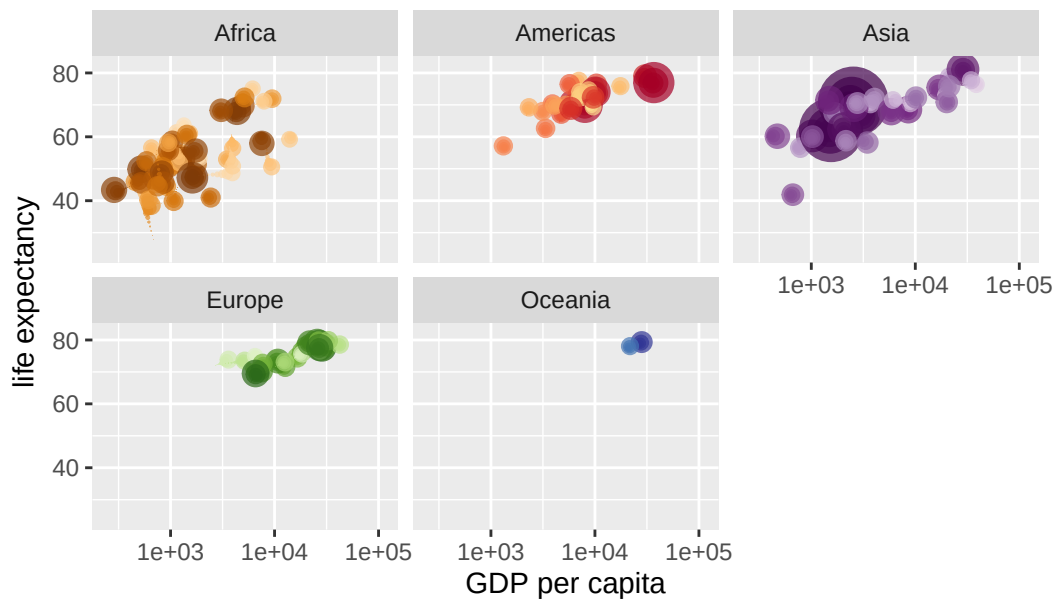
Year: 1998



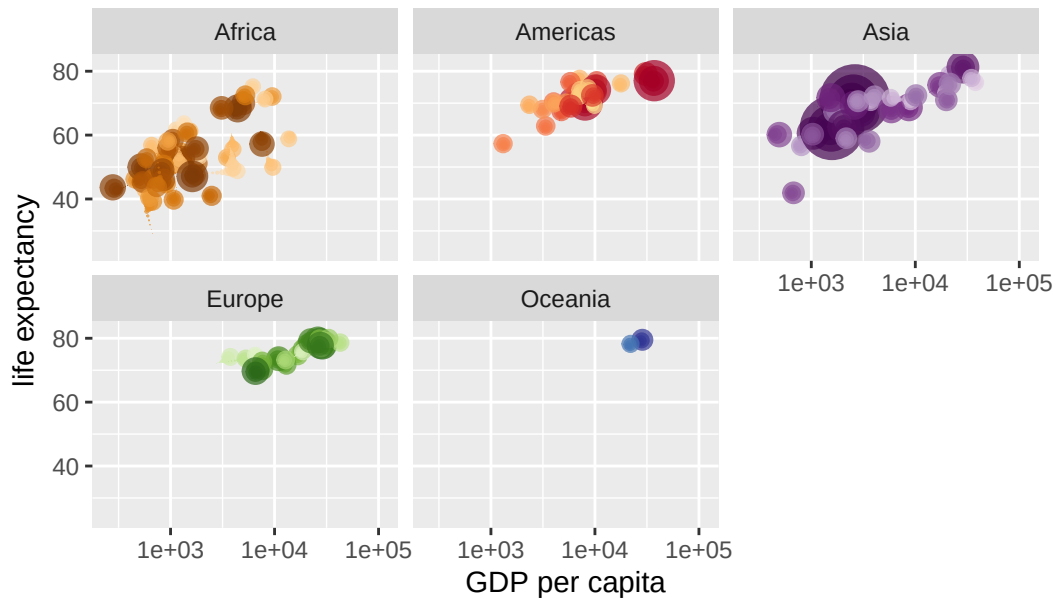
Year: 1998



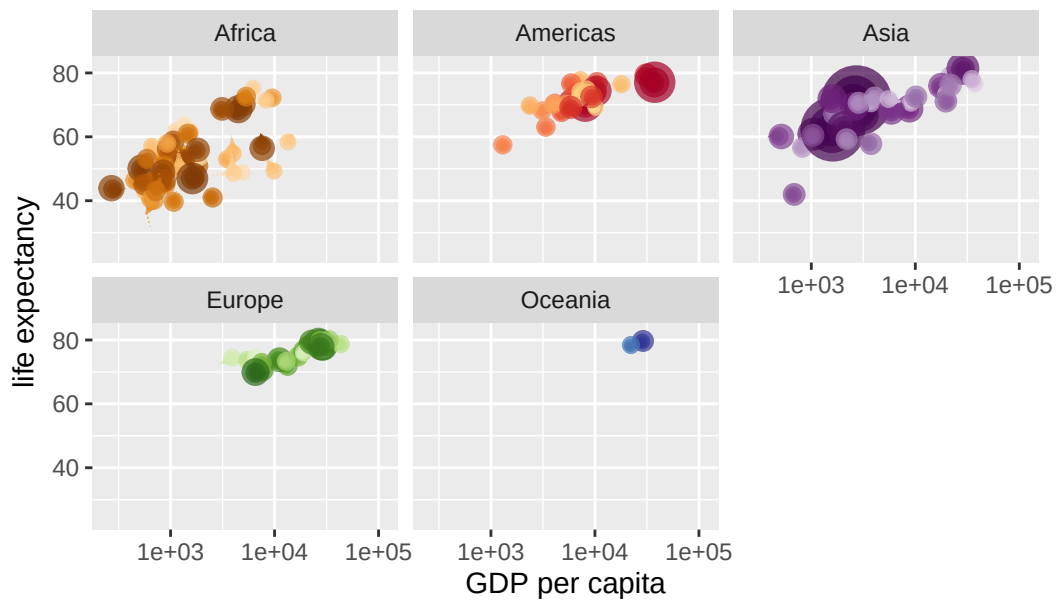
Year: 1999



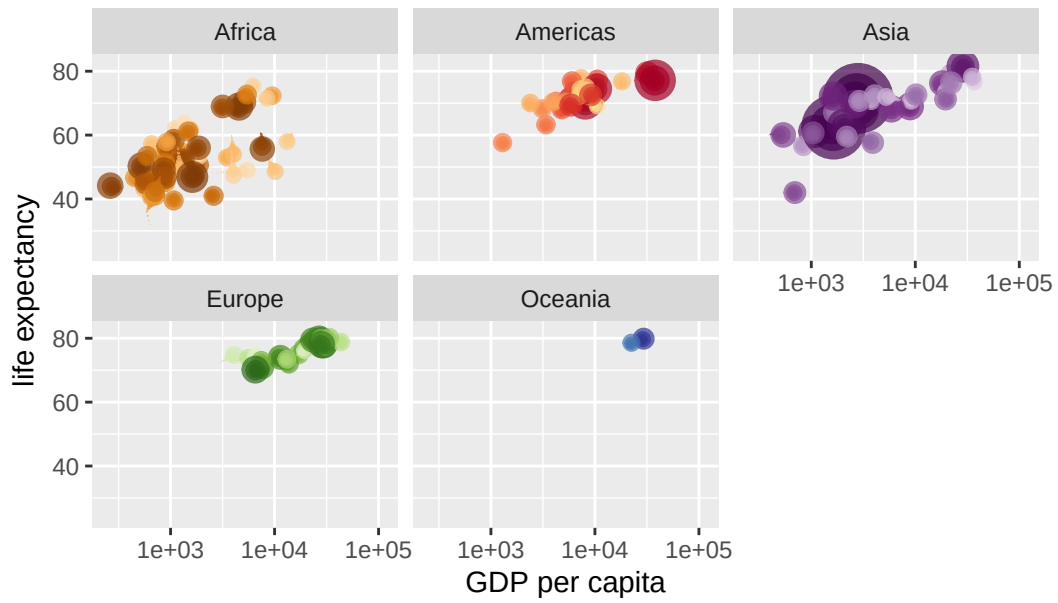
Year: 1999



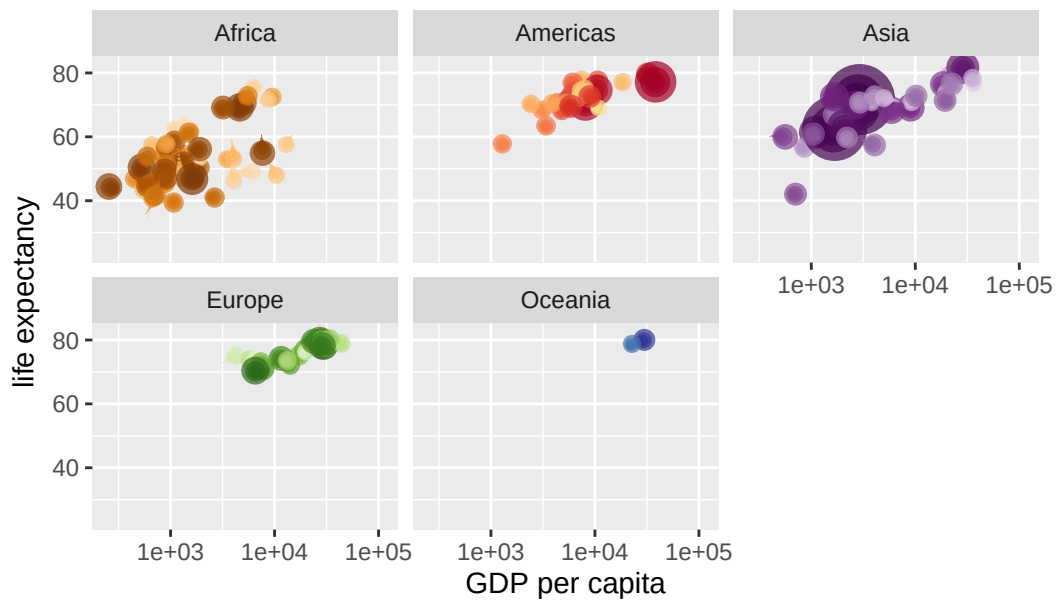
Year: 2000



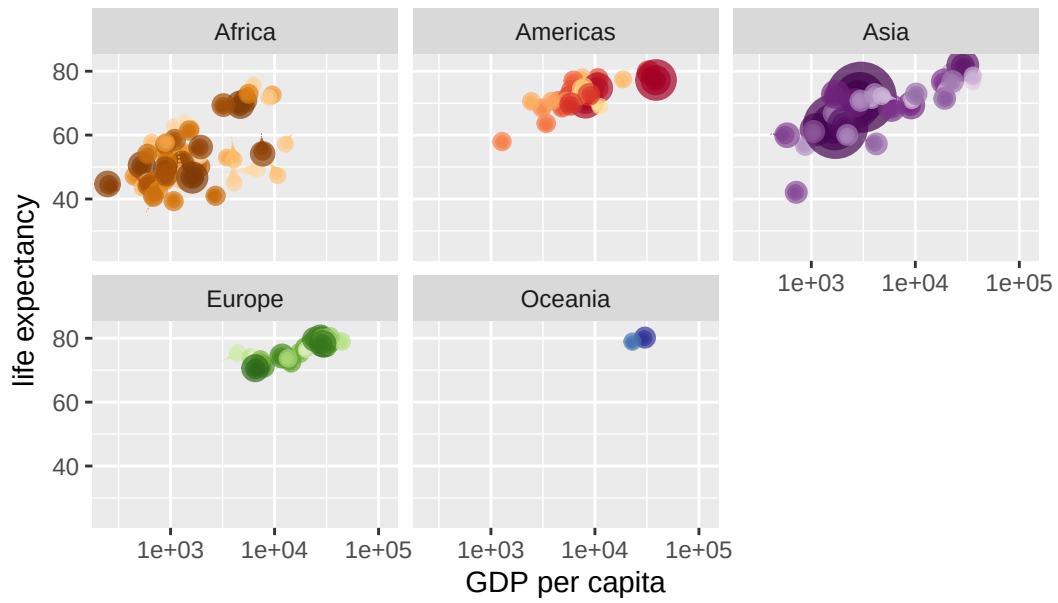
Year: 2000



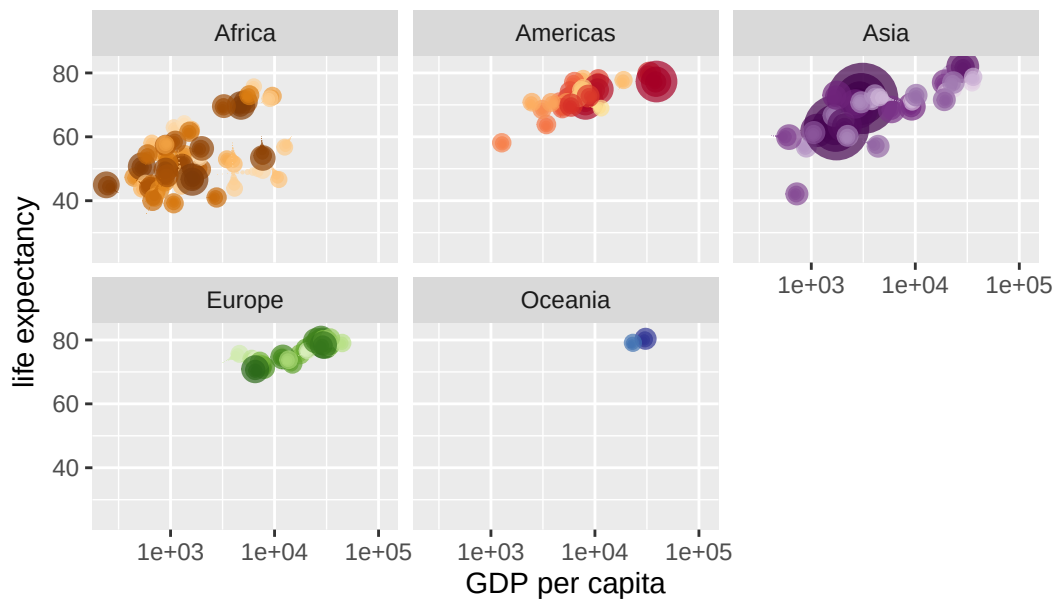
Year: 2001



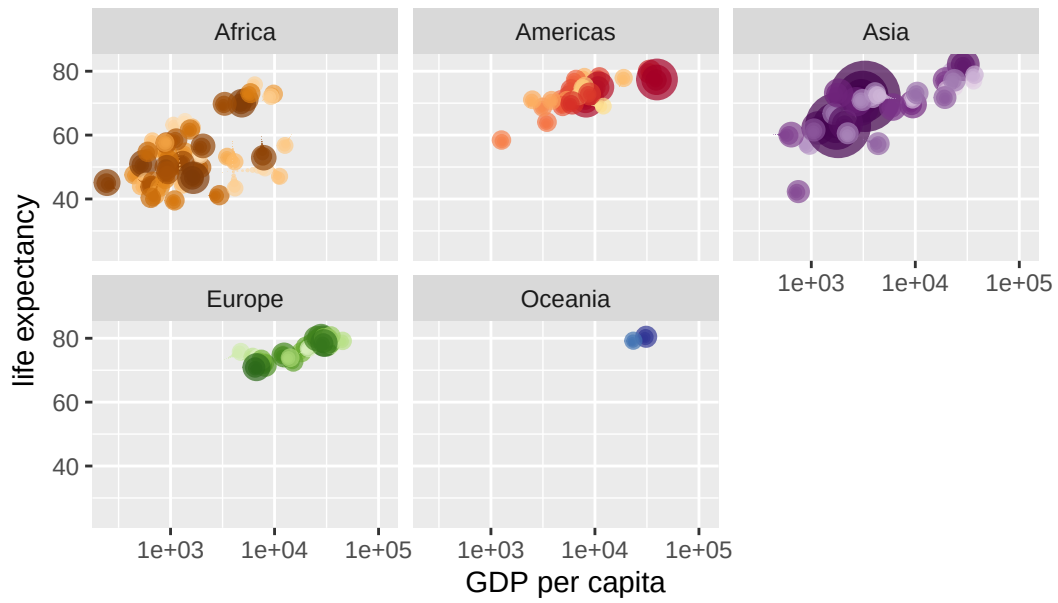
Year: 2001



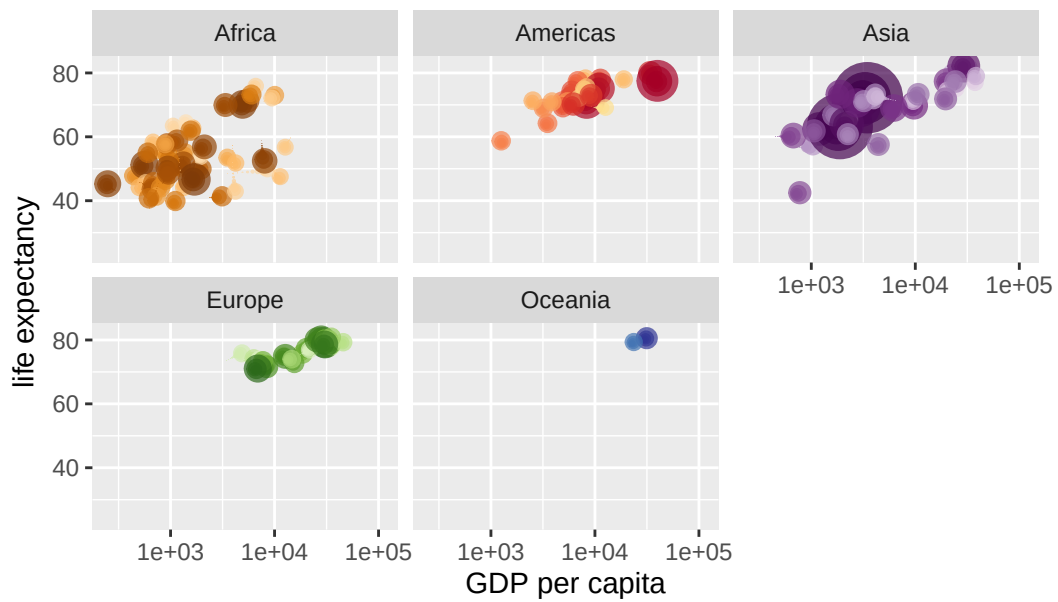
Year: 2002



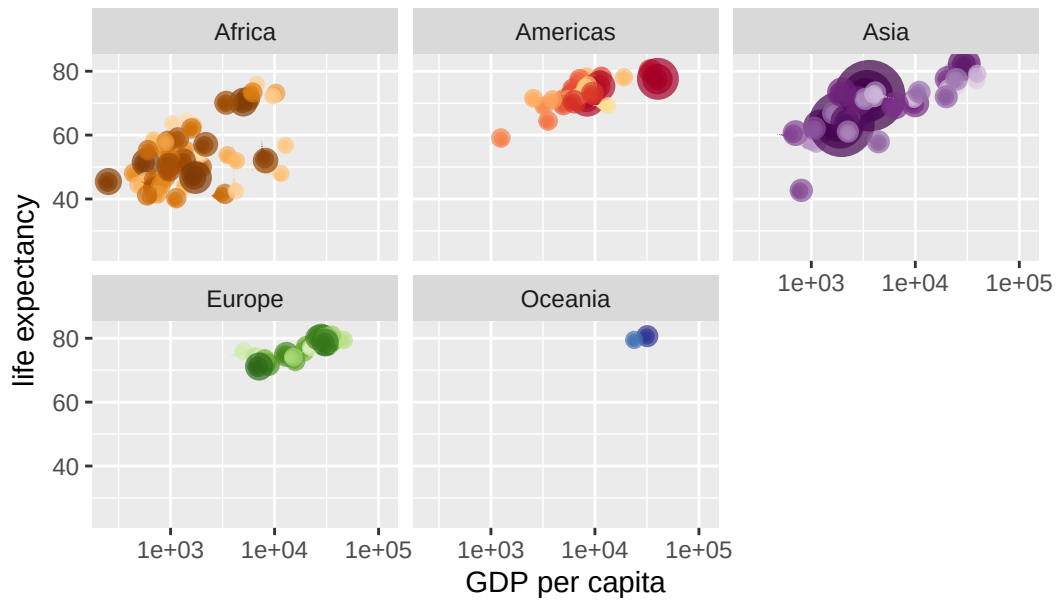
Year: 2003



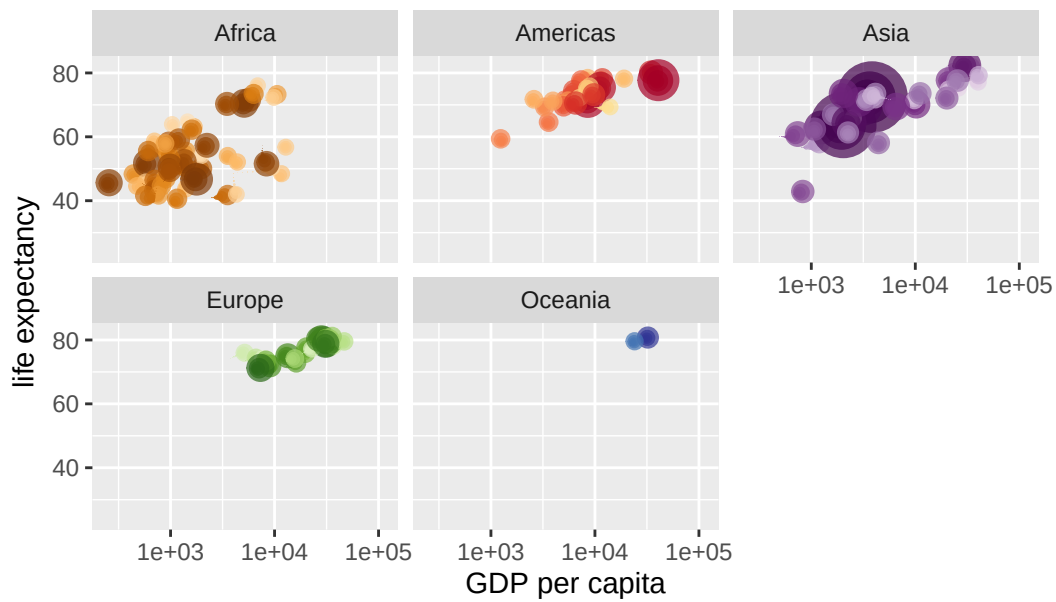
Year: 2003



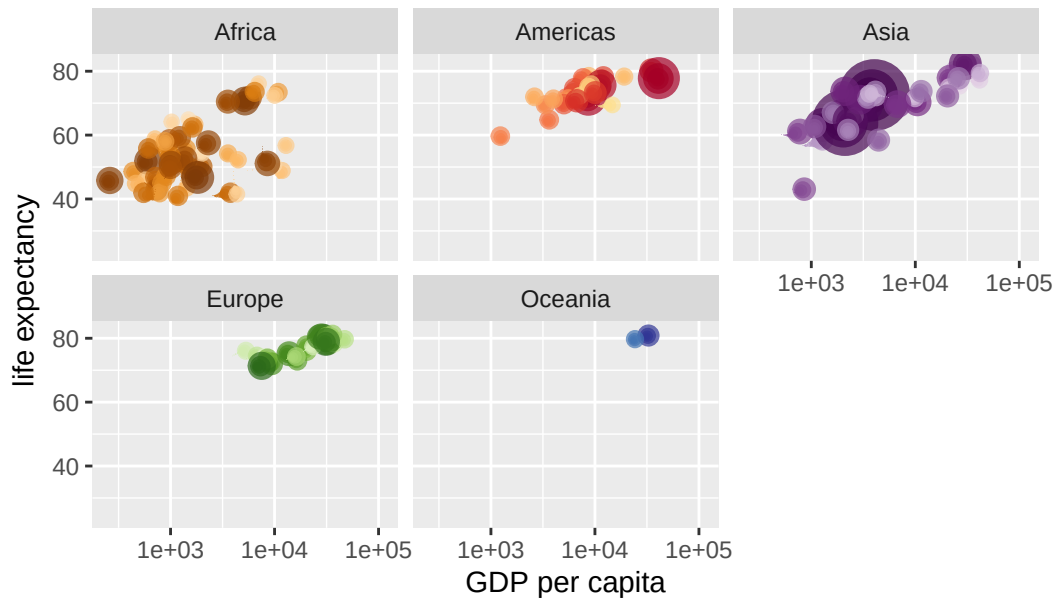
Year: 2004



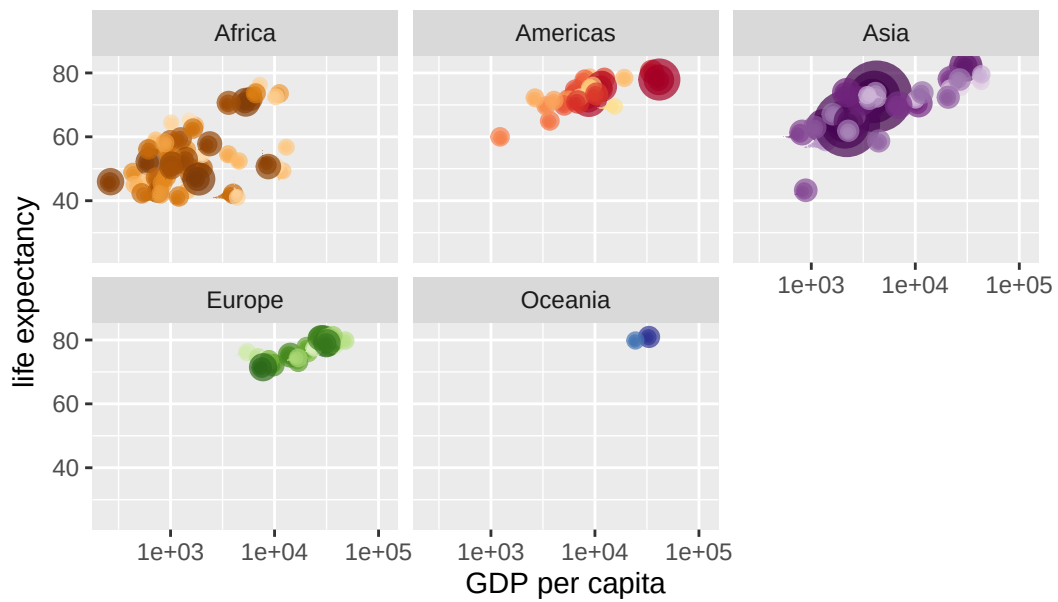
Year: 2004



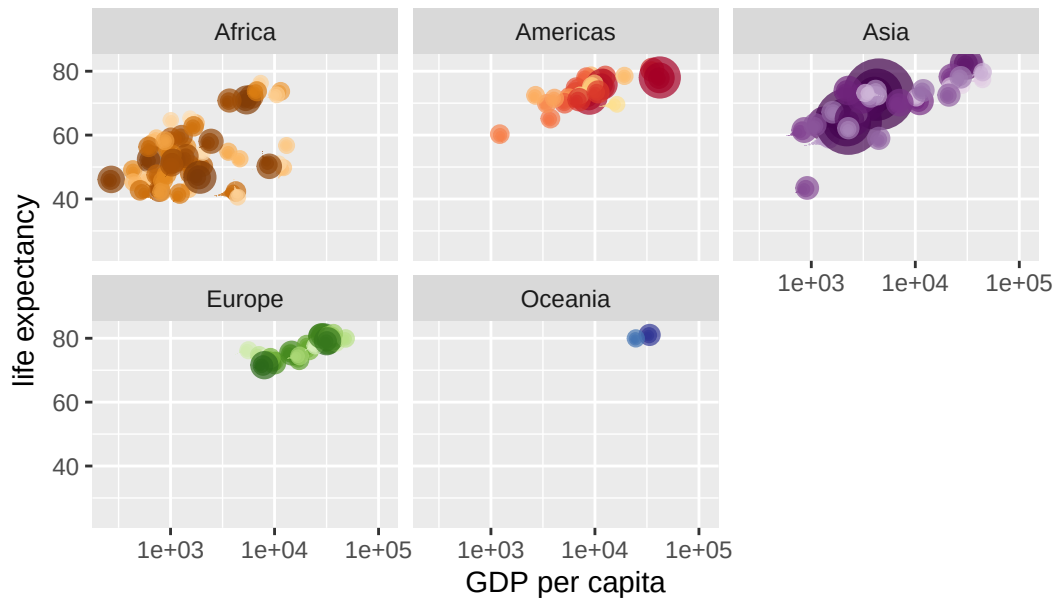
Year: 2005



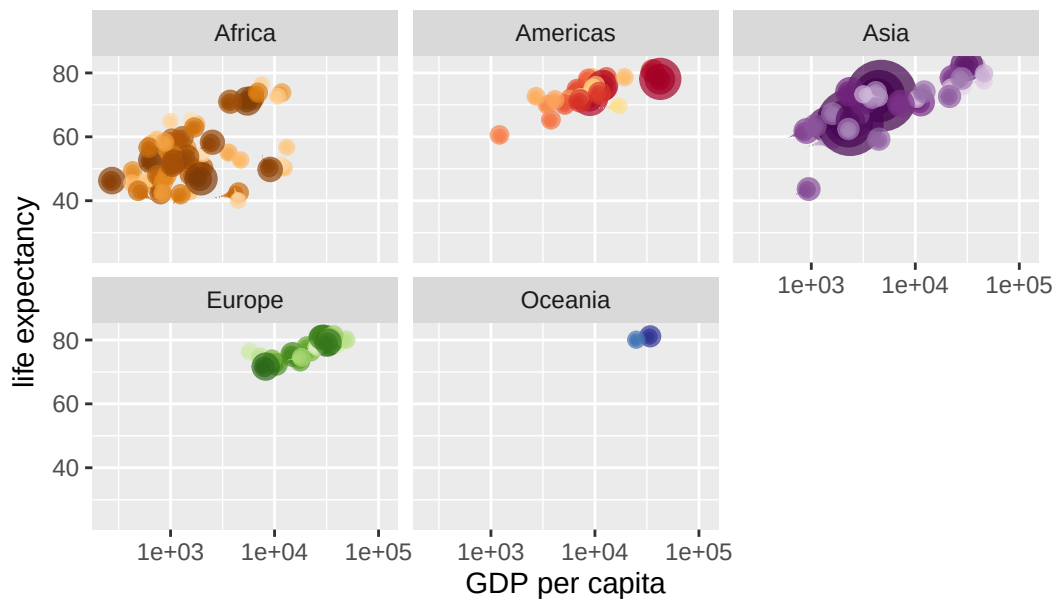
Year: 2005

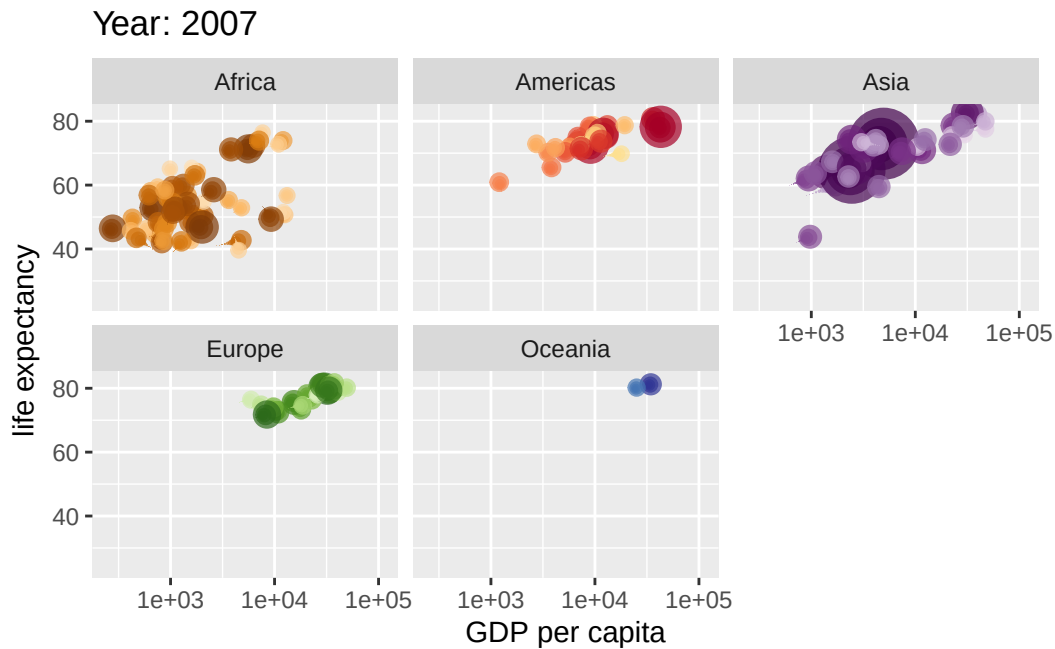


Year: 2006



Year: 2006





```
#adding a trail to the dots
```

Section 10: Combing plots

```
#using patchwork to combine all teh plots into a multi panel figure so we can see evevthing a

#installed patchwork in console

library(patchwork)

# Setup some example plots
p1 <- ggplot(mtcars) + geom_point(aes(mpg, disp))
p2 <- ggplot(mtcars) + geom_boxplot(aes(gear, disp, group = gear))
p3 <- ggplot(mtcars) + geom_smooth(aes(displ, qsec))
p4 <- ggplot(mtcars) + geom_bar(aes(carb))

# Use patchwork to combine them here, and setting up where i want the plots
(p1 | p2 | p3) /
  p4
```

``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'

